

LATIN AMERICA &
CARIBBEAN REGION

Environment &
Water Resources
OCCASIONAL PAPER SERIES



Framework for Conducting Benefit-Cost Analyses of Investments in Hydro-Meteorological Systems

June 2014



Framework for Conducting Benefit-Cost Analyses of Investments in Hydro-Meteorological Systems

June 2014

Arun S. Malik

Gregory S. Amacher

Jason Russ

Enos E. Esikuri

Keiko Ashida Tao

© 2014 International Bank for Reconstruction and Development / The World Bank
1818 H Street NW
Washington DC 20433
Telephone: 202-473-1000
Internet: www.worldbank.org

This work is a product of the staff of The World Bank with external contributions. The findings, interpretations, and conclusions expressed in this work do not necessarily reflect the views of The World Bank, its Board of Executive Directors, or the governments they represent.

The World Bank does not guarantee the accuracy of the data included in this work. The boundaries, colors, denominations, and other information shown on any map in this work do not imply any judgment on the part of The World Bank concerning the legal status of any territory or the endorsement or acceptance of such boundaries.

The Environment & Water Resources Occasional Paper Series is a knowledge product of the Environment Unit in the Latin America and Caribbean Region (LCSN) of the World Bank. The publications in this Series were designed and produced under the supervision of Emilia Battaglini, Acting Sector Manager, Keiko Ashida Tao, Environmental Specialist, and Anna-Lena Sauer, Consultant (LCSN).

A list of the most recent papers is located at the end of this publication. For electronic copies of all our LAC Environment & Water Resources Occasional Paper Series please visit our web site at <http://www.worldbank.org/LAC/LCSNSeries>.

Rights and Permissions

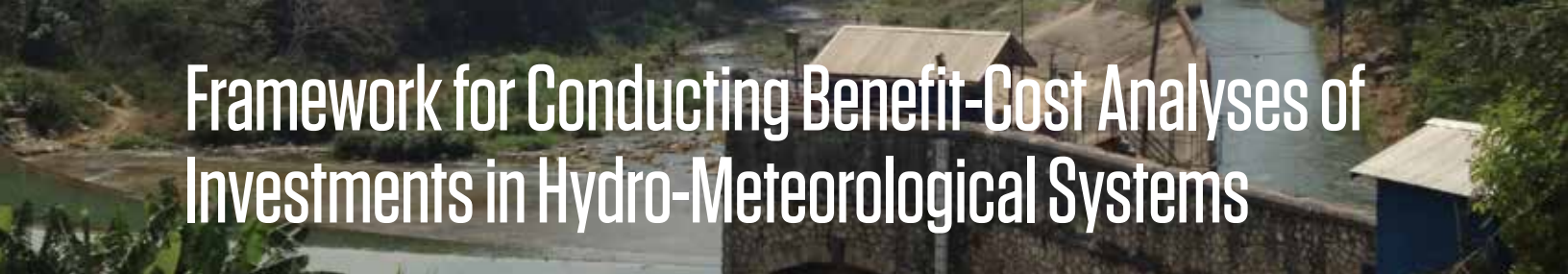
The material in this work is subject to copyright. Because The World Bank encourages dissemination of its knowledge, this work may be reproduced, in whole or in part, for noncommercial purposes as long as full attribution to this work is given.

Any queries on rights and licenses, including subsidiary rights, should be addressed to the Office of the Publisher, The World Bank, 1818 H Street NW, Washington, DC 20433, USA; fax: 202-522-2422; e-mail: pubrights@worldbank.org.

Photos courtesy of Rachel Pasternack.

Front cover: Doppler Weather Radar, Coopers Hill, Jamaica.

Back cover: Rio Cobre Flood Warning System, St. Catherine, Jamaica.



Framework for Conducting Benefit-Cost Analyses of Investments in Hydro-Meteorological Systems

Foreword

The Latin America and Caribbean (LAC) region has a unique mix of qualities and challenges when it comes to the environment. It is exceptionally endowed with natural assets—diverse ecosystems including the world’s greatest carbon sink in the Amazon, globally significant biodiversity such as the Mesoamerican Barrier Reef, and valuable crops. At the same time, the region registers the highest rates of urbanization in the developing world, water and natural resources overuse, and increased pollution, with detrimental consequence for the environment and the health of people, especially the poor.

Over the past twenty years, the LAC region has made impressive gains in addressing these issues. It leads the developing world in biodiversity conservation, natural and water resource management, and is at the forefront in reducing urban pollution. The World Bank often has the privilege to partner with countries in the region to pioneer innovative environmental policies and initiatives. Such initiatives include improvement of fuel and air quality standards in Peru, carbon emission reduction in Mexico, payment for ecosystem services in Costa Rica, participatory and integrated water resources management in Brazil, and new approaches to irrigation management in Mexico.

The Environment & Water Resources Occasional Paper Series, a publication of the Environment Unit of the Sustainable Development Department in

Latin America and Caribbean Region (LCSN) was launched in 2013. The objective of the Series is to contribute to global knowledge exchange on innovation in addressing environmental issues and the pursuit of greener and more inclusive growth. The papers seek to bring to a broader public—decision-makers, development practitioners, academics and other partners—lessons learned from World Bank-financed projects, technical assistance and other knowledge activities jointly undertaken with our partners. The Series highlights issues relevant to the region’s environmental sustainability agenda such as biodiversity conservation, natural and water resources management, irrigation, ecosystem services, environmental health, environmental policy, pollution management, environmental institutions and governance, environmental financing, and climate change and their linkages to development, growth and shared prosperity.

The cases presented in the Series show how the LAC region continues to make its growth more environmentally sustainable and inclusive. We hope that this Series will make a contribution to knowledge sharing among a wider audience within the LAC region and globally.

Emilia Battaglini
Acting Sector Manager
Environment Unit
Sustainable Development Department
Latin America and Caribbean Region

Table of Contents

Foreword	iii
Abbreviations	vi
Acknowledgements	vii
Executive Summary	viii
1. Introduction	1
2. Benefits of Hydromet Investments and Their Expected Development Impacts	3
3. Rationale for Hydromet Investments by Public Sector and World Bank's Involvement	7
4. Factors That Influence Forecast Value	9
5. Approaches to Estimating Benefits of Routine Climate for Specific Users/Sectors	12
6. Approaches to Estimating Country-Level Net Benefits of Hydromet Investments	16
7. Costs of Hydromet Investments	22
8. Proposed Framework for Estimating Net Benefits of Investments in Hydromet Systems	26
9. Interim and Ex-Post Evaluations of Hydromet Investments	35
10. Conclusions and Recommendations	38
Annex 1: Economic Value of a Forecast – An Illustrative Example	39
Annex 2: The Value of Hydromet Forecasts to the Insurance and Financial Sector	44
Annex 3: The Value to the Hydropower Sector of Improved Forecasts of Routine Climate	48
Annex 4: The Value of Hydromet Forecasts to the Transport Sector	50
Annex 5: Economic Analysis of Improving Climate Data and Information Management Project—Jamaica	54
References	70

List of Figures

Figure 1	Simplified Representation of How Hydromet Systems Generate Social Benefits	3
Figure 2	Overview of Framework for Estimating Net Benefits of Hydromet Investments Expected Net Benefits of Hydromet Investments	28
Figure A-1	Deriving the Expected Value of a (Perfect) Forecast.	41

List of Tables

Table 1	Examples of Benefits from Forecasts	5
Table 2	Comparison of Country-Level Approaches	17
Table 3	Benchmarking – First Stage Parameter Values	18
Table 4	Comparison of Results of Assessment of Hydromet Modernization Using Benchmarking and Sectoral Approaches	20
Table 5	Hydromet Modernization Cost Components from Selected Studies	25
Table 6	Expected Cost Example (Euros Million).	33
Table A-1	Payoffs for Alternative Harvesting Decisions	39
Table A-2	Benefits to Hydropower on the Missouri River from Improvements in Forecasting Ability	48
Table A-3	Estimated Annual Total Benefits (Euros) Per Capita and Per Driver from Weather Forecasts and Warnings Under Two Hydromet Assumptions	52
Table A-4	Examples of Benefits from Improved Climate Forecasts	55
Table A-5	Major Damage-Causing Meteorological Events, 2000–2010	57

Table A-6	Average Damages from Hurricanes by Sector (USD Million, Constant 2010)	61
Table A-7	Estimates of Percentage Losses Avoided with Project.	63
Table A-8	Expected Annual Benefits of Improved Forecasts of Extreme Hazards and Early Warning Systems, Baseline Year (USD Million, Constant 2010)	64
Table A-9	Expected Annual Benefits of Improved Forecasts of Routine Climate for Selected Sectors (Million USD, Constant 2010)	65
Table A-10a	Project Benefit-Cost Ratios Assuming No Climate Change for Alternative Benefits Estimates.	68
Table A-10b	Project Benefit-Cost Ratios Assuming Moderate Climate Change for Alternative Benefits Estimates	68
Table A-10c	Project Benefit-Cost Ratios Assuming High Climate Change for Alternative Benefits Estimates	69
Table A-11a	Project Benefit-Cost Ratios Assuming No Climate Change and Reduced Damage Due to Mitigation for Alternative Benefits Estimates	69
Table A-11b	Project Benefit-Cost Ratios Assuming Moderate Climate Change and Reduced Damage Due to Mitigation for Alternative Benefits Estimates	69
Table A-11c	Project Benefit-Cost Ratios Assuming High Climate Change and Reduced Damage Due to Mitigation for Alternative Benefits Estimates	69

Abbreviations

ADM	Administrative Cost	GFDRR	Global Facility for Disaster Reduction and Recovery
AHDL	Advisors to the High Level Group on Infrastructure Charging	HLC	Hospital Care Cost
ASEAN	Association of Southeast Asian Nations	NLP	Net Lost Production
BFFWC	Bangladesh Flood Forecasting and Warning Centre	PDV	Property and Material Damage Cost
CAT	Catastrophe	PLA	Percentage Loss Avoided
CCRIF	Caribbean Catastrophe Risk Insurance Facility	PLC	Price Level Coefficient
DHMZ	Meteorological and Hydrologic Service of Croatia	PPCR	Pilot Program for Climate Resilience
ENSO	El Niño Southern Oscillation	SPCR	Strategic Program for Climate Resilience
EPNF1	Expected Payoff for Option One	UNISDR	United Nations International Strategy for Disaster Reduction
EPNF2	Expected Payoff for Option Two	U.S.	United States
EPPF	Expected Payoff with Perfect Forecast	USD	United States Dollar
EUR	Euro	VOSL	Value of Statistical Lives Saved
EVPF	Expected Value of Perfect Forecast	WRMA	Weather Risk Management Association
GDP	Gross Domestic Product	WTP	Willingness To Pay
		WMO	World Meteorological Organization

Acknowledgements

This paper was prepared by Arun S. Malik (Professor, Department of Economics and School of Public Policy and Public Administration, George Washington University), Gregory S. Amacher (Julian N. Cheatham Professor of Natural Resource Economics, Department of Forest Resources and Environmental Conservation, College of Natural Resources, Virginia Polytechnic Institute and State University), Jason Russ (Adjunct Professor, George Washington University), Enos E. Esikuri (Task Team Leader and Sr. Environmental Specialist, World Bank), and Keiko Ashida Tao (Environmental Specialist, World Bank). This work was led by the World Bank

task team comprised of Enos E. Esikuri, Richard Damania, and Keiko Ashida Tao. The team would like to acknowledge the comments and the guidance received from the reviewers, which included Dugkeun Park, Rohan Longmore, and Momoe Kanada (World Bank).

Funding for this work was made possible through the Jamaica Improving Climate Data and Information Management Project which is financed by the Pilot Program for Climate Resilience (PPCR) under the Strategic Climate Fund of the Climate Investment Funds.

Executive Summary

The importance of hydro-meteorological (hydromet) services in the form of weather, climate, and hydrological forecasts is in many ways self-evident. Across the world, hundreds of thousands of weather forecasts, severe weather warnings, and climate predictions are issued each year. These forecasts are used by myriad users, ranging from households to firms to government agencies.

Determining the appropriate level and type of investment in hydromet systems requires a comparison of the social benefits and costs associated with different levels and types of investment. In recent years, tight government budgets have resulted in renewed efforts to identify and estimate the net social benefits of the services provided by hydromet agencies. The call for such estimates is especially strong when large-scale investments to improve or maintain hydromet services are being contemplated.

This whitepaper provides a survey of the issues involved in estimating the (expected) net social benefits of investments in hydromet systems, and it presents (in Section 8) a framework that can be used to obtain first-cut estimates of these net benefits at a country level by those tasked with evaluating hydromet investments. Specifically, the whitepaper:

- provides an overview of the types of benefits associated with hydromet investments, the process by which the benefits are generated, as well as their expected development impacts;
- explains the rationale for public sector investment in hydromet systems and for involvement by the World Bank;
- discusses the wide range of factors that influence the magnitude of benefits generated by hydromet systems;

- surveys approaches that have been used to estimate the value of improved forecasts to specific user groups or sectors of an economy, as well as approaches that have been used to estimate the net benefits of improved forecasts at the country level;
- lays out a framework for estimating the expected net benefits of hydromet investments at a country level without onerous data or analytical requirements; and
- describes data that can be collected to conduct interim and *ex-post* evaluations of hydromet investments that can potentially enhance the net benefits yielded by current and future hydromet investments.

The proposed framework draws on existing approaches with some important modifications and additions. In particular, it attempts to capture the benefits from improved forecasts of extreme meteorological events as well as the benefits from improved forecasts of routine climate. To the extent possible, estimates of the first type of benefits are based on historical data on losses from extreme events. The proposed framework makes extensive use of benefits transfer. However, the results of benefits transfer should be subject to review and revision by sector experts.

Given the high degree of uncertainty regarding the precise magnitude of benefits and costs associated with hydromet investments, conducting a wide range of sensitivity analyses is an essential component of the proposed framework. Sensitivity analyses should be conducted using alternative estimates of benefits and costs, alternative time horizons and discount rates, alternative assumptions about the consequences of climate change for the frequency and magnitude of extreme

events, and alternative assumptions about the ability of households and enterprises to adapt to these events.

Not all of the benefits of hydromet investments can be quantified and/or monetized. Although these benefits fall outside the framework of a traditional economic evaluation (in which all benefits and costs are monetized), they should not be ignored. Examples of such benefits are improved budgeting and contingency planning for extreme meteorological events, and improved ability to develop higher-resolution climate change scenarios that can enhance climate-resilience planning and decision-making.

Benefits of this type that are likely to be large in magnitude should be identified and characterized even if they cannot be quantified or monetized.

Supplementing *ex-ante* evaluation of a hydromet investment with interim and *ex-post* evaluations offers a range of benefits beyond the obvious ability to assess the validity of the *ex-ante* evaluation and refine it. In particular, priorities and allocations of funds for the investment being evaluated can be revised, and design of future hydromet investments can be improved. Accordingly, collection of data for conducting interim and *ex-post* evaluations plays an important role in enhancing the net benefits from hydromet investments in the long run, in addition to enhancing the ability to evaluate the investments *ex-ante*.



Framework for Conducting Benefit-Cost Analyses of Investments in Hydro-Meteorological Systems

1. Introduction

The importance of hydro-meteorological (hydromet) services in the form of weather, climate, and hydrological forecasts is in many ways self-evident. Across the world, hundreds of thousands of weather forecasts, severe weather warnings, and climate predictions are issued each year. These forecasts are used by myriad users, ranging from households to firms to government agencies. For the U.S. alone, it is estimated that households collectively make use of 300 billion weather forecasts each year (Lazo et al. 2009), and that approximately one-seventh of the country's economy is weather sensitive (National Research Council 1998).

These observations, striking though they are, tell us little about the appropriate level of investment by society in hydromet systems (Pielke and Carbone 2002). Determining the appropriate level and type of investment requires a comparison of the social benefits and costs associated with different levels and types of investment. In recent years, tight government budgets have resulted in renewed efforts to identify and estimate the net social benefits of the services provided by hydromet agencies (e.g., Pielke and Carbone 2002, Morss et al. 2008, Mills 2010). The call for such estimates is especially strong when large-scale investments to maintain or improve hydromet services are being contemplated.

In this white paper we provide a survey of the issues involved in estimating the (expected) net social benefits of investments in hydromet systems, and we lay out a tentative framework that can be used to estimate the net benefits of these investments

at a country level without onerous data or analytical requirements.

The whitepaper is organized as follows:

- Section 2 provides an overview of the types of benefits associated with hydromet investments, the process by which the benefits are generated, and their expected development impacts;
- Section 3 explains the rationale for public sector investment in hydromet systems and involvement by the World Bank;
- Section 4 discusses the wide range of factors that influence the magnitude of benefits generated by hydromet systems, in particular the value of weather/climate forecasts. The discussion is supplemented by a stylized example presented in Annex 1;
- Section 5 provides an overview of approaches that have been used to estimate the value of improved forecasts of routine climate to specific user groups or sectors of an economy;
- Section 6 then turns to an overview of approaches that have been used to estimate the net benefits of hydromet investments at the country level. The primary benefits estimated by these approaches are those associated with improved forecasts of extreme meteorological events;
- Section 7 contains a discussion of the costs of hydromet investments, with particular attention given to the challenges faced in estimating these costs in developing countries;
- Section 8 lays out a framework for estimating the expected net benefits of hydromet

investments at a country level. The proposed framework builds on existing approaches and is designed to be used with data available from secondary sources. This section will be of central interest to those tasked with conducting economic evaluations of hydromet investments;

- Section 9 describes data that can be collected to conduct interim and *ex-post* evaluations of hydromet investments that supplement and refine *ex-ante* evaluations of these investments; and
- Section 10 offers conclusions and recommendations.

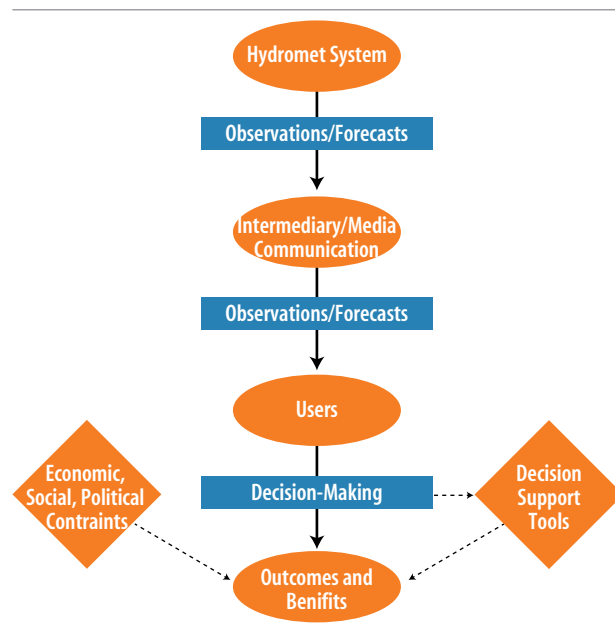
2. Benefits of Hydromet Investments and Their Expected Development Impacts

The benefits of hydromet systems stem from the services that they provide. Efforts to estimate the benefits of hydromet services date back at least 40 years. At this point there is a small, and still growing, literature that examines the use and value of weather forecasts.¹ Recent assessments of the current state of knowledge reveal that despite these efforts many challenges remain. Our understanding of the value of forecasts is still patchy and incomplete (e.g., Pielke and Carbone 2002, and Morss et al. 2008). In large part, this is a function of the very diverse set of users that make use of forecasts and the multitude of contexts in which forecasts are used. To accurately determine the value of forecasts to each of these users requires a fairly detailed understanding of how each user interprets forecasts and combines them with other information in its decision-making processes, and how the outcomes of the decision-making processes are altered by the forecasts. For instance, the manner in which a small farmer uses a forecast is far different from the manner in which a hydro-electric generator does so. The farmer's decision-making process is likely to be informal whereas the hydro-electric generator's is likely to make use of sophisticated decision support tools. Moreover, the changes induced by the forecasts in the outcomes of their decision-making processes are likely to vary markedly in both character and magnitude.

Even for a single farmer, the types of forecasts used by the farmer and the decision contexts can vary. For instance, the farmer's decision on which crop varieties to plant will depend on seasonal forecasts, such as forecasts of whether rainfall during a coming monsoon will be above or below average, whereas the farmer's decision on when to plant will depend on near-term forecasts of precipitation.

A stylized, yet informative representation of how forecasts result in social benefits is presented in Figure 1. The figure is adapted from Morss et al. 2008 and Lazo et al. 2008. The box labeled “Hydromet System” at the top of the figure encompasses the infrastructure of hydromet systems, including weather satellites, Doppler radar, radio-sondes, ocean buoys, river gauges, and weather stations, along with the data assimilation, numerical modeling and other activities that are undertaken to generate forecasts. These forecasts are communicated to users either directly or indirectly through intermediaries that repackage the forecasts (such as local news channels) or add value to them (such as specialized forecasts for the aviation industry).

Figure 1 Simplified Representation of How Hydromet Systems Generate Social Benefits



Source: Authors.

¹ Surveys of this literature can be found in Katz and Murphy (2005) and Katz and Lazo (2011).

A diversity of users combine these forecasts with other information in their decision-making processes to make choices. In the case of larger users, decisions are likely to be made with the aid of decision support tools. Users' choices together with actual meteorological conditions determine outcomes. These outcomes range from individuals not getting wet when it rains, to higher crop yields, to fewer heat-wave casualties. These outcomes determine the change in expected net economic benefits, broadly defined, enjoyed by users, or society, as a result of the forecasts. The economic value of (better) forecasts is the aggregate increase in expected (net) economic benefits enjoyed by all users.

Classification of Benefits. A wide range of benefits have been attributed to investments in hydromet systems and to improving dissemination of information about meteorological conditions and hazards. The benefits are associated with improved forecasts of extreme meteorological events such as hurricanes, storms, floods, and droughts, as well as improved forecasts of routine weather and climate conditions.

Improved forecasts of extreme meteorological events and effective dissemination of information about their effects and appropriate responses to them can substantially reduce economic losses caused by the events. Improved forecasts of routine climate can result in increased enterprise profits (or reduced costs) and improved decision-making by households. We classify benefits into three categories:

- benefits from improved forecasts of *extreme meteorological events*;
- benefits to *enterprises* of improved forecasts of *routine climate*; and
- benefits to *households* of improved forecasts of *routine climate*.

On an economy-wide level, the value of improved forecasts varies across sectors. Some sectors of the economy are more climate sensitive than others, and are therefore more likely to benefit from improved forecasts. The sectors generally considered to be climate sensitive are: agriculture, aviation, construction, surface and water transportation, water resources, energy, fisheries, forestry, health, and tourism and recreation (Houston et al. 2004 and World Bank 2008).

The Spectrum of Benefits. Table 1 presents examples of the use of different types of forecasts in different sectors of the economy and the variety of benefits that they generate. The table emphasizes benefits associated with forecasts of routine climate, which are often less apparent. The examples are drawn from a wide variety of sources. Following convention (e.g., Sene 2010), short-range forecasts generally extend 2 days from when the forecast is issued, whereas medium-range forecasts extend from 3 to 10 days into the future. Seasonal forecasts are descriptions of average weather parameters over the next 3 to 6 months, excluding individual events. The examples illustrate the pervasive importance of climate forecasts to economic activity.

Importance of Unquantifiable Benefits. An important subset of the benefits of hydromet investments is likely to be difficult to quantify. Ignoring such benefits is justifiable if, and only if, they are likely to be small in relative magnitude. If they are likely to be large in magnitude, they should be identified and described even if they cannot be quantified. Examples of benefits of hydromet improvements that are difficult to quantify but that can be large in magnitude include:

- Improved budgeting and contingency planning for extreme meteorological events. Better forecasts enable governments to develop better plans for coping with extreme events and

Table 1 Examples of Benefits from Forecasts

Sector	Period of Forecast	Benefits from Forecasts
Agriculture	Short-Range Forecasts	Information on daily precipitation is vital to pesticide application decisions, as heavy rains can wash away recently applied pesticides.
	Medium-Range Forecasts	Accurate medium term precipitation forecasts inform farmers whether or not they need to irrigate, and how much.
		Ideal seeding rates are sensitive to the weather conditions in the days and weeks following planting.
		Timing of planting and harvesting decisions can be improved.
	Seasonal Forecasts	Having more accurate data on the seasonal climate can aid farmers in determining which crops will yield more value.
		Crop insurance programs can benefit from reduced uncertainty of weather patterns.
		Agencies can anticipate food shortages earlier with better seasonal forecasts.
Household	Short-Range Forecasts	Weather forecasts are used to make everyday decisions, such as what to wear, or whether or not to take an umbrella.
	Medium-Range Forecasts	Early warnings of major storms can signal the need for a household to stock up on essentials in case of power outages or road closures, or for potentially life-saving evacuations in cases of extreme events.
	Seasonal Forecasts	Seasonal forecasts can inform households on many decisions, ranging from whether or not and how much insurance to purchase, to what type of household fortifications and improvements to undertake.
Energy	Short-Range Forecasts	Daily weather patterns have an effect on peak energy use patterns, and more accurate weather forecasts can inform power plants when to increase or decrease electricity production.
	Seasonal Forecasts	Hydro-electric generators benefit from improved streamflow forecasts.
Recreation and Tourism	Short-Range Forecasts	Golf course management, recreational fishing, and other outdoor and marine-based activities benefit from more accurate temperature and precipitation forecasts.
	Seasonal Forecasts	Tourist resorts make staffing and investment decisions based on expected tourists in a given season, which can be very sensitive to seasonal weather.
Transportation and Shipping	Short-Range Forecasts	Routing decisions of trucks, ships, and airplanes can be improved with better forecasts of daily weather conditions.
	Medium-Range Forecasts	Cargo ships can use better forecasts to minimize costs and delivery delays in maneuvering around unexpected storms.
Water Resource Management	Seasonal Forecasts	Improved forecasts can lead to more efficient reservoir operations and savings from avoiding groundwater pumping to augment reservoirs.
Fisheries	Medium-Range Forecasts	Fishing vessels rely on weather forecasts to determine when to set sail, how long to stay at sea, and where to navigate to avoid adverse weather conditions.
Emergency Response	Short-Range Forecasts	Adequate warning before extreme weather disaster can significantly reduce losses of life. First responders and emergency rescuers can pre-position emergency response assets to places where they will be most effective for rescue operations. More accurate forecasts can reduce the costs associated with “false alarms,” including the costs of evacuating communities.

Source: Authors.

develop better estimates of the sums needed to cope with such events;

- Increased ability to develop higher-resolution climate change scenarios that can enhance climate-resilient planning and decision-making;
- Improved ability of individuals, enterprises, and government agencies to adapt to changes in climate given better information about climate trends and future changes;
- Diminished likelihood of maladaptations because of poor or inaccurate information about climate trends and future changes in climate; and
- Facilitated development of weather insurance instruments that can help users manage the large risks associated with extreme meteorological events. Further detail on this benefit is provided in Annex 2 at the end of this document.

Monetized vs. Non-Monetized Benefits. An economic evaluation of a project should ideally monetize all benefits and costs, i.e., all benefits and costs should be expressed in terms of monetary (e.g., dollar) values. This allows for a clear-cut comparison of benefits and costs since they are

expressed in common units. In general, monetizing costs is not problematic given that they are easily observed and measured. In contrast, monetizing some benefits of hydromet improvements can be problematic. A necessary condition for monetizing benefits is the feasibility of quantifying the benefits in physical terms. For example, benefits of improved forecasts for agriculture can be quantified in terms of the magnitude of expected increases in annual crop yields. This quantified benefit can then be monetized by applying suitable prices for the crops. The examples in the preceding paragraph indicate that in some cases it may not be feasible to quantify benefits. In other cases, it may be possible to quantify benefits, but monetizing them may be difficult or controversial because identifying a suitable price (or economic value) to attach to the quantified change may be difficult. Examples of such benefits are expected reductions in mortality and morbidity due to more timely, or more accurate, forecasts of extreme meteorological events. Although economic values can be attached to these benefits, the precise values to be attached can be subject to disagreement. In such cases, benefits can be expressed in their “native” units instead of being monetized.

3. Rationale for Hydromet Investments by Public Sector and World Bank's Involvement

Hydromet services are typically divided into two broad categories: (i) basic services and (ii) special services. Both of these services are derived from the basic data collection and processing infrastructure that make up a hydromet system, often referred to as *basic systems*. Basic services comprise the weather and climate forecasts made available to the public through mass media. In addition, basic services generally include forecasts designed for large community sectors, such as farming, fishing and sailing. In contrast, special (or value-added) services are tailored to specific users or small user groups, such as construction, shipping and aviation. The specialized forecasts differ from basic forecasts in the frequency of the forecasts, their spatial location and resolution, and the set of weather variables that are forecast.

Basic services are traditionally provided by the public sector through a national meteorological service, as part of a government's obligation to protect the life and property of citizens and to enhance their welfare. These basic services, and the basic systems on which they are based, have long been viewed as best provided by a national government to avoid the wasteful duplication of infrastructure that would arise if the services were provided by competing private enterprises (Zillman and Freebairn 2001).

A further rationale for public sector provision of basic services is that they have the characteristics of "pure public goods". The defining characteristic of a public good is that provision of the good to one group (of individuals or enterprises) does not diminish the ability of another group to simultaneously benefit from the good. For example, if households benefit from a forecast of the likelihood of precipitation tomorrow, this does not diminish the ability of enterprises to benefit from the same

forecast. The distinguishing characteristic of *pure* public goods is that it is infeasible or prohibitively expensive to exclude one group from benefiting from the good if it is made available to some other group. For example, if basic weather forecasts are made available to households, it is infeasible to exclude enterprises from learning of these forecasts and benefiting from them. The pure public good nature of basic weather forecasts implies that it is infeasible for the private sector to provide them, or to provide them at a socially desirable level. Attempts by a private forecast provider to collect payments for the forecasts would be thwarted by the ability of users to free-ride off forecasts purchased by others. In general, pure public goods, such as basic weather forecasts, are therefore provided by the public sector and paid for out of general revenues.

Specialized (or value-added) weather forecasts, as opposed to basic forecasts, are provided successfully by the private sector, or by the public sector, for a fee. As noted, these forecasts are tailored to specific users or small user groups. The tailored nature of these forecasts implies that forecasts prepared for one user group are unlikely to be of high value to another user group, diminishing the free-riding problem. For example, specialized forecasts prepared for offshore oil and gas producers are likely to be of little value to inland construction companies.

Building climate resilience is critical for the World Bank Group's twin goals to end extreme poverty and build shared prosperity. Climate change will have huge impacts on poor and marginalized populations, who commonly live in the highest-risk areas and have the least ability to recover from extreme and/or recurrent low-intensity events. Many low income countries are also those with the least capacity to prepare for, and absorb, the effects of climate events. Unless development projects address

their preparedness and adaptation to current and future climate and disaster risks, physical investments made would be repeatedly wasted after extreme climate events and disasters occur. With increasing change in the climate and uncertainty of natural disasters, building climate and disaster resilience must assume a central role in development and in national plans. Risk reduction and better preparedness to deal with climate and disaster impacts can substantially decrease the cost of disasters. Early warning systems, better preparedness, and improved safety codes have proven to be cost effective, save human lives, and protect public and private investments. Climate and disaster resilient development, therefore, makes sense both from poverty alleviation and economic perspectives (World Bank/GFDRR 2013).

Weather, climate and hydrologic monitoring and forecasting are essential to inform decision-making for climate resilience and provide critical inputs to early warning systems. The World Bank currently

supports hydro-meteorological investments close to USD 500 million, including integrated support in Central Asia, Mozambique, Nepal, and Yemen. In 2011, the Global Facility for Disaster Reduction and Recovery (GFDRR) launched a hydro-meteorological initiative to support and leverage World Bank investments to strengthen weather, climate and hydrological services, and ensure that World Bank investments support and contribute to international norms, standards, systems and efforts under the auspices of the World Meteorological Organization (WMO) and the Global Framework for Climate Services (World Bank/GFDRR, 2013). The Pilot Program for Climate Resilience (PPCR) has helped channel more than USD 95 million in investments and USD 50 in co-financing through the Strategic Programs for Climate Resilience (SPCRs) into hydromet related projects in Cambodia, Grenada, Nepal, Niger, Saint Vincent and the Grenadines, Samoa, Tajikistan, Zambia, Mozambique, Jamaica, Yemen, and Bolivia.

4. Factors That Influence Forecast Value

As the examples in Table 1 earlier suggest, the magnitudes of benefits associated with improved forecasts will vary across users and decision contexts. In addition, the benefits depend on the precise characteristics of the forecasts being evaluated. The forecast characteristics that are of importance are (e.g., Mjelde et al. 1989, Stern and Easterling 1999, Blench 1999, Houston et al. 2004, and Teisberg and Weiher 2009, Lazo and Waldman 2011):

- time frame (span) of the forecast, e.g., forecasts can be made of tomorrow's weather or the coming summer's average weather;
- lead time of the forecast, i.e., the length of time between the issuance of a forecast and the time of the event forecasted;
- spatial resolution of the forecast;
- set of weather parameters forecast, e.g., rainfall, temperature, etc.; and
- perceived and actual accuracy of the forecast—perceived accuracy can differ from actual accuracy, especially if past forecasts have been wrong.

Longer lead times for forecasts are obviously beneficial if they are not accompanied by marked increases in uncertainty. For example, a very accurate flood forecast with limited lead time might enable households to flee a flood zone, but might not give them time to move their belongings to higher ground. The importance of lead times can be more pronounced in poor communities: individuals who do not have access to transport or are reliant on animal-powered transport cannot move themselves and their belongings as quickly as those who have access to motorized transport.

Ability of Users to Understand and Incorporate Forecasts in Decision-Making. For forecasts to be of value to users, they must be presented in a

manner that users can understand and incorporate in their decision-making processes. It has traditionally been assumed that providing more accurate forecasts at higher spatial and temporal resolutions will enable users to make better decisions, which in turn will result in larger benefits (Millner 2009). However, as has already been emphasized, users differ widely. Some may be able to process and benefit from improved forecasts, but others may not and may instead be overwhelmed when provided with more detailed and disaggregate forecasts (Morss et al. 2005, 2008).

In countries that have fairly modern, well-developed hydromet systems, investments in improving how forecasts are communicated and incorporated into users' decision-making process may yield higher benefits than investments in more sophisticated infrastructure. As Mills (2010) notes, "...it may be more effective to change the message or the medium rather than invest in a new monitoring technology or supercomputer. Often this involves greater consideration of users and the decisions they face".

In countries with outdated or poorly-developed hydromet systems, investments in infrastructure are a necessary condition for providing basic meteorological services. However, investments in infrastructure alone will not ensure that the potential benefits of providing these services are realized. Investments in systems that effectively communicate the forecasts to users and help them take advantage of the forecasts is critical. This is especially true for "small" users, such as individual households or small enterprises, and it is likely to be especially true in the case of small users in low-income countries who have limited education.

Even "large" users may be overwhelmed by more detailed and disaggregated forecasts. In a recent

paper, Webster (2012) observes that: “Paradoxically, as forecasts become better and their resolution grows, it becomes more difficult for developing countries to access them.” He notes that global weather forecasts prepared by the National Centers for Environmental Prediction in the United States are posted daily on the Internet. However, downloading the large amounts of information associated with these forecasts and extracting the relevant regional component in a timely manner requires reliable, high-speed internet access. In developing countries, this is problematic given often costly and slow internet connections. Thus, for the benefits of investments in improved hydromet systems to be fully realized, investments in associated communications infrastructure may well be needed.

In addition to this data acquisition problem, there is the issue of processing the data to generate informative local forecasts in a timely manner. Webster (2012) cites the case of the 1 to 10-day flood forecasting system for Bangladesh created in 2007 by the European Centre for Medium-Range Weather Forecasts and other parties. In 2009, in an effort to boost regional capacity, the flood forecast modules were handed over to the Bangladesh Flood Forecasting and Warning Centre (BFFWC). However, the large volume of data generated by these modules proved to be too difficult for the BFFWC to handle. As a result, responsibility for using the models to prepare flood forecasts was handed over to an international non-government entity, the Regional Integrated and Multi-Hazard Early Warning System.

Conveying Forecast Uncertainty. As noted in the above list of forecast characteristics, users must trust the accuracy of forecasts if they are to act on them (e.g., Stern and Easterling 1999, Chapter 4; Blench 1999, World Bank 2008, Chapter 2). What is of importance is the perceived accuracy of forecasts. A forecast’s perceived accuracy

will generally differ from its actual accuracy, especially if past forecasts have been wrong. For example, in South Florida, there was limited response to warnings issued in advance of Hurricane Wilma in 2005. One of the explanations offered for the limited response is that almost all of the six hurricane warnings issued in the preceding two years had been false alarms (Meyer 2006).

Strikingly, most forecasts issued today are single-valued: they provide a predicted value of a meteorological parameter with no objective information about the accuracy of the forecast. As a result, any information about the forecast’s accuracy must be derived by the user, and will therefore be subjective. This is true even though forecasts are invariably subject to uncertainty, with the uncertainty typically increasing with the lead time of the forecast. There has been a push in recent years to convey forecast uncertainty to users, so that users are able to make better use of the forecasts in their decision-making (National Research Council 2006, American Meteorological Society 2011). In principle, conveying forecast uncertainty is also likely to engender greater long-run confidence in forecasts among users, because they will be explicitly aware of the probability that a forecast is wrong.

However, simply conveying uncertainty about forecasts is not sufficient. Many, perhaps most, users may not have the skills needed to properly incorporate the uncertainty in their decision-making (Morss et al. 2008). Some users may in fact prefer the seeming certainty of single-valued forecasts to the explicit uncertainty of probabilistic forecasts. Accordingly, calls for forecasts to include information about uncertainty have been accompanied by recommendations that users be educated on how to interpret and make use of this information (e.g., American Meteorological Society 2011). This is a significant challenge given the large number of

users who make use of forecasts and the diversity of their decision contexts, and it is a challenge in both developing and developed countries.

Costs of Forecast Inaccuracy. The inaccuracy of forecasts can result in costs being incurred that would not be incurred in the absence of the forecast. For example, a forecast of a hurricane can prompt evacuation of communities. If the forecast proves to be incorrect, the costs of evacuation cannot be recouped (they are sunk). For the U.S., a back-of-the-envelope estimate of the average annual cost of false alarm evacuations is USD 1 billion. More accurate forecasts offer the potential for reducing the costs of such false alarms (Regnier 2008).

Ability of Users to Respond to Forecasts. In terms of the benefits that forecasts generate, no less important than communicating the forecasts effectively is ensuring that users are aware of how to make use of the forecasts and respond to them. In the case of flood forecasts for Bangladesh, Webster (2012) observes that to promote the effective use of the forecasts, village and community leaders were trained to interpret forecasts and to recommend appropriate actions for communities, such as harvesting crops, sheltering animals, storing clean water, and securing belongings.

In addition to knowing how to make use of forecasts and respond appropriately to them, the ability of users to actually do so will determine the value that they derive from the forecasts. Users often face economic, social, and political constraints that limit their ability to take appropriate action. This is clearly true in developing countries, but it also true in developed countries. Consider the case of Hurricane Katrina. Despite repeated and early warnings of the hurricane, approximately 100,000 residents of New Orleans did not have the means to leave the city. Moreover, the city was unprepared to

provide suitable shelter for this number of people (Meyer 2006).

In a unique study of flood risk management in Colorado, Morss et al. (2005) note that probabilistic forecasts may complicate the already difficult job faced by practitioners because they do not have the time or resources for the more complex analyses necessitated by such forecasts. In addition, Morss et al. (2008) find that local flood plain managers “must also balance the needs and desires of multiple constituencies, including local elected officials, private businesses, and the diverse populations at risk, and respond to these constituencies’ decisions and demands”.

The above discussion makes clear that what happens at the bottom of the chain from forecast generation to outcomes in Figure 1 is critical to economic benefits being realized from forecasts. Providing forecasts does not by itself ensure that net benefits will be realized from them, no matter how detailed, accurate and timely the forecasts may be. Similarly, an understanding of what happens at the bottom of the chain is critical to accurately estimating the economic value of forecasts. Annex 1 presents a simple economic example that illustrates the importance of a number of the factors described above that influence the economic value of forecasts.

Maximizing Forecast Value. Figure 1 suggests a uni-directional flow from hydromet systems to outcomes and benefits. However, in recent years analysts have noted that a feedback loop from outcomes to hydromet systems is desirable. Better information about how users make use of different forecasts and about the magnitudes of benefits derived can help guide decisions on the types of hydromet services to provide or enhance, with an emphasis on high-value or “high-impact” weather forecasts in order to maximize the net benefits generated by hydromet systems (Lazo et al. 2008 and Morss et al. 2008).

5. Approaches to Estimating Benefits of Routine Climate for Specific Users/Sectors

A survey of the literature indicates that there have been several dozen efforts to empirically estimate the value of forecasts to specific users or to specific industries/sectors. These studies typically estimate the value of current forecasts or the increase in value from improving one or more dimensions of current forecast quality, such as accuracy of the forecast, its frequency, its lead time, or the medium through which it is conveyed. Surveys of these studies can be found in Nichols (1996), Anaman et al. (1998), Stern and Easterling (1999, Chapter 5), Houston et al. (2004), Katz and Murphy (2005), Weiher et al. (2005), Teisberg and Weiher (2009), Rogers and Tsirkunov (2010), Mills (2010), and Katz and Lazo (2011).² Not surprisingly, the studies have focused on users in the most climate-sensitive sectors of the economy, in particular agriculture, but they encompass a wide range of sectors. A preponderance of the studies are of the value of forecasts in the United States.

A variety of analytical approaches have been used in these studies. Following Katz and Murphy (2005) and Katz and Lazo (2011), the approaches can be divided into four categories:

- prototype decision-making models;
- prescriptive (or normative) decision-making studies;
- descriptive decision-making studies; and
- stated preference methods.

The first three approaches rely on simulating a user's decision-making process and the role that forecasts play in it. Prototype decision-making models are abstract and theoretical in nature, whereas the prescriptive and descriptive decision-making

studies attempt to replicate real-world decision-making contexts more faithfully. In contrast, stated preference methods rely on individuals' responses to hypothetical survey questions to infer the value users place on forecasts; there is no attempt to simulate a user's decision-making process. A brief description of each approach is provided below, along with examples of applications of each, where appropriate.

Prototype Decision-Making Models. As their name suggests, these models are highly simplified versions of real-world situations that capture only a few critical features of the situation being modeled. The simplification is motivated by the desire to obtain analytical results about the relationship between the value of a forecast and characteristics of the forecast or the situation being modeled. The approach is prescriptive, or normative, in nature, in that it prescribes how a rational decision-maker that wants to maximize its welfare should make use of a forecast.³ The objective is to obtain general insights that ideally extend to more complex real-world situations.

Prescriptive Decision-Making Studies. These studies share the prescriptive, or normative, character of prototype decision-making models, but incorporate much more detail about the specific, real-world decision context being modeled. Unlike prototype decision-making models, the objective is to obtain numerical estimates of the value of forecasts (or improved forecasts) in the situation being modeled rather than general analytical insights.

For example, Mjelde et al. (1988) study corn production in Illinois and assess how characteristics

² Details on some, more recent studies can be found online at <http://www.isse.ucar.edu/staff/katz/esig.html#cases>.

³ More formally, the models employ a Bayesian decision-theoretic approach and assume that the user maximizes expected utility.

of the agricultural producer's environment as well as characteristics of the forecast, namely accuracy and lead time, influence the value of a forecast. The producer decisions modeled include the timing of nitrogen fertilizer application, selection of the hybrid variety to plant, planting density, and the timing of both planting and harvesting. Seasonal forecasts are examined with variations in the lead time of the forecast and its accuracy. They find that forecasts are of value only if they alter the producer's decisions. The estimated value of a forecast depends on its accuracy and lead time, as well as on the price of corn; estimates obtained range from tens of cents per acre per year to nearly USD 30 per acre per year. Interestingly, they also find that a less accurate forecast received earlier can have greater value than a more accurate forecast received later.

As is true for nearly all studies of the value of forecasts, Mjelde et al. use a "partial-equilibrium" framework. Specifically, they assume that changes in yields induced by better forecasts do not affect the market price of corn. But improved weather forecasts are likely to benefit a large portion of a country's agricultural producers and are therefore likely to result in shifts in the supply of agricultural commodities, leading to changes in market prices. Solow et al. (1998), building on earlier work by Adams et al. (1995), take these "price effects" into account when estimating the value of the El Niño Southern Oscillation (ENSO) forecasts for U.S. agriculture in general. The study links together scientific and economic models that trace the effect of ENSO forecasts on crop yields, which in turn affect producers' cropping decisions, the supply of crops, and thus market prices. The value of the forecast is calculated in terms of the change in economic welfare as measured by the change in the sum of producers' and consumers' surpluses. The estimated value of ENSO forecasts range from USD 240 million to USD 323 million per year, depending on the

assumed quality of the forecast. The authors note that these numbers may be underestimated because they are derived from the assumption that producers alter only their cropping decisions in response to the forecasts. In reality, producers could also take advantage of the forecasts by altering their input use and harvesting decisions. As is true for all prescriptive studies, there is also a possibility that forecast value is overestimated because the analysis assumes that producers make optimal use of the forecasts.

Descriptive Decision-Making Studies. Like prescriptive decision-making studies, descriptive decision-making studies incorporate much more detail about the situation being modeled than prototype decision-making models. However, unlike prescriptive decision-making studies, they attempt to model how users actually behave and make use of forecasts instead of assuming ideal, optimizing behavior. Descriptive studies can range from simple surveys of forecast users, to in-depth interviews of users, to monitoring of users in the course of making use of actual or hypothetical forecasts. These studies often do not develop numerical estimates of forecast value. Instead, they are used to complement and inform prescriptive studies.

For example, Stewart et al. (1984) conduct a descriptive study of how fruit growers in Washington state use frost forecasts to make decisions about protecting their orchards. The study follows up on an earlier prescriptive study (Katz et al. 1982) of the same set of users. The prescriptive study assumed that growers only used the frost forecasts in their decision-making. In contrast, the descriptive study revealed that growers also made use of dew point and temperature observations obtained after the frost forecasts were issued. Growers used this additional information to make a series of decisions on whether to protect their orchards over the course of a night, as opposed to the single,

irreversible decision early in the night that was assumed in the prescriptive model. Stewart et al. modify the prescriptive analysis to obtain more realistic estimates of the value of frost forecasts. The modifications resulted in a small drop of USD 45 per acre in the estimated value of frost forecasts from the original estimate of USD 808 per acre.

Stated Preference Methods. These methods rely on surveys of samples of users to directly or indirectly obtain estimates of the value users place on forecasts (or improvements to forecasts). Stated preference methods can take the form of stated value or stated choice surveys. In the case of stated value surveys, users are directly asked what they are willing to pay for a forecast or an improvement to a forecast. In the case of stated choice surveys, users are asked questions that require them to make choices among alternative forecasts with different costs associated with each. Users' preferences indirectly reveal the value they place on the forecast characteristics.

For example, Lazo and Chestnut (2002) used both stated value and stated preference methods to estimate the value that U.S. households place on (then) current forecasts, as well as the value they place on improvements to the forecasts. A total of 391 individuals in 9 cities across the U.S. were surveyed. The survey results indicated that on average each household was willing to pay USD 109 per year for current forecasts. The average value placed by households on improving multiple dimensions of current forecasts, specifically, their frequency, accuracy, and geographic detail, to their maximum possible level was estimated to be USD 16 per year. In a more recent study, Lazo et al. (2009) rely on stated value surveys of a much larger number of individuals (1,520) and estimate the median value placed by U.S. households on current weather forecasts to be USD 240 a year.

Some economists have expressed skepticism of the value estimates derived from stated preference surveys, arguing that they are likely to be biased upwards (e.g., see Diamond and Hausman 1994). However, this criticism is based on their use in valuing environmental goods and services. In the case of valuing forecasts, the direction of bias is more ambiguous. As Smith (2008) points out, stated preference surveys may understate benefits if respondents are not fully aware of how they directly and indirectly benefit from forecasts. For example, he questions whether individuals are fully aware of how forecasts improve winter weather road maintenance.

Policy Relevance of Estimates. The studies described above provide some foundation for understanding the determinants of forecast benefits and the magnitudes of these benefits. However, the studies are fragmented and vary considerably in their coverage, sophistication, and realism. Millner (2009) cautions: "Traditional models that have attempted to gauge forecast value have focused on a best-case scenario, in which forecast users are assumed to be statistically sophisticated, hyper-rational decision-makers with perfect knowledge and understanding of forecast performance. These models provide a normative benchmark for assessing forecast value, but say nothing about the value that actual forecast users realize. Real forecast users are subject to a variety of behavioral effects and informational constraints that violate the assumptions of normative models." Millner's caution regarding the likely upward bias of existing estimates based on models of user decision-making is a valid one, however, from a policy perspective, what is of critical importance is the magnitude of the bias. Small biases are not of great concern, but large biases are. Unfortunately, the current state of knowledge does not allow for any general assessment of the magnitude of biases.

Thus, existing studies are in nearly all cases partial equilibrium analyses that ignore price effects. This can be a significant shortcoming if the estimates derived from the studies are used to value the benefits of investments in hydromet systems at the country level. As Katz and Lazo (2011) observe, efforts

to account for price effects, as in the work of Adams et al. (1995) and Solow et al. (1998), are noteworthy, but the gain in the economic realism of these studies comes at the cost of less attention paid to modeling the effect of weather on yields and the manner in which producers make use of forecasts.

6. Approaches to Estimating Country-Level Net Benefits of Hydromet Investments

A number of alternative approaches have been used to estimate the net benefits of hydromet investments or modernization at the national or country level. These were developed to estimate the benefits of hydromet investments on a large regional scale, in addition to being applicable where systematic recording of damages and losses has not been undertaken for sectors and populations within a country. The various approaches are complementary, and often the recommendation is that more than one method be used in order to corroborate results or obtain a range of net benefits from hydromet investments (GFDRR 2009).

This section discusses three such approaches, the benchmarking method, the sectoral approach, and the conditional probabilities approach.⁴ These three approaches are similar in that they attempt to quantify the benefits from hydromet investments in terms of the changes in the economic well-being of climate-sensitive sectors in a country.

A general comparison of the three methods is provided in Table 2. All three approaches have a common sequence of analysis. An initial monetary loss with and without investment is estimated for each industry or sector in the economy, adjustments are made to these using expert opinions, and a final benefit estimate is computed across all sectors in the economy. A key similarity among the methods is that they arrive at a country-level benefit estimate that has been refined using as much local information and expertise as possible. The refinement stage is critical given that the ultimate success

of improved forecast information depends on the structure by which information is disseminated and used locally. Compared to formal benefit-cost analyses of a specific sector discussed earlier, rapid expert assessments take the place of model-based or survey-based estimation of benefits.

Referring to Table 2, it is worth noting that the three approaches are closely related. Sectoral analysis sometimes makes use of Gross Domestic Product (GDP) to define an initial damage reduction from hydromet investment. The conditional probability approach is very similar to sectoral analysis with the added improvement of specifying a prior distribution of weather-related damages, and then adjusting this prior distribution (i.e., using conditional probabilities) to reflect the effects of hydromet modernization using local information. Changes in expected damages then depend on both an assessment of the reductions in losses resulting from hydromet modernization, and changes in the probability of losses. Thus, conditional probability approaches allow for some uncertainty in terms of the effects of weather when arriving at a benefit estimate. Benchmarking, on the other hand, essentially assumes a deterministic effect of hydromet modernization on weather hazards through the use of GDP-based estimates.

Benchmarking Method. The benchmarking method has been the most widely applied approach, especially in the Southern Caucasus (Azerbaijan, Armenia, Georgia), the Balkans (Albania and Serbia), and other parts of the former Soviet Union, namely,

⁴ There are also sectoral-like approaches that rely on conjoint analysis, although these have been applied infrequently to estimate country-level benefits of hydromet modernization (an example is Leviäkangas et al. 2007). Conjoint analysis uses the survey method common in contingent valuation but also an additional survey where users of forecast information and experts in selected sectors are asked to rank alternatives or attributes associated with different hydromet investment scenarios and outcomes. The results of these expert assessments are then used to adjust benefits estimated by contingent valuation, and the results are used to extend the contingent valuation estimates of willingness to pay for users from a specific sector to other sectors. Conjoint analysis is therefore a combination of the contingent valuation approach with further sociological surveys aimed to refine and map the use of better forecast and climate information for the country in question.

Table 2 Comparison of Country-Level Approaches

Method	Losses (with and without modernization)	Adjustment	Net Benefit Estimate of Hydromet Modernization
Benchmarking	Based on GDP	Expert opinion used to adjust losses	Increase in GDP
Sectoral	Based on benefits transfer*	Expert opinion plus surveys of affected agents used to adjust losses	Decrease in estimated monetary losses
Conditional Probabilities	Based on benefits transfer*	Expert opinion used to adjust losses and assign conditional probabilities	Decrease in expected monetary losses**

Source: Authors.

*Benefits estimated in other studies in regions judged similar to the one in question.

**Decrease in expected monetary losses = [probability of losses without modernization x losses without modernization] – [conditional probability of loss with modernization x losses with modernization].

Belarus, Ukraine, and Kazakhstan (GFDRR). It has also been applied in other parts of Europe and in Asia (World Bank 2008). A similarity among these countries is a lack of data on actual total and sector-specific economic losses from hydro-meteorological hazards.

The benchmarking method was developed to estimate country-level economic benefits from hydro-meteorological modernization. It is based on national official macroeconomic and sector-specific statistics, namely GDP, assessments of the weather dependence of the economy, an assessment of how vulnerable the country is to weather events based on meteorological statistics for temperature (minimum and maximum), relative humidity, precipitation and wind, the current status of hydromet equipment and facilities, and the quality of hydromet service provision to citizens in the country. In many cases, key parameters are derived through expert opinion or studies undertaken in other countries. The main advantage of the benchmarking approach is its simplicity and the fact that it does not require time-consuming surveys or complicated quantitative analysis, yet it is able to inform policy makers concerning the value of preventable losses in a country from weather events and the extent to which these losses can be reduced with hydromet investments.

Benchmarking is undertaken in a two-stage process. Referring to Table 3, the two most important components it estimates are the level of annual direct economic losses (as a share of GDP) caused by hydro-meteorological hazards (L), and the level of annual preventable losses that can be achieved through improved forecast information or warnings as a result of hydromet modernization (P). Preventable losses are an agreed-upon percentage of the total direct economic losses with the existing hydromet system. The first stage of benchmarking begins with a given set of initial values (benchmarks) for these components. In the second stage these benchmarks are adjusted following assessment by experts. The second stage is what distinguishes one country from another, in that experts familiar with the local situation refine the initial values using specific country parameters such as weather and climate, economy diversification and structure, and status of the current hydromet system.

As Table 3 shows, the first stage of benchmarking involves rules for initial values that are common regardless of the country being studied. Initial values are set for: 1) average annual losses from dangerous weather events (ranging from 0.1–5% of GDP); 2) average annual preventable weather losses, measured as 40% of the losses that would have occurred without hydromet investments computed

in the first component; 3) an estimate of how climate sensitive the economy under consideration is based on the share of sectors in the economy that are judged to depend on weather (such as agriculture, transportation, electricity generation, and tourism)—the initial level is set at 50% of GDP; 4) the share of GDP due to agriculture (set initially at 15%); meteorological vulnerability (assumed to be average initially); and the status of hydromet service provision. The last component is a measure of how effectively hydromet investments can be integrated into the country's forecasting information services and is initially assumed to be "satisfactory". This therefore accommodates nuances in infrastructure and educational capital necessary to implement improved forecasts and disseminate information. In the second stage, expert opinion is used to adjust the initial entries in the second column of Table 3 for local circumstances. If, for example, there are barriers to disseminating information, as would be the case if infrastructure or education in the country is below average, then the value in the second row would be adjusted downward. In developing economies that are largely agrarian, the value in the fourth row is adjusted upward.

The annual benefit from hydromet modernization is computed very simply from Table 3. After all

Table 3 Benchmarking – First Stage Parameter Values

Parameter	Initial Estimate
Average annual loss from weather hazards (L)	0.1–5% of GDP
Average annual preventable loss from weather hazards (P)	40% of annual loss
Weather dependence of economy (W)	50% of GDP
Share of GDP due to agriculture (A)	15% of GDP
Effectiveness/ability of country to make use of hydromet service modernization	Satisfactory

Source: World Bank (2008).

adjustments are made to the numbers in the second column, the annual incremental benefit from hydromet modernization is estimated to equal: $P \times L$. Then, information in the last three rows is used to further adjust this incremental benefit estimate upward or downward.

Advantages and Limitations of Benchmarking.

The main advantage of benchmarking is that all hydromet modernization benefits are based on readily available, generally reliable, estimates of GDP. Benchmarking can be accomplished quickly and for many countries in a region at once, and does not require time consuming study of specific sectors except to understand the extent to which they contribute to GDP. The use of local experts is also appealing as a means of tailoring estimates of benefits from hydromet investments.

Given its inherent simplicity, benchmarking is not without limitations. The main problem is common to all currently used country-level approaches: benefits that have been assessed in the historical literature can be valued, but this is problematic because the literature may not focus on losses specific to local situations. Second, in many countries GDP may not be reliably reported or computed—this has in fact prevented benchmarking from being applied in Turkmenistan (see below). Third, the benchmarking method is too aggregate to evaluate specific benefits of information that may be important to hydromet targeting or location. For example, the safety benefits to households or the nuances concerning infrastructure that are important factors in the benefits of hydromet investments may differ considerably in their contribution to GDP estimates, and yet the primary way to evaluate these is with expert opinion.

Sectoral Approach. The sectoral approach aims to assess the economic desirability of hydromet investments by comparing the costs of hydromet

investments to the resulting losses prevented by the investments. As is true for the benchmarking method, the benefits are computed across all climate-sensitive sectors of the economy that generate a significant share of GDP.

The sectoral approach relies on estimation of two key parameters for each sector. The first is R_i , the percentage of losses that are potentially preventable in sector i . The second is S_i , the percentage of potentially preventable losses in sector i that are avoided with the hydromet improvement. The product of these two, $R_i \times S_i$, gives the percentage of losses that are avoided in sector i with the hydromet improvement. Applying these percentages to estimates of losses for each sector yields estimates of the reduction in losses from the improvement. The sum of these reductions across sectors equals the benefits of the improvement. The benefits are compared to an estimate of the increase in costs, Δ , associated with the improvement. All of these parameters, R_i , S_i and Δ , are determined through surveys of experts.

Expert assessment and surveys are the most time consuming part of sectoral approaches.⁵ These surveys must assess both general information concerning the scale and quality of the hydromet information used, and the two key parameters needed to estimate benefits. Thus, these assessments usually proceed in two stages. First, experts are polled to obtain estimates of the two key parameters for each sector of the economy that is considered to be climate sensitive (R_i and S_i). If a “scenario” analysis is deemed suitable, then these experts also recommend a range of values to represent

the best and worst case outcomes, with the mean value used as the base case. Second, experts assess the direct and indirect costs of hydromet modernization with an eye toward how modernization should be targeted (this determines Δ). In so far as possible, actual statistics are used to back the range of values picked by experts, but this is not always possible.

Advantages and Limitations of Sectoral Approach. Nearly all of the caveats for benchmarking apply to sectoral approaches. However, the sectoral approach does have the advantage of tailoring loss information to specific sectors, rather than simply linking all losses to changes in GDP as benchmarking does. Sectoral approaches also benefit from the potential to consider the validity of statistics concerning weather effects using expert judgment. In principle, this approach should provide a more “local” refinement of benefits and costs of hydromet modernization. However, again the quality of expert opinion and availability of outside information that can be transferred to the country in question remains a major obstacle to the sectoral approach.

Comparisons of Benchmarking and Sectoral Approaches. A recent study compares the results of using the benchmarking and sectoral approaches in the Southern Caucasus (GFDRR 2009). A summary of this comparison is found in Table 4. In the table, benchmarking estimates appear in parentheses, while the other numbers represent estimates using the sectoral approach. Notice that average annual preventable losses (assessed by experts) are common across the approaches.

⁵ Hallegatte (2012) proposes a specific approach, risk management analysis, for loss transfer and modification. His work details a procedure to use losses from European countries in terms of saved lives and reduced economic losses from weather data, and then transfer this information to estimate losses in developing countries. He is therefore able to provide a layer of additional sophistication and data quality to estimating benefits of hydromet modernization in developing countries that lack sufficient data for analysis. His procedure is based on asset losses, and he also details several necessary modifications to making sectoral analysis more rigorous in these contexts. Finally, he argues that benefit estimation must be sensitive to five needs in developing country contexts: local observation systems, forecasting capacity, interpretation capacity, communication tools, and user decision-making capabilities. Finally, this approach compares only annual benefits and costs and therefore does not depend on a discount rate.

Table 4 Comparison of Results of Assessment of Hydromet Modernization Using Benchmarking and Sectoral Approaches*

Estimated Parameter	Kyrgyz Republic	Republic of Tajikistan
Average annual losses from weather hazards (USD million)	(24.9) 39.6	(24.9) 37.0
Average annual losses (percent of GDP)	(1.0) 1.5	(1.04) 1.6
Average annual preventable losses (USD million)	10.1	5.8
Expected average annual benefits due to hydromet modernization (USD million)	(2.9) 3.8	(1.7) 3.1
Investment efficiency (%)**	(244) 318	(199) 357

Source: GFDRR (2009).

*Benchmarking estimates are in parentheses. Turkmenistan was not judged to have reliable GDP data and was not included in the comparison.

**This is an imputed internal rate of return that based on the stream of benefits and costs over time.

These results show that benchmarking yields lower estimates of the net benefits from hydromet investments. For example, in the Kyrgyz Republic, the benefits of hydromet investments range from a benchmarking estimate of USD 2.9 million per year to a sectoral approach estimate of USD 3.8 million per year. In this study, it is worth noting that the sectors studied (water management, agriculture, and electricity) are responsible for about 75% of GDP and are all highly climate sensitive. In Tajikistan, once again benchmarking yields lower estimates than the sectoral approach. The estimates range from USD 1.7 to 3.1 million per year in benefits for benchmarking and sectoral analyses, respectively. As a result, the computed investment efficiencies in the last row of the table are considerably different even though the expenditure on hydromet modernization was equal across approaches for the same country. While these two approaches yield a relatively wide range of values for hydromet benefits, both give a consistent, bottom-line economic efficiency assessment.

Conditional Probabilities Approach. This approach estimates the relationship between the state of the hydromet system and the frequency distribution of losses experienced due to meteorological

events. For each sector, a distribution of the likelihood of occurrence of losses and their magnitude in the absence of hydromet investment is first estimated. This baseline distribution is based on available data on the magnitude and frequency of losses, as well as expert opinion. In a second step, the changes to this distribution resulting from improvements to the hydromet system are estimated using available information and filling in with expert opinion where necessary. The benefits of hydromet improvement are then calculated based on the resulting change in the distribution of losses.

The data required for the conditional probabilities approach are more extensive than for the benchmarking approach, but less than for formal benefit-cost analyses of individual climate-sensitive sectors. It relies on historical data on losses suffered from extreme weather events in climate-sensitive sectors in the economy, perceptions from the community and experts concerning social, economic, and governance aspects of weather events and information from improved forecasts, identification of hydromet modernization impacts, and recommendations from experts concerning preventative actions arising from dissemination of improved forecasts.

The conditional probabilities approach was recently used to evaluate modernization of Mexico's hydromet system (World Bank 2012, Project Appraisal Document). Estimates of the frequency distribution of baseline losses were based on data on insurance payments for property damage, loss of livestock and agricultural output, as well as government expenditures on rebuilding and repairing infrastructure, and government assistance to low-income households affected by extreme events. The change in the distribution of losses due to hydromet modernization was largely derived using expert opinion.

Advantages and Limitations of Conditional Probabilities Approach. The conditional probabilities approach has an advantage over the benchmarking and sectoral approaches in that not only preventable losses are assessed, but a probability distribution is applied over potential reductions in these losses from modernization. Moreover, the adjustment to probabilities from prior to conditional distributions takes explicit account, in principle, of changes in decision-making, barriers to dissemination of information, and opportunities to use forecast information in the country in question.

This said, the limitations of the conditional probabilities approach are similar to the limitations of the other approaches, in large part because it too relies heavily on expert opinion to estimate the benefits of improved forecasts. One important difference, however, is the added complexity of having to rely on expert opinion for both an assessment of preventable losses, as well as an assessment for how probability distributions change. Indeed, even

when incremental benefits and incremental costs can be reliably obtained or inferred, the magnitude of expected total loss reduction is highly dependent on the adjustment in probabilities. Further, the underlying incremental costs and benefits attributable to improved forecasts are not assumed to change with hydromet investments, and so like the benchmarking and sectoral approaches, the conditional probabilities approach does not specifically incorporate information on changes in decisions that agents make. Thus, while there is a degree of refinement beyond both the sectoral and benchmarking approaches, this refinement is still based on expert opinion and surveys rather than actual information from decision-makers who use forecast information.

Limitation of Country-Level Approaches. All of the country-level approaches to estimating the benefits of hydromet investments focus largely on losses avoided from extreme meteorological events. As such, these approaches generally need to be supplemented with sector-level estimates of the benefits yielded by improved forecasts of routine climate. The approaches for deriving these estimates were described in Section 5 above. It is important, however, to ensure that benefits of hydromet investments are not double counted: if benefits from improvements in forecasts of routine climate are captured for a sector in the country-level analysis, then adding an estimate of these benefits from a sector-level analysis would constitute double counting of benefits. However, the two estimates could be compared, and if substantially different, they could be used as alternative values of the benefit in a sensitivity analysis.

7. Costs of Hydromet Investments

The costs of hydromet modernization are multidimensional and include more than simply installing new radar stations or upgrading aging equipment and software so that better forecasts can be made. The total costs encompass a variety of other investments and activities necessary for successfully preparing and delivering forecasts. These include implementation and service associated with hydromet upgrades, additional weather service and information delivery infrastructure, and investments in training and hiring professionals to interpret data and generate warnings and forecasts. The costs of implementation and service activities are sometimes the largest component of total costs, and they are often paid over an extended period of time, while the bulk of equipment installation costs are paid up front in the project life cycle. In the literature, implementation and delivery costs are sometimes referred to as capacity or institution building costs.

Any country-wide hydromet system includes many components, such as automatic monitoring stations, telecommunication networks, and computers to store software and data. The requirement of “nowcasting”, or real time monitoring and warning of extreme events, means that the establishment of telecommunication networks and their subsequent management are often a large part of the total cost of any modernization over the hydromet project life cycle. Kokko and Vaisala (2005) argue that ongoing maintenance, calibration, and periodic update/upgrade of hydromet systems represent a significant cost that is sometimes overlooked. Network and maintenance costs can be thought of as implementation costs. Further, a part of the life cycle costs of hydromet systems are “contingency” based, that is, they are not known with certainty at the time hydromet systems are installed or upgraded. Contingency costs imply that some future

hydromet costs may not be foreseen with certainty at the time of modernization. This uncertainty is nearly always the rule given that the average service lifetime of newly purchased radar and modeling software ranges between 10–15 years.

Classification of Economic Costs. Broadly defined, and regardless of whether installation or implementation of hydromet systems is considered, there are two types of costs associated with hydromet investments: fixed costs and variable costs. Fixed costs of modernization are those that do not depend on the amount of information generated by the forecasting agency but are necessary for the operation to exist. Examples are costs associated with purchases of new equipment and software, and the building of new roads or communication networks and towers to disseminate information. Variable costs are those that are a function of the extent of forecasts produced and the speed of information delivery using the new hydromet system, that is, they are a function of the volume and type of information generated with the new system. Examples of variable costs are the number of additional forecasters and other staff required to deliver new and improved weather information from an expanded or upgraded hydromet network, and new monitoring and warning activities that are required given additional information generated by an expanded or upgraded hydromet network.

Relative Magnitudes of Costs. For complex and multi-year investments such as hydromet modernization, the relative magnitude of fixed and variable costs can vary widely. In some cases, the fixed costs are the larger component of total costs, but this is not the rule. For example, Subbiah et al. (2008) estimate for hydromet modernization in Bangladesh that up-front fixed costs equaled USD 2.1 million, while variable costs amounted to

over USD 400,000 annually over a project life cycle of more than 10 years. The largest components of variable costs (50%) in this study were capacity building of national and sub-national institutions for translation, interpretation, and communication of probabilistic forecast information.

The largest fixed cost components are typically early warning system technology purchase and installation, and high performance computing system investment. Tsirkunov (2011) reports that in the Republic of Tajikistan, improvement in hydromet observation networks, an activity with a large fixed cost component, account for about USD 8.9 million of the estimated USD 13 million required for modernization in that country. A similar pattern is found in the Kyrgyz Republic, with upgrading of the hydromet observation networks estimated to account for USD 3.8 million of the total USD 6 million required for modernization.

In mountainous regions, i.e., regions with very varied topography, a denser network of monitoring stations and radar are typically needed. This can result in fixed costs accounting for a larger fraction of total costs than in regions with less varied topography.

Hallegatte (2012) observes that developing country hydromet modernization projects can differ markedly from those in developed country projects, with many of the differences reflected in costs. He argues that, in general, the total costs and experiences from developed countries should not simply be applied to determine modernization costs in developing countries. He notes that developing country hydromet systems are characterized by poor conditions with regard to staffing, buildings, observation sites, and instruments. The costs of upgrading such systems can be many times higher than modernization in a country that already has high quality infrastructure.

Cooperation and Economies of Scale. The magnitude of costs associated with upgrading a country's hydromet systems can also depend on the presence or absence of cooperation with neighboring countries. United Nations International Strategy for Disaster Reduction (UNISDR 2008) reports that in a seven country region of southeastern Europe, financing to modernize national hydromet services amounts to 90 million EUR without regional cooperation and 63 million EUR with cooperation, with those costs attributable to measurement, upper-air sounding, radars, communication, and delivery of forecast warnings. Cooperation reduces costs through data sharing across countries, which decreases the required extent of hydromet equipment installation, or through sharing of weather service delivery networks.

The differences between cooperation and no-cooperation scenarios suggest there may be economies of scale in modernization for developing countries. Economies of scale means that the costs of a larger integrated system can be lower per unit of forecast information produced (however, this is defined) than smaller projects that still require a large upfront investment in equipment.

Cost Estimation Caveats. There are four important caveats with regard to hydromet cost estimation relevant to evaluating hydromet investments in developing countries:

- counting only incremental costs;
- treatment of joint costs and benefits;
- cost uncertainty in long-lived projects; and
- potential for cost savings.

Each of these caveats is discussed in turn below.

Counting only incremental costs. When evaluating an investment, the relevant costs are only those that are attributable to the investment. These

incremental costs are, in principle, straightforward to estimate, as they are attributable to activities associated with modernization, such as purchase and installation of additional equipment, networks, software, and hiring of additional staff.

Joint costs and benefits. A second, more challenging issue has to do with the joint nature of some of the fixed and variable costs associated with hydromet improvements. These joint costs are costs that are incurred in the course of hydromet improvements but they also impact other non-weather related activities that have value in the economy.⁶ Examples of joint costs are those associated with improvements in infrastructure, including roads (to monitoring stations) and communication networks (for collecting and disseminating information from monitoring stations as well as forecasts). In studies of hydromet modernization, joint costs often fall under the category of capacity improvements deemed necessary for improved forecast information to be utilized by the population. Such improvements can yield benefits that extend far beyond those related to hydromet systems.

Dealing with joint costs and benefits is challenging. In principle, if all joint costs are attributed to the hydromet investment, then all the joint benefits should be as well. However, quantifying all the joint benefits is likely to be difficult and beyond the scope of the hydromet investment analysis. In such cases, a fraction of the joint costs could be attributed to the hydromet investment, with the fraction based on a rough estimate of the fraction of joint benefits that are hydromet related.

Cost uncertainty. The common need to upgrade or put in place associated infrastructure when improving hydromet systems in developing countries often

implies greater uncertainty about the magnitude of project costs. This is especially true of long-lived projects that include contingency-based costs. Estimating costs with precision is further complicated if there is little historical data on the types of costs associated with a project. Relying on cost transfer from modernization efforts in developed countries, where such data are generally available, adds an additional component of uncertainty to estimates of costs in developing countries.

Potential cost savings. An upgraded hydromet system can reduce some of the costs of operating a hydromet system. For example, new radar equipment can eliminate the often high costs of maintaining old, outdated systems for which spare parts must be custom made. Further, in modern hydromet systems many monitoring and information reporting functions are automated (Kokko and Vaisala (2005)), reducing the need for staff resources to be devoted to collecting information from monitoring stations.

Table 5 presents a breakdown of the types of specific costs that may be associated with hydromet modernization listed by the activity required by the investment. These activities are drawn from a survey of selected recent hydromet projects, including those described in Tsirkunov (2011), Subbiah et al. (2008), GFDRR (2009), and World Bank (2012). While all of these activities are not always required in modernization efforts, depending on the current system's capabilities and the type of improvements undertaken, they are illustrative of the types of costs that may need to be estimated. The table indicates the likely nature of the costs in each category. Some of the categories encompass many sub-activities. For example, strengthening institutional capacity and communication and

⁶ Another type of jointness in costs refers to the fact that one activity required of hydromet modernization may be synergistic to another activity, so that paying for one may reduce the cost of another.

information networks include activities such as educational investments related to gathering and interpreting climatological information, and developing means for communications with users that fit the country's political and geographic features. Capacity building can include development of institutional, legal, and regulatory authority within a country. This has been true, for example, in the case of hydromet modernization in the Caucasus (Tsirkunov 2009). Modernizing observation infrastructure and technical design/engineering services include rehabilitation and expansion of radar networks, development of meteorological products and services, enhancement of computational capacity both in terms of modern high speed computer systems and software required to produce meteorological information, and establishment of quality control mechanisms for forecasts. Communication and information networks include means to develop a regional climate information capability and require activities such as design and maintenance of telecommunication systems, and development of early warning and nowcasting procedures.

As noted in Table 5, some costs can have both fixed and variable components. For example, the level and amount of capacity building needed could depend on the planned extent to which weather information is generated and delivered. A last point is that many of these costs are related, in that activities in one category can affect costs in another (i.e., there are cost-related synergies across activities). For example, the type of radar chosen will affect

the degree of automation and human calibration needed over time. Infrastructure improvements will affect the costs associated with building communication networks to deliver forecast warnings. The precise nature of these synergies, and therefore the costs associated with all of them, is also critically dependent on a country's social, political, and economic state. This highlights the recommendation by Hallegatte (2012) to not simply transfer cost information from developed countries to developing countries. It is also for this reason that the scale of the project may matter critically in estimation of total costs for hydromet modernization.

Table 5 Hydromet Modernization Cost Components from Selected Studies

Category	Fixed	Variable	Joint
Technical Design, Engineering Services	X	X	
Building Communication and Information Networks (Regional Capacity)	X		X
Strengthening Institutional Capacity	X	X	X
Ongoing System Maintenance/Calibration		X	
Modernizing Observation Infrastructure	X		
Non-Meteorological Infrastructural Investments	X	X	X
Monitoring and Forecast Staffing Enhancement		X	

Source: Tsirkunov (2011), Subbiah et al. (2008), GFDRR (2009), and World Bank (2012).

8. Proposed Framework for Estimating Net Benefits of Investments in Hydromet Systems

In this section we present an outline of a proposed framework for estimating the expected net benefits of investments in hydromet systems at the country level. The framework draws on the existing approaches described in the previous sections, but with some important changes and additions. It bears emphasis that the proposed framework yields first-cut estimates of net benefits, as is true for existing approaches. The discussion in Section 4 and Annex 1 makes clear that obtaining precise estimates of the benefits enjoyed by all users/sectors from investments in hydromet systems is very difficult given the current state of knowledge and data available.

Given the greater difficulty of identifying and quantifying the benefits of hydromet improvements, as opposed to their costs, the focus of the proposed framework is on benefits estimation. Considerable detail on the costs of hydromet improvements is provided in the preceding section.

Benefits Transfer. The proposed approach makes use of benefits transfer. As noted earlier, benefits transfer entails taking benefits estimates derived in one setting to estimate benefits in another, similar setting. The technique has been widely used to value the benefits of environmental improvements. Benefits transfers must be performed with care, and benefits estimates derived using this procedure should be viewed as indicative rather than exact. For the transfer to be valid, the original setting in which benefits were estimated and the new setting to which they are being applied must be at least roughly comparable.

Incremental vs. Total Benefits An important point when conducting benefits transfers in the context of valuing the benefits of investments in hydromet systems is the distinction between the

total benefits of hydromet systems (and the forecasts they generate) and the incremental benefits resulting from improvements to (or investments in) hydromet systems. This distinction between total and incremental benefits is emphasized by Freebairn and Zillman (2002). Much of the existing literature on the benefits of forecasts focuses on estimates of total benefits. A small subset of the literature focuses on estimating the benefits of improvements in hydromet systems. However, this body of literature estimates the incremental benefits associated with improved forecasts resulting from hydromet investments, instead of estimating the total benefits of existing, or hypothetically perfect, forecasts.

The objective of the framework proposed here is to estimate the benefits of moving from the current hydromet system to an improved hydromet system. Therefore incremental benefits are the relevant benefit measure. Total benefits would be the relevant measure if the objective of the analysis were to estimate the benefits of having a hydromet system, i.e., the benefits of going from a scenario with no hydromet system to one with a hydromet system.

Time Horizon of the Analysis. A key question when evaluating any project or investment is the time horizon of the analysis. In principle, the appropriate time horizon corresponds to the time period over which the project is supposed to generate any costs or benefits. In the case of physical infrastructure investments in hydromet systems, the appropriate time horizon could be dictated by the life of the physical infrastructure. For example, investment in a Doppler weather radar system could call for a time horizon of 15–20 years, which is likely to be the typical useful life of the system. However, the benefits of broader investments in hydromet

systems, such as improvement of early warning systems, could well extend beyond the life of the physical infrastructure. This is especially true if the improvements encompass better climate resilient planning and better decision support tools in climate-sensitive sectors. Such improvements can yield benefits that extend well beyond one to two decades. The common problem with using long time horizons is the difficulty of estimating benefits or costs with reliability more than two to three decades into the future.

With these caveats in mind, we turn to a description of the proposed framework. The description is presented as a sequence of steps to be carried out. This approach borrows from that in Lazo et al. (2008). In practice, the analysis is likely to be iterative in nature, rather than purely sequential, with earlier steps in the process being revisited. To make the proposed approach more tangible and concrete, in places the following description refers to its application to evaluate investments in hydromet systems in Jamaica. A full description of the Jamaica analysis is contained in Annex 5. An overview of the framework is provided in Figure 2.

Step 1: Identifying the Full Range of Benefits

As discussed in Section 5, the benefits from improving a country's hydromet system depend on a number of broad factors. Among them are:

- the state of the existing hydromet system;
- the nature and magnitude of the improvements being contemplated;
- the country's susceptibility to extreme meteorological events; and
- the structure of the country's economy, and its dependence on climate-sensitive sectors, namely, agriculture, aviation, construction, surface and water transportation, water resources, energy, fisheries, forestry, health, and tourism and recreation.

A first step in estimating the benefits of investments in hydromet systems is identifying the types of benefits the investments will likely generate. The existing literature and past efforts to estimate benefits of hydromet investments clearly provide useful guidance when identifying types of benefits. However, as noted in Section 2, these sources may not adequately consider benefits associated with more recent hydromet investments intended to promote climate resilience and climate adaptation.

The output of this first step is a comprehensive listing of benefits likely to be generated by the hydromet investments.

Step 2: Screening Benefits

The benefits identified in Step 1 will invariably differ widely in character. The objective of the second step is to screen benefits and classify them into one of three categories:

- **Ignorable Benefits.** Benefits that can be ignored in the analysis because they are likely to be small in magnitude;
- **Unquantifiable Benefits.** Benefits that are likely to be relatively large in magnitude but cannot be quantified (given existing methods/knowledge); and
- **Quantifiable Benefits.** Benefits that are likely to be relatively large in magnitude and can be quantified.

There is likely to be some iteration between this step and the next step in which benefits are quantified and monetized. Determining which benefits are quantifiable will inevitably require consideration of the approaches that can be used to quantify benefits.

Benefits that cannot be quantified should be omitted from further analysis only if they are likely to be small in relative magnitude, i.e., they are ignorable.

Benefits that are likely to be large in magnitude should be identified and described even if they cannot be quantified. Where possible, these benefits should be described in some detail and qualitative assessments of their magnitude (e.g., small, moderate, large) provided, as well as an indication of the likelihood of the benefits being realized (e.g., unlikely, likely, very likely).

The output of this second step is a classification of the benefits identified in the first step. Subsequent analysis is restricted to the benefits classified as quantifiable.

Step 3: Selecting Approaches for Estimating Quantifiable Benefits

Given a list of quantifiable benefits, the next step is to select methods to quantify the benefits. Following the classification scheme described earlier in Section 2, quantifiable benefits are likely to be of one of the following three types:

- benefits of improved forecasts, and associated early warning systems, of extreme meteorological hazards (**Type I benefits**);
- benefits to enterprises of improved forecasts of routine climate and dissemination of these forecasts (**Type II benefits**); and
- benefits to households of improved forecasts of routine climate and dissemination of these forecasts (**Type III benefits**).

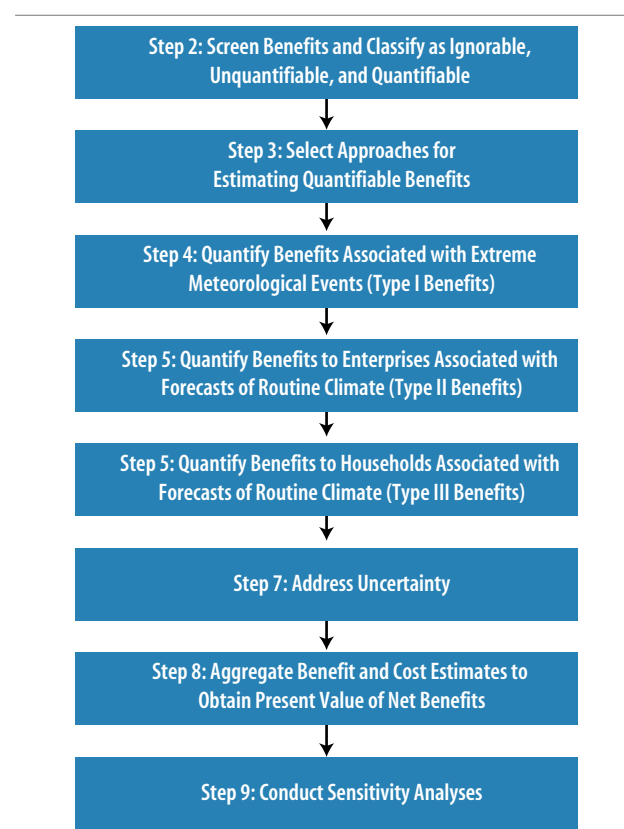
For brevity, these benefits are referred to as Type I, II, and III benefits. Type II and III benefits are similar in that they are both associated with improved forecasts of routine climate, however the techniques used to estimate them differ, hence a distinction is made between them.

Type I benefits, i.e., benefits from improved forecasts of extreme meteorological events and

effective dissemination of information about their effects and appropriate responses to them, can substantially reduce economic losses caused by the events. The magnitude of these benefits will be especially large for countries that are susceptible to extreme meteorological events, such as countries in the Caribbean.

Type II and III benefits, i.e., benefits from improved forecasts of routine climate can result in increased enterprise profits (or reduced costs) and improved decision-making by households. These benefits may be small at the individual household or

Figure 2 Overview of Framework for Estimating Net Benefits of Hydromet Investments Expected Net Benefits of Hydromet Investments



Source: Lazo et al. (2008).

individual enterprise level, but when multiplied by an entire population at risk they can be substantial.

To estimate Type I benefits we prescribe an approach that is similar to the sectoral approach described in Section 4, but differs in that it makes use of historical data on losses, by sector, from extreme meteorological events to establish a baseline for the analysis. We describe the approach further below. In the absence of adequate historical data on losses, two alternative approaches can be pursued: (i) use expert opinion to develop estimates of historical losses, or (ii) employ the benchmarking method described in Section 4.

For estimating the benefits of improved forecasts of routine climate—Type II and Type III benefits—simple benefits transfer is likely to be the most suitable approach, given likely time and budget constraints that render infeasible primary user/sector-specific studies of the type described in Section 5. We illustrate the use of benefits transfer to estimate these types of benefits below.

Step 4: Quantifying Benefits Associated with Extreme Meteorological Events (Type I Benefits)

Specifying an Appropriate Baseline. The approach used to estimate Type I benefits entails estimating the reduction in losses resulting from hydromet investment relative to the baseline. Accordingly, the first task is to quantitatively identify the appropriate baseline, i.e., the expected losses that would be experienced from extreme meteorological hazards in the absence of the investment over the time horizon of the analysis.

This baseline should take the form of sector-by-sector estimates of expected losses in the absence of the investment. By disaggregating expected losses by sector, it is possible to arrive at a more accurate estimate of total expected benefits. Each sector

in an economy is affected differently by meteorological hazards, depending on the type of meteorological event, the sector's exposure to weather, and the value of its assets. Furthermore, the benefits yielded by the hydromet investment will differ across sectors. For instance, a hotel (in the tourism sector) in a developing country is likely to be a sturdy structure that will experience limited damage from a storm even with little advance warning. In contrast, a rural home will be much more vulnerable to strong winds and rain, and a small increase in the lead time of a storm warning could enable the household to better fortify the house and move possessions to safer ground.

To a first approximation, the baseline is the current level of expected losses if there is no reason to believe that expected losses will change over the time horizon of the analysis (absent the project). In practice, for the time horizons likely to be considered when evaluating hydromet investments, this is unlikely to be true, for at least four reasons:

- the magnitude of assets and populations at risk is likely to rise over time, increasing losses;
- existing physical hydromet infrastructure will deteriorate/depreciate, reducing the quality of services provided, thereby increasing losses;
- households, enterprises, and government agencies may autonomously adapt to extreme events, mitigating the losses that they cause; and
- for sufficiently long time horizons, the effects of climate change on the frequency or severity of extreme events may result in departures from “current” experience, likely increasing losses.

The first reason is self-explanatory; it is worth noting though that depending on the time horizon being considered and the pace of change in the country's economy, for some sectors (e.g., agriculture), assets at risk may decline if the composition of the

economy is changing. For time horizons of 10–15 years or more, deterioration of existing hydromet infrastructure can be significant and could result in a non-trivial reduction of the quality of hydromet services provided, increasing expected losses from extreme events.

The possibility of autonomous adaptation merits some explanation. Autonomous adaptation refers to adaptation to climate change or extreme weather events that takes place even without government intervention (e.g., see Malik and Smith 2012). For example, in the face of increasing frequency or severity of hurricanes, households in developing countries could mitigate the damages they cause by shifting from dwellings made of wood to dwellings made of brick or concrete. If this adaptation has already occurred to a large degree, then it is presumably already reflected in estimates of current losses. However, if the adaptation is in progress, and is likely to continue over the time horizon of the analysis, then estimates of expected losses need to be adjusted downward to reflect the effects of this adaptation. The available empirical literature is largely silent on the magnitude of the losses avoided from this type of adaptation. One notable exception is a recent study of adaptation to tropical cyclones using data from across the world (Hsiang and Narita 2012). It indicates that only about 3% of the losses from incremental changes of countries' current tropical cyclone climates are “adapted away” in the long run.

Turning to the last reason in the list above, for relatively short time horizons of 5–15 years, the consequences of climate change for the frequency or severity of extreme events are likely to be fairly small. But for longer time horizons, depending on projections for the country in question, they may have a significant impact on expected losses.

The above discussion makes clear that the appropriate baseline must reflect *future* conditions in

the absence of the investment; future conditions can differ appreciably from current conditions. Given the nature of extreme meteorological hazards, it is impossible to reliably forecast their number and severity years into the future. To derive estimates of expected losses from such hazards in the future, as a first approximation it can be assumed that expected annual losses from extreme hazards in future years are equal in magnitude to average annual losses over the recent past. The period comprising the “recent past” must be sufficiently long to be representative. These “preliminary baseline loss estimates” are then adjusted to capture the effects of the four factors listed above.

In the case of the Jamaica analysis, sector-level data on losses from extreme events over the 11-year period 2000–2010 were used to estimate an average expected loss by sector and event type. For events for which sectoral decompositions of losses were not available, sector-by-sector losses were estimated by taking average loss percentages by sector and hazard type for events for which loss decompositions were available and applying these percentages to estimates of total losses.

A total of 10 climate-sensitive sectors were selected for analysis and three event types (hurricanes/storms, floods, and droughts). These preliminary baseline loss estimates were then adjusted to reflect the effects of autonomous adaptation, the likely increase in the frequency and intensity of extreme meteorological hazards due to climate change, and the expected increase in the values of assets at risk. The latter two adjustments were made using interpolation of available year 2030 estimates for three different climate change scenarios developed for the insurance industry (no, moderate, and high climate change). To correct for reductions in losses due to autonomous adaptation, losses in all sectors were reduced by 10% each year. This liberal estimate of the effects of

autonomous adaptation was employed to avoid potentially overstating the benefits of the proposed investment. No adjustment was made for depreciation of the existing hydromet infrastructure given its already inferior state.

Estimating Percentage Loss Avoided (PLA). Having estimated expected annual losses in the baseline, by sector, for each hazard type, the reduction in losses with the investment must be estimated. This is accomplished by first developing estimates of the percentage loss avoided (PLA) with the project. The percentage loss avoided will generally vary across sectors and hazard types. Because of this heterogeneity, a separate PLA estimate should be developed for each sector and hazard type. The PLA is equivalent to the product $R_i \times S_i$ used in the sectoral approach (see Section 5), where R_i is the percentage of losses that are potentially preventable in sector i , and S_i is the percentage of potentially preventable losses in sector i that are avoided with the hydromet improvement. The product of these two, $R_i \times S_i$, gives the percentage of losses that are avoided in sector i with the hydromet improvement.

Estimates of PLA by sector are determined by some combination of benefits transfer⁷ and expert opinion. For example, for the Jamaica analysis, absence of Jamaica-specific information resulted in a reliance on benefits transfer, i.e. use of PLA estimates derived in similar studies of similar countries. In cases where PLA estimates for similar studies/countries could not be found, available PLA estimates were reduced by 50% to be conservative and to acknowledge differences in climate, geography, economic structure, and hydromet systems.

PLA estimates drawn from the literature were also subject to sensitivity analyses. For each sector/

event type, three PLA estimates were considered—low, middle, and high values. If only one PLA estimate was available in the literature for a given sector/hazard, this value was treated as a “high” value. The “low value” was taken to be 50% of the high value. Where multiple PLA estimates were available, the lowest value was taken to be the low value, and the highest value was taken to be the high value. The middle value was the average of the latter two values. Alternatively, if more than two PLA estimates are available for a given sector/hazard type, the median value of the estimates can be used as the middle value.

When feasible, PLA estimates derived using benefits transfer should be reviewed by in-country sector experts to ensure that the estimates are plausible and reflect local conditions and the specifics of the hydromet investments being considered.

Estimating Expected Losses Avoided. Multiplying the PLA estimates by the baseline expected annual losses for each sector/hazard yields estimates of the expected annual reduction in losses from extreme meteorological hazards. Summing these estimates over all sectors and hazard types for each year yields an estimate of expected benefits for that year.

Correcting for Forecast Inaccuracy. As noted in Section 4, forecast inaccuracy implies that forecasts generate costs in addition to benefits. Percentage loss avoided estimates in the literature generally do not appear to account for these costs. Therefore, some correction for forecast errors is appropriate in order to guard against overstating the benefits of hydromet investments. Subbiah et al. (2008) present a simple method for doing so, absent specific information on the costs of forecast errors. The

⁷ For example, estimates of percentage loss avoided for improvements in hydromet systems and associated early warning systems can be found in Subbiah et al. (2008), Bangladesh Water Development Board (2006), and Rogers et al. (2009).

key assumption underlying the approach is that the cost of an incorrect forecast is equal in magnitude to the benefit of a correct forecast. Letting A denote the average accuracy of the relevant forecast, expressed as a percentage, expected annual benefits need to be multiplied by a correction factor of:

$$\begin{aligned} &\text{Correction Factor for Forecast Inaccuracy} \\ &= (\text{Probability Forecast Correct}) \\ &\quad - (\text{Probability Forecast Incorrect}) \\ &= A - (1-A) \end{aligned}$$

This correction factor reflects the fact that benefits of improved forecasts are enjoyed $A\%$ of the time, while costs of incorrect forecasts are incurred $(1-A)\%$ of the time. The assumption that benefits from correct forecasts are equal in magnitude to the costs of incorrect forecasts allows for use of this simple correction factor equal to the difference in probabilities of the two events.

The output from this fourth step is annual estimates of Type 1 benefits from the hydromet investment over the time horizon of the analysis.

Step 5: Quantifying Benefits to Enterprises

Associated with Routine Climate (Type II Benefits)

The approach used to estimate the benefits of improved forecasts of routine climate is qualitatively different from that used to estimate the benefits associated with extreme events. Instead of estimating reductions in losses, it relies on estimates of increases in benefits or GDP by sector. However, the approach also makes use of benefits transfer supplemented by expert opinion. The precise application of benefits transfer will depend on the sector being examined. Annexes 3 and 4 illustrate how estimates from existing studies can be transferred to other settings. The examples presented in the annexes are for two important climate-sensitive sectors—energy and transportation. Additional

examples can be found in the Jamaica hydromet study mentioned earlier.

The output from this step is a time series of annual benefits for the time horizon of the analysis for a number of climate-sensitive sectors of the economy.

Step 6: Quantifying Benefits to Households

Associated with Routine Climate (Type III Benefits)

Estimates of Type III benefits, i.e., benefits to households from improved forecasts of routine climate, generally rely on stated preference methods. As discussed in Section 5, these methods rely on surveys of samples of users to directly or indirectly obtain estimates of the value users place on improvements to forecasts. Stated preference methods take the form of stated value or stated choice surveys. In stated value surveys, users are directly asked what they are willing to pay for an improvement to a forecast. In stated choice surveys, users are asked questions that require them to make choices among alternative forecasts with different costs associated with each. Users' preferences indirectly reveal the value they place on forecast characteristics.

Annex 5 provides an illustration of how stated—preference-based estimates of the value of forecast improvements in the U.S. can be transferred to a developing country.

The output from this step is a time series of annual benefits for the time horizon of the analysis for the household sector.

Step 7: Dealing with Uncertainty

An inevitable feature of any economic analysis of large and long-lived projects such as hydromet investments is uncertainty about the magnitude of benefits and costs. There are a number of ways of dealing with this uncertainty. The simplest and conceptually

most straightforward approach is to conduct a sensitivity analysis on benefits and costs when computing the present value of net benefits (described in the next step). Expert opinion or ranges of estimates for transferred benefits and costs can be used to determine high, middle, and low values for a wide set of benefit and cost estimates. This is the approach advocated here and discussed further in Step 9.

Another approach is to adjust upwards the discount rate used to calculate the present value of benefits and/or costs to acknowledge uncertainty about their magnitude. This approach is conceptually less desirable since it subjects all costs and benefits to the same adjustment for uncertainty.

A third, and conceptually more defensible, approach is to compute expected values for benefits and costs using alternative estimates of the possible values of benefits and costs along with estimates of the probability of each value being realized. This approach is in essence a means of aggregating the results from sensitivity analyses of benefits and costs using probabilities as weights.

The expected value approach can be illustrated by returning to an example based on the UNISDR study discussed in Section 7 (under Cooperation and Economies of Scale). Recall that coordination between countries was argued to lower hydromet modernization costs. Suppose experts determine there is a 70% probability of cooperation (30% of not cooperating) in forecast generation and warnings that would reduce the present value of costs according

to the estimates in UNISDR (2008). The relevant data are presented in Table 6. Assume that the level of hydromet services delivered to the population in all countries is equivalent under the cooperation and no cooperation scenarios. Further suppose, for simplicity, that the present value of benefits from hydromet modernization is certain and equal to 80 million EUR. The expected present value of costs in the third column of the table 6 is computed by weighting each scenario by its likelihood of occurrence: $0.70 \times 63 + 0.30 \times 90 = 71.1$ million EUR. This gives an expected net present value of modernization equal to $80 - 71.1 = 8.9$ million EUR. Notice that if sensitivity analysis alone is used to compute the present value of expected net benefits without determining the expected present value of costs, then the net present value is positive under cooperation (second row and column) but negative under no cooperation (second row, third column).

While the expected value approach to dealing with uncertainty is conceptually straightforward, the challenge is determining the probabilities to assign to possible values of uncertain benefits and costs. One approach is to rely on expert opinion. Another is to simply assign equal probabilities to possible values of an uncertain cost (or benefit).

Step 8: Aggregating Benefit and Cost Estimates to Obtain Present Value of Net Benefits

Given that the benefits and costs associated with hydromet investments occur over a multi-year time horizon, benefits and costs must be expressed in

Table 6 Expected Cost Example (euros million)

	Cooperation	No Cooperation	Expected PV
PV of Total Costs	63	90	71.1
Net Present Value	17	-10	8.9

Source: Based on data from UNISDR (2008), and assuming a 70% probability of cooperation and present value of benefits equal to 80 million EUR.

present value terms using an appropriate discount rate. Economic theory does not clearly specify the choice of discount rate. However, (annual) discount rates commonly used for evaluating publicly funded projects range from 4 percent to 12 percent in real terms. Once all benefits and costs are expressed in present value terms, they can be aggregated to obtain an estimate of the present value of net benefits of the project.

The choice of discount rate will generally influence both the magnitude and sign of the present value of net benefits. As noted in Section 7, a substantial portion of the costs of hydromet improvements are incurred in the initial years of a project. In contrast, the majority of benefits are likely to accrue with a lag given the time needed to bring new equipment online, train personnel, establish means of effectively communicating forecasts, and educating users on how to make use of forecasts. This difference in the time profiles of costs and benefits implies that higher discount rates will result in a larger reduction in the present value of benefits than in the present value of costs, lowering the present value of expected net benefits.

The output of this step is an estimate of the present value of expected net benefits from the investment. This estimate should be accompanied by a characterization of the unquantifiable benefits described in Step 2.

Step 9: *Conducting Sensitivity Analyses*

Sensitivity analyses are an important component of any economic evaluation. This is especially true in the context of estimating the net benefits of investments in hydromet systems, given the high degree of uncertainty regarding the precise magnitude of benefits and costs. The robustness of conclusions drawn can be assessed by conducting sensitivity analyses using:

- alternative values of benefits estimates;
- alternative values of cost estimates;
- alternative time horizons;
- alternative real discount rates;
- alternative assumptions about the increase in expected losses from extreme meteorological hazards over time due to climate change; and
- alternative assumptions about adaptation by households and enterprises that reduces damages from more frequent, or more intense, extreme meteorological hazards even without the project.

The sensitivity analyses allow the analyst to identify the set of values that are critical to the conclusions drawn from the economic evaluation of the investment. Further effort can then be devoted to refining estimates of these values if desired.

9. Interim and *Ex-Post* Evaluations of Hydromet Investments

The framework described in the previous section provides an *ex-ante* evaluation of a proposed hydromet investment. If the investment is undertaken, interim and *ex-post* evaluations can also be conducted. These evaluations can yield multiple benefits. They provide an opportunity to:

- assess the validity of the *ex-ante* evaluation and refine it;
- collect information that can be used to modify project priorities and allocation of funds in order to increase the net benefits yielded by the investment;
- enhance the design of future hydromet investment projects; and
- improve the accuracy of future *ex-ante* evaluations of hydromet investments.

Interim Evaluations. Data collected as part of an interim evaluation of an investment can be used to improve *ex-ante* estimates of the expected net benefits of the investment. Referring to the framework described in the previous section, three types of information would be particularly valuable:

- *Data on losses from extreme meteorological events.* If data on losses from extreme events have traditionally not been collected, acquiring data on these losses would allow for better estimates of the magnitudes of losses that could be avoided with hydromet investments. Information on such losses could be drawn from data on: (i) private and government insurance payments for property damages, loss of livestock and agricultural output, etc.; (ii) data on private and government expenditures on rebuilding and repairing infrastructure; and (iii) government assistance to households affected by extreme events;

- *Improved estimates of percentage loss avoided (PLA).* Better data on losses from extreme events would allow for improved estimates of the percentage loss avoided used to estimate the benefits of improved forecasts of extreme events. PLA estimates could be further refined using in-country assessments of the types of losses avoided with the improved forecasts generated by the investment, and through aggregation of expert opinion from various user groups;
- *Data on the costs of hydromet investments.* As noted in Section 7, data on the fixed costs of hydromet investments can be estimated fairly accurately *ex-ante*, however, the variable costs associated with operating and maintaining improved hydromet system are more difficult to estimate, especially in developing countries. In the course of implementing a hydromet project, data on both variable and fixed costs, such as actual costs of purchasing and installing hardware, upgrading communication networks, training staff, and hiring new staff, should be collected as these costs are incurred and then compared to *ex-ante* estimates.

***Ex-Post* Evaluations.** The types of information collected to conduct interim evaluations are also useful for conducting *ex-post* evaluations. Ideally, an *ex-post* evaluation would use data on the realized benefits and costs of the hydromet investment for a few years after implementation of the investment to derive a revised estimate of expected net benefits. However, given the stochastic (or random) nature of climate and extreme events, such an analysis would need to be conducted with care. For example, a comparison of losses due to extreme events before and after the hydromet investment would be meaningful and informative only if the

pre- and post-investment extreme events were similar in character, in terms of the type of event, its timing and severity, and the area affected. Similarly, comparisons of agricultural output pre- and post-investment would allow for benefits of the hydromet investment to be reliably estimated only if meteorological (and economic) conditions a few years after implementation were similar to conditions before the investment.

Meaningfully comparing losses from similar extreme events before and after an investment is more likely to be feasible than comparing economic productivity or output (e.g., crop yields). A comparison of productivity or output could reasonably be conducted only several years after implementation of the investment. Only then would there be a sufficiently long time series of data to reliably estimate average changes over time. Using data for only one or two years post investment would render the comparison susceptible to bias due to the vagaries of weather in a given year. For example, crop production from one year to the next could vary simply because of changes in the quantity or timing of rainfall. The effects of these changes in rainfall would be difficult to disentangle from the effects of the hydromet investment.

A pitfall of waiting several years to conduct the ex-post evaluation, in order for sufficient data to be collected, is that holding constant the effects of other changes in the economy becomes more difficult. If other changes are not held constant, the effects of the hydromet investment per se cannot be reliably identified and estimated.

The above issues imply that conducting a traditional ex-post evaluation of hydromet investments is problematic and may not always be feasible. An alternative approach, one that is more generally applicable and that yields valuable information, is to assess the extent to which the characteristics of

forecasts and their uses have changed as a result of the hydromet investment. In particular, information could be collected on:

- *Changes in forecast characteristics.* Important characteristics of forecasts are their accuracy, frequency, lead time, and spatial resolution;
- *Perceptions of households regarding changes in forecast characteristics and the value of these changes.* Household surveys could be conducted to learn whether households have noticed changes in forecast characteristics, and how they have made use of perceived changes in forecasts and benefited from the changes. The surveys should ideally be conducted for a cross-section of socio-economic groups and regions. In addition, stated preference surveys could be conducted to estimate the value placed by households on the perceived changes; and
- *Perceptions of enterprises regarding changes in forecast characteristics and the value of these changes.* As in the case of households, enterprises and specific user groups could be surveyed to determine whether they have observed changes in forecast characteristics and how any perceived changes have benefited them. In addition, surveys could be conducted to learn what additional changes in forecast characteristics and dissemination would be of most value to various user groups.

This information would allow for an assessment of the extent to which improved forecasts have been effectively communicated to users and the ability of users to benefit from them. As emphasized in Section 4, effective communication of forecasts to users, their ability to understand the forecasts and incorporate them into their decision-making, and their ability to respond to forecasts are critical to generating value from forecasts. Furthermore, an understanding of how various user

groups make use of forecasts, and of the forecast characteristics that are of most importance to them, can help guide decisions on the types of hydromet services to provide and enhance. As discussed in Section 4, identifying and prioritizing high-value or “high-impact” forecasts enables the net benefits generated by hydromet systems to be increased.

As indicated earlier, information gained from interim and *ex-post* evaluations, including the findings of the surveys described above, can be used to inform future hydromet projects in other countries, thereby generating project spillovers. Specifically, these data collection efforts and accompanying analyses can potentially make future hydromet investment projects cheaper and more effective.

10. Conclusions and Recommendations

Hydromet investments can generate a wide range of benefits that are enjoyed by a broad swath of a country's economy. The benefits take the form of reduced damages from extreme meteorological events because of better forecasts, and increased profits, or reduced costs, due to improved decision-making by users whose day-to-day economic welfare is affected by weather, such as households, farmers, and energy producers.

Estimating the benefits, and hence the net benefits, of hydromet investments with precision is a challenging task given the wide variety of forecast users and the very varied decision contexts in which forecasts are used. The task is further complicated by the fact that the magnitude of benefits depends on the precise characteristics of the forecast, the effectiveness with which forecasts are communicated to users, and the ability of users to understand and make appropriate use of the forecasts.

The framework proposed here for estimating the (*ex-ante*) expected net benefits of hydromet investments is intended to yield first-cut estimates of net benefits at the country level, without onerous data or analytical requirements. The framework draws on existing approaches with some important modifications and additions. In particular, it attempts to capture the benefits of improved forecasts of extreme meteorological events as well as the benefits from improved forecasts of routine climate. To the extent possible, estimates of the first type of benefits are based on historical data on losses from extreme events. The proposed framework makes extensive use of benefits transfer. However, the results of benefits transfer should be subject to review and revision by sector experts.

Given the high degree of uncertainty regarding the precise magnitude of benefits and costs associated with hydromet investments, conducting a wide range of sensitivity analyses is an essential component of the proposed framework. Sensitivity analyses should be conducted using alternative estimates of benefits and costs, alternative time horizons, alternative assumptions about the consequences of climate change for the frequency and magnitude of extreme events, and alternative assumptions about the ability of households and enterprises to adapt to these events.

Not all the benefits of hydromet investments can be quantified and monetized. Although such benefits fall outside the framework of a traditional economic evaluation (in which all benefits and costs are monetized), they should not be ignored. Benefits that are likely to be large in magnitude should be identified and characterized even if they cannot be quantified or monetized.

Supplementing *ex-ante* evaluation of a hydromet investment with interim and *ex-post* evaluations offers a range of benefits beyond the obvious ability to assess the validity of the *ex-ante* evaluation and refine it. In particular, priorities and allocations of funds for the investment under study can be revised and design of future hydromet investments can be improved. Accordingly, collection of data for conducting interim and *ex-post* evaluations plays an important role in enhancing the net benefits from hydromet investments in the long run, in addition to enhancing the ability to evaluate the investments *ex-ante*.

Annex 1: Economic Value of a Forecast – An Illustrative Example

To develop an understanding of some of the key determinants of the economic value of forecasts and the types of information needed to estimate these values, we present a simple example of the benefits generated by an improved, or better, forecast.⁸ The example is based on an agricultural setting, but the insights derived are generally applicable.

Decision Context. Consider a farmer who must decide when to harvest his crop in the face of uncertainty about heavy rains that would damage the crop. The farmer is certain that it will not rain today but believes there is some chance that there will be (heavy) rain tomorrow. Suppose the farmer has two options:

1. Option 1: harvest the entire crop today, at a total cost of USD 10,000
2. Option 2: harvest half the crop today and half tomorrow, at a cost of USD 3,000 per day

The higher cost associated with Option 1 could be due to having to pay labor overtime. If it rains tomorrow and the farmer has chosen Option 2, he will only be able to harvest half his crop (the portion harvested today); the other half will be damaged and rendered worthless. Each half of the crop can be sold for USD 15,000. The “payoff matrix” facing the farmer, which shows the farmer’s profits for the two options with and without rain tomorrow, is presented in Table A-1.

If the farmer harvests the entire crop today, whether it rains tomorrow is irrelevant. With or without rain, the farmer’s profit or payoff is USD 20,000 (USD 30,000 in revenues minus USD 10,000 in

Table A-1 Payoffs for Alternative Harvesting Decisions

Farmer’s Option	Weather Tomorrow (USD)	
	No Rain	Rain
Option 1: harvest all today	20,000	20,000
Option 2: harvest ½ today, ½ tomorrow	24,000	12,000

Source: Based on Quirk (1976) and Macauley (1997).

costs). If the farmer harvests only half the crop today, then his payoff is only USD 12,000 if it rains tomorrow because the remaining crop will be ruined (USD 15,000 in revenues minus USD 3,000 in costs). But if it doesn’t rain tomorrow, he will also be able to harvest and sell the remaining half and will earn USD 24,000 in total.

If we let p denote the probability that it rains tomorrow as perceived by the farmer, we can calculate the farmer’s expected payoff for each of the two options:

- (1) EPNF1 = Expected Payoff for Option 1 without (Better) Forecast

$$= p(\text{USD } 20,000) + (1-p)(\text{USD } 20,000)$$

$$= \text{USD } 20,000$$
- (2) EPNF2 = Expected Payoff for Option 2 without (Better) Forecast

$$= p(\text{USD } 12,000) + (1-p)(\text{USD } 24,000)$$

$$= \text{USD } 24,000 - (\text{USD } 12,000)p$$

The value of p perceived by the farmer could be determined by a currently available forecast of tomorrow’s weather, or it could be entirely subjective, i.e., based on the farmer’s prior experience and beliefs regarding weather.

If we assume, for simplicity, that the farmer is risk neutral, he will choose the option that yields the

⁸ The example builds on one in Quirk (1976) and Macauley (1997).

higher expected payoff. This preferred option clearly depends on the value he attaches to the probability p . For example, if he believes with certainty that it will rain tomorrow ($p = 1$), then Option 1 yields the higher expected payoff (USD 20,000 vs. USD 12,000). On the other hand, if he believes there is no chance of rain tomorrow ($p = 0$), then Option 2 is preferred (USD 24,000 vs. USD 20,000).

The expected payoffs for the two options are depicted in Figure A-1. The solid, horizontal line is the expected payoff with Option 1, and the downward-sloping, solid line is the expected payoff with Option 2. The two lines intersect at $p = 1/3$. This is the value of p at which the expected payoffs for the two options are equal. Notice that the preference for one option over the other (i.e., the gap between the two lines) grows larger as the value of p moves away from $1/3$ in either direction. The hashed portions of the two lines show the expected payoff when choosing the preferred option for each value of p .

Value of a Perfect Forecast. To estimate the value of a better forecast in the simplest setting possible, let us assume that the improved forecast is a *perfect* forecast—the forecast will specify with certainty, and complete accuracy, whether or not it will rain tomorrow. This is obviously a best-case scenario of an improvement in forecasts. What would the economic value of such a forecast be to the farmer? Following economic convention, the value of the forecast is the maximum amount the farmer would be *willing to pay* for it. This amount would equal the difference between the farmer's expected payoff with the forecast and his expected payoff without it,

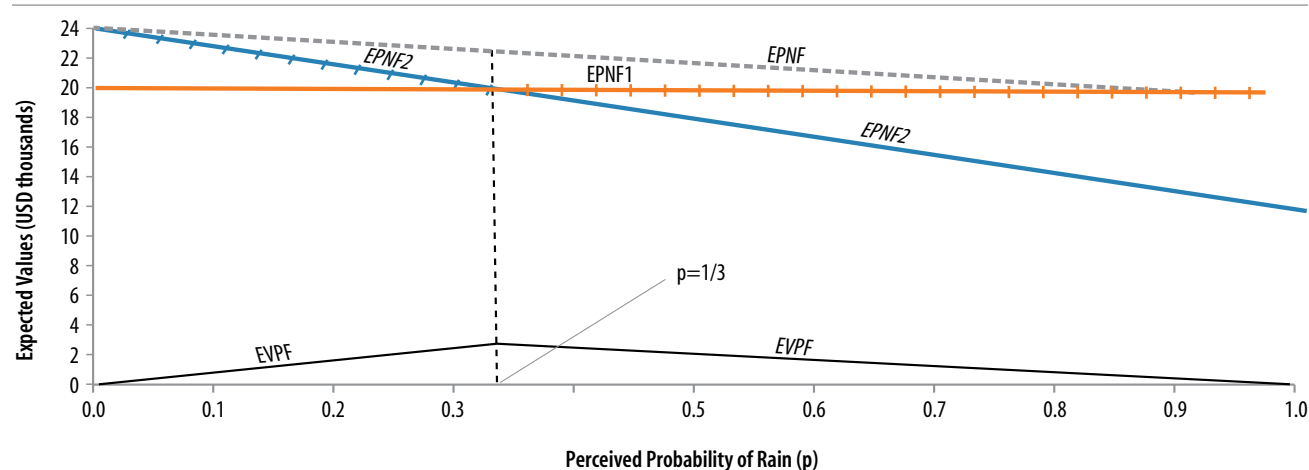
$$\begin{aligned} (3) \text{ EVPF} &= \text{Expected Value of Perfect Forecast} \\ &= \text{Expected Payoff with Perfect Forecast} \\ &\quad - \text{Expected Payoff without (Better) Forecast} \end{aligned}$$

We have already determined the farmer's expected payoff without the forecast (captured by the kinked, hashed line in Figure A-1). Therefore, we only need to derive the farmer's expected payoff with the forecast. That we are computing the *expected* payoff with the forecast merits some explanation. The farmer does not know what the forecast is (rain or no rain) until he obtains it. Therefore, *ex-ante*, i.e., before the forecast is available, we can only compute his expected payoff with the forecast. This expected payoff is calculated using the farmer's *ex-ante* assessment of the probability that it rains tomorrow. If the forecast reveals that it will not rain tomorrow, the farmer will choose Option 2, since it yields a larger payoff in the event of no rain (USD 24,000 vs. USD 20,000). *Ex-ante* the farmer believes that the probability of the forecast revealing that it will not rain tomorrow is equal to $(1-p)$, which is simply his assessment of the probability that it will not rain tomorrow. Correspondingly, if the forecast reveals that it will rain tomorrow, the farmer will choose Option 1 (payoff of USD 20,000 vs. USD 12,000 with Option 2), and *ex-ante* the farmer attaches a probability of p to receiving this forecast. Thus,

$$\begin{aligned} (4) \text{ EPPF} &= \text{Expected Payoff with Perfect Forecast} \\ &= p(\text{USD } 20,000) + (1-p)(\text{USD } 24,000) \\ &= \text{USD } 24,000 - (\text{USD } 4,000)p. \end{aligned}$$

This expected payoff is shown by the dashed line at the top of Figure A-1. Note that the expected payoff with the perfect forecast is higher than the expected payoff without the forecast (hashed line) except for the extreme cases where the farmer believes there is no chance of rain tomorrow ($p = 0$) or he is certain that there will be rain tomorrow ($p = 1$). For either of these two extreme cases, the expected payoff with the perfect forecast is the same as the expected payoff without the forecast. Thus, if the farmer is certain of tomorrow's weather, even if his beliefs are mistaken, he will not attach any value *ex-ante* to the forecast.

Figure A-1 Deriving the Expected Value of a (Perfect) Forecast.



Source: Based on Quirk (1976) and Macauley (1997).

The expected payoff to the farmer with no forecast depends on the option he chooses. EPNF1 is the payoff with Option 1 and EPNF2 is the payoff with Option 2. The option that yields the higher expected payoff depends on the probability of rain perceived by the farmer (p). The kinked, hashed line represents the highest payoff without a forecast. The expected payoff with a perfect forecast is given by EPPF. The expected value of the perfect forecast is obtained by subtracting the highest payoff without the forecast from EPPF. The kinked line labeled EVPF shows the resulting expected value of the perfect forecast.

The vertical distance between the dashed line and the hashed line corresponds to the difference on the RHS of equation (3). Hence it captures the expected value of the perfect forecast. This value is shown by the kinked, solid line near the bottom of Figure A-1. As the figure makes clear, the value of the forecast depends on the value of p . As noted above, the forecast is of no value to the farmer if he does not perceive any uncertainty—he is certain that it will not rain tomorrow ($p = 0$) or he

is certain that it will ($p = 1$). If the farmer believes there is some chance of rain tomorrow ($0 < p < 1$), the forecast is of value. Notice from the figure that the expected value of the forecast is highest when p takes on a value of $1/3$.⁹ This is the value of p for which the farmer would choose one of the two options by a coin toss (in the absence of the forecast), because they yield the same expected payoff. The value of the forecast diminishes from this maximum when the value the farmer attaches to p moves away from $1/3$, in either direction, inducing the farmer to have a stronger preference for one option over the other in the absence of a forecast.

Determinants of Forecast Value. The above observations lead us to the following insights about the value of better forecasts:

- The expected value of a better forecast to a user depends on the user's prior beliefs about the uncertainty he faces. If the user does not believe there is uncertainty about future weather, better forecasts have no value to the user.

⁹ The expected value of the forecast when $p = 1/3$ can be calculated as follows: $EVPF = EPPF - EPNF1 = EPPF - EPNF2 = (24,000 - 4,000p) - 20,000 = 4,000 - (4,000)(1/3) = 2,666.67$.

- The value of a better forecast is larger, the smaller is the difference in expected payoffs (in the absence of the forecast) across a user's alternative choices; conversely, the value of a better forecast is smaller, the stronger is the user's preference for one choice over another in the absence of the improved forecast.

Note that prior beliefs about the uncertainty will likely differ across farmers. Thus, even if the better forecast has no, or little, value for some farmers, it may have high value for other farmers. Given the public good nature of forecasts, the total value of a better forecast is the sum of all users' willingness to pay for the forecast.

Although the economic value of forecasts is routinely expressed in terms of their *ex-ante* expected value, it is instructive to briefly consider the *ex-post* value of forecasts. *Ex-post*, a forecast is of value only if it increases the farmer's actual (rather than expected) payoff. For example, suppose that the farmer perceives the probability of rain to be 0.2. From Figure A-1 the expected value of the perfect forecast is seen positive. In the absence of the forecast, the farmer would choose Option 2 since it yields a higher expected payoff. Now suppose that the farmer obtains the perfect forecast. If the forecast indicates that it will rain tomorrow, he will choose Option 1 instead of Option 2 and the forecast will make him better off: his payoff will be USD 20,000 instead of USD 12,000. However, if the forecast indicates that it will not rain tomorrow, he will be no better off with the forecast: both with and without the forecast he would choose Option 1. This example demonstrates that even though a forecast has a positive value *ex-ante*, this may not be true *ex-post*.

The economic circumstances facing the farmer will also influence the value the farmer places on a better forecast. For example, suppose the cost

of harvesting the entire crop in one day is lower, and equal to the total cost of harvesting it over two days, USD 6,000. Then the expected payoff with Option 1 would be USD 24,000. In Figure A-1, the horizontal line labeled EPNF1 would now have a vertical intercept of USD 24,000. The farmer would now always choose Option 1, except when he believes that there is no chance of rain tomorrow ($p = 0$), in which case he would be indifferent between the two options. A forecast would now be of no value to the farmer. The forecast is rendered valueless because one option is preferred to the other regardless of the value of p . It can therefore be concluded that:

- A better forecast is of no value to a user if it does not alter the choices made by the user.

If, instead, the cost of harvesting the crop in one day was somewhat higher, say, USD 7,000 instead of USD 6,000, then the expected payoff with Option 1 would be USD 23,000. It is not difficult to verify that in this case the perfect forecast would have a positive value, but its value would be lower than that depicted in Figure A-1, because the difference in the maximum payoffs for the two options would only be USD 1,000 instead of USD 4,000. This leads to:

- The value of a forecast to a user is larger the larger are the differences in the maximum payoffs across the choices available to the user.

This can be stated somewhat differently,

- The value of a forecast is larger the larger is the loss in expected payoffs from making the wrong choice.

The example has assumed that the farmer has two options. Suppose, instead, that it is impossible for the farmer to harvest the entire crop in one

day—labor or capital constraints prevent him from doing so. So the only option available to the farmer is to harvest the crop over two days. In this case, the forecast is of no value to the farmer because he is unable to modify his actions to take advantage of the forecast. More generally, we have:

- A better forecast is of value to a user only if the user is able to alter his actions in a manner that allows him to earn a higher payoff.

In the above analysis, it has been implicitly assumed that the better forecast is made available to the farmer in time for him to alter his choices. Specifically, it has been implicitly assumed that the farmer receives the forecast before he has decided whether to harvest the crop over one or two days. Suppose, instead, that he receives the forecast at the end of today. He would then not be able to switch from Option 2 to Option 1, and the forecast would have no value to him: if it rains tomorrow, there is nothing he can do, and if it does not rain, he will simply harvest the remaining crop, if any. The forecast does not allow him to increase his payoff. Now suppose that the farmer has started to exercise Option 1, and after harvesting half his crop he obtains a forecast indicating that it will not rain tomorrow. He then may still be able to switch to Option 2, which would allow him to increase his payoff (given the original numbers in the example). It can therefore be stated that:

- A better forecast is of value only if it is provided with sufficient lead time for the user to modify his choices and increase his payoff.

A general insight provided by the above example and the accompanying analysis is that the value of a better forecast to a user is determined by the user's beliefs about the uncertainty he faces and by the agent's ability to use the forecast to modify his choices and increase his expected payoff. This ability is determined by, among other factors, the lead time of the forecast, the choices available to the agent, and the economic circumstances facing the user.

The above example makes the simplifying assumptions that the forecast user is risk neutral, rather than risk averse, and that the forecast, when available, is perfect. A few observations are in order about the effects of relaxing these assumptions. It can be shown using formal and more general modeling of decision-making by a forecast user, that the value of a forecast is higher the better, or higher quality, is the forecast (Katz and Lazo 2011).¹⁰ This result is consistent with intuition. Two other results are not. Intuition suggests that the value of a forecast should be higher the more risk averse is the user. But formal analysis indicates that this is not necessarily true. Similarly, it is plausible to think that the value of a forecast is higher if the user has more flexibility, i.e., the user has a larger set of options from which to choose. However, this too is not necessarily true. Although additional options always make a user better off (or at least no worse off), they may not increase the value of the forecast, because the additional options also make the user better off in the absence of the forecast (see Katz and Lazo 2011).

¹⁰ "Higher quality" has a precise definition in this context that is related to the statistical notions of "sufficiency" and "encompassing." Measures of quality that do not satisfy the "sufficiency" criterion can lead to so-called quality/value reversals with value falling as quality increases. See Katz and Lazo (2011) for more information.

Annex 2: The Value of Hydromet Forecasts to the Insurance and Financial Sector

The multi-billion dollar global insurance and financial sector has expanded in the face of catastrophes since the 1970s, but most of this growth has been in developed countries. Hydromet systems are closely linked to the protection afforded by the financial and insurance sector, as up to date weather information is needed to design contracts between insurers and insured, verify losses, and initiate compensation. A recent example is weather indexing, where compensation is provided for recorded and verified weather events that result in parameters such as temperature or rainfall outside of normal limits set by insurance contracts. These instruments are not yet present in developing countries save for a few small-scale pilot projects involving agriculture. Thus, there is at present very little quantitative information concerning the benefits of hydromet modernization to the financial and insurance sector in developing economies. As a result, quantifying the benefits to this sector from hydromet improvements is difficult at present. However, the benefits could be large for some countries. If so, they should not be ignored even though they are currently unquantifiable.

Scope of Losses and Importance of Insurance.

The scale and scope of potential losses from extreme weather events as well as periodic cycles and fluctuations in climate have large effects on country level GDP and per capita wealth. Between 1974–2006, losses in terms of drought, flood, and tropical storms accounted for between 0.3%–11.98% of GDP in aggregate losses over that period among developing economies comprising Southeastern Europe alone (Pollner et al. 2010). Very few of these countries possess a well-developed insurance sector and nearly all have largely insufficient government budgets with which to deal with losses of this

scale. Most of the losses are therefore completely absorbed by individuals and firms.

Types of Weather Insurance Instruments. There are clear synergies between weather events, attempts to forecast these events, and contracts in the private insurance and financial sector. At a countrywide scale, there are two types of insurance instruments closely related to hydromet systems and data: industry or individual agent based (micro), and macro based. These are also sometimes referred to as ‘parametric insurance’ instruments as they are based on measurements of critical weather and forecast predictions and variables. Macro based instruments focus on mitigation to government budgets due to the adverse effects of disasters by allowing governments to hedge against potentially large fiscal losses such as infrastructure repair. Examples are insurance offered from large philanthropic investors, major insurers, or other countries that provide development aid in the event of major disasters. Micro based instruments, for producers or consumers, are risk mitigation mechanisms aimed at preventing damage to specific assets and are most commonly provided by private insurance firms. Examples are insurance purchased by farmers that guarantee an annual return in the event of rainfall or crop yields below an agreed-upon (by contract) level, degree-day contracts with power companies that provide compensation for temperatures above or below levels that greatly increase costs of energy production, and wind power insurance in Europe that provides compensation to owners for periods without sufficient wind.

Importance of Hydromet System Performance.

Improved weather forecasts from hydromet systems have a large impact on the development of micro and macro instruments in developing countries in several ways. Insurers can use accurate monitoring and weather information to target instruments

where the probability of risk is highest by aiding in the determination of premiums and underwriting of risk sharing contracts, sell contracts to outside investment firms that include these instruments in their portfolios (e.g., reinsurance), and to avoid potentially large losses that the private sector has trouble compensating and that require government intervention. Further, the insurance sector is multilayered, with lower layers having direct insurance firms collecting adequate premiums to cover losses, but higher layers required for extreme events with a much larger capital level at risk—at higher layers, the premium to expected loss ratio can be as large as eight times higher than lower layers (WRMA 2002). The higher layers also operate under greater uncertainty concerning arrival of extreme weather events and the damage that these events cause. For both layers, but especially the higher ones, the contribution to profits of firms in the sector from more capable hydromet systems can be considerably high. Further, there are indirect benefits of hydromet data that can be difficult to quantify, but are important nonetheless. For example, more risk but better information about that risk implies new and greater opportunities for a hedge fund industry to build portfolios with assets relevant to this risk. The increased forecast accuracy from a modernized hydromet system would undoubtedly make decisions regarding investments from this industry more informed and therefore more profitable.

Potential Benefits of Weather Insurance Instruments to Developing Economies. Both micro and macro insurance instruments are still rarely used in developing economies. As far as macro insurance instruments go, Central Asia is a representative example of the challenges faced. Here, every country government aside from Turkmenistan devotes some part of their annual budget appropriation to emergency weather events. However, this budgetary protection is often as small as a few percentage

points of the amount actually needed once an extreme weather event arrives (UNISDR 2009). UNISDR reports that in all countries but Kazakhstan the economic losses from a two hundred year catastrophic event are in the range of 100–200 times the annual planned emergency budgetary allocations. Here, better insurance targeting through improved weather information has great potential to increase wealth and ensure continued economic growth, as losses from natural disasters are at present funded by borrowing or reallocating existing budget money.

The Current State of the Insurance Sector in Developing Economies. There are a few examples of micro insurance instruments in parts of Asia, Ethiopia, India, and Malawi where crop insurance contracts exist based on drought (e.g., see Suarez et al. 2008 and ASEAN 2012). However, insurance penetration is still low; for example, in Central Asia, a region that has garnered recent interest in hydromet modernization, UNISDR (2009) finds that penetration is currently as low as 1.5 percent of GDP at risk; this is roughly an average protection of only 12 USD per capita, most of which is not attributable to real property. The UNISDR report identifies the following reasons for this observation:

- insurance regulators lack expertise and information to accurately assess risk exposure;
- firms lack underwriting and actuarial support needed to form contracts;
- there is a lack of reliable weather information to assess true risk exposures to both insurers and insured; and
- the majority of insurer firms do not have reinsurance or excess-of-loss protection options.

The reasons identified by UNISDR exist in part because accurate indexing against weather information is not possible, but they also follow from lower than optimal premiums. It is not surprising that

the insurance sector in developing countries is very underdeveloped with available insurers vulnerable to extreme weather events. Better weather information will allow governments to obtain adequate market based insurance in order to reduce contingency budget exposure, receive immediate liquidity after weather damage, increase fiscal stability and economic growth, and allow real estate owners to protect assets and firms to protect earnings. Such insurance will also reduce the costs of borrowing at all levels in the economy.

A New Frontier for Hydromet Investments. Most of the barriers in the insurance sector could be lessened with more accurate weather data recording and forecast information with which to design contracts, for all parties involved including individuals, governments, insurers, and re-insurers. The World Meteorological Organization has called the development of insurance sectors and markets a new frontier in hydromet investments (Golnaraghi 2009). Golnaraghi argues that the benefits of hydromet modernization would derive from the availability and accessibility of historical and real-time data; data quality assurance, reliable and timely data for contract design and settlement; and forecasts for forward looking futures investments. Indeed, she argues that improved weather information is the reason for the rapid development of weather-indexed micro insurance and investment instruments in developed countries.

Dutton (2002) describes weather insurance contracts as “weather derivatives.” A common set up for these contracts in the U.S. and Europe involves use of high/low degree or high/low wind days as a signal of climate variability-based damages. For example, a utility could arrange to receive payment

for days above a certain temperature up to some limit on the total payout, in return for a premium paid to the insurance company that is based on a percent of the limit. In a more recent discussion, ASEAN (2012) reviews and reports several different types of insurance instruments relevant to weather derivatives in Asian countries. This report notes that weather derivatives offer many advantages to the typical catastrophic insurance arrangements (such as CAT bonding),¹¹ including: tailor made compensation and premium indices, multi-annual protection, flexibility with regards to the geography of protection, and rapid payout as parameters indexed are quickly and clearly measured and observed. The report mentions the main disadvantage of using these instruments as insufficient historical data measurements that impede their adoption in developing countries. Regardless of the type of weather derivative, hydromet information is critical, and these parametric contracts cannot exist without accurate forecasts needed to define the premium payment schedule and event payout limits.

A multiplier type value of hydromet investments to the insurance sector can also be realized through greater private investor interest in weather derivatives, such as from hedge fund managers and futures traders. Futures traders could purchase contracts from insurance companies, obtaining a high return through collected premiums in periods that extreme events do not occur. Futures trading would serve to pool risk and generate additional returns both in the financial sector and for those affected by extreme weather events. When this interaction between financial investors and insurers is considered, the size of the potential monetary benefit attributable to better forecasts could be very

¹¹ Catastrophe bonds are security instruments designed to reduce risk for potentially affected insurers by transferring that risk to private investors. First created in the U.S. in response to hurricanes and earthquakes in the 1990s, they were designed so that insurance companies would continue to cover individuals in regions where the value of potential damages exceed premium collections. These bonds are issued through investment companies and sold to investors, who receive potentially high returns in periods where catastrophes do not occur.

large. The Weather Risk Management Association (WRMA 2002) estimated that in 2001 the value of weather derivatives was more than USD 4 billion per year in the U.S., and in Europe weather derivatives were worth, in terms of total risk transferred, USD 1.5 billion.

Another encouraging fact for hydromet investments is that the scope of meteorological information required to structure weather derivatives may be low in cost to obtain once hydromet systems are modernized. This would increase the net benefits estimated for hydromet investment through economic evaluations. For example, Zeng (2003) and Jewson and Caballero (2003) have shown how forecasts can be used in the pricing of weather derivatives for heating degree days, finding that accurate calculations of expected payoffs in linear contracts require only accurate mean temperatures over the contract period, while nonlinear contracts require only accurate forecasts of both the mean and the distribution of temperatures, but not the dependence between temperature distributions on different days. Carriquiry and Osgood (2012) also find that agricultural insurance contracts when properly designed can exploit important synergies between forecasts, insurance, and effective input use. Further, as noted in ASEAN (2012), the critical need for hydromet information in weather derivatives is for development of hazard rate functions for use

by the insurance sector. This study notes that, for Asian cases, there are lower operating costs for weather derivatives than for traditional insurance, because weather based (parametric) arrangements do not require costly monitoring and there is no costly moral hazard or adverse selection typical in the insurance industry since weather events are observed by all parties involved.

Summary of Expected Benefits from Hydromet Modernization. The collective message from the literature and the reality of weather risk exposure in developing economies—the large potential losses, and a large percent of the population exposed—imply that weather based instruments, growth in the financial and insurance sector, and modern hydromet systems are closely linked. Although it is conjecture, the size of the weather dependent insurance sector in developed countries means that, depending on the size of the economy, expected benefits to a developing country from modernizing hydromet systems could range from hundreds of millions USD to over 1 billion USD once the insurance and corresponding financial sectors were fully integrated around the new data generated from improved forecasts. Moreover, the resulting future growth in the financial sector and GDP would lead to additional government resources to ameliorate the effects of extreme weather events on budgets, services provided, and victims of disasters.

Annex 3: The Value to the Hydropower Sector of Improved Forecasts of Routine Climate

Energy producers frequently operate in an environment of high uncertainty. Demand and supply of energy are constantly in flux, and can be difficult to predict. Many types of energy generating equipment require significant amounts of time to “warm up” and “cool down,” delaying the speed at which plant operators can react to changes in the energy marketplace. Continuously maintaining sufficient power to the grid, while also minimizing excess power production, therefore requires both constant monitoring and rebalancing of energy production, and also a sophisticated model to predict short-term fluctuations in energy demand. In addition, extreme meteorological events in the form of heat waves, droughts, tropical storms, and blizzards, can cripple vital portions of the energy infrastructure at a time when having reliable power is most critical.

While electrical energy is too expensive to store, hydroelectric plants have the ability to store water in their reservoirs. During times of low energy demand, water is collected in the reservoir. When demand increases, that water is released from the reservoir and the current it generates is used to spin turbines, which generate electricity. This

system allows plant operators to better align increased energy production with times of peak energy demand. However, if a reservoir is allowed to get too full, flooding can occur, which will cause both a lost opportunity for energy production and possible damage to the surrounding area. If the reservoir level is allowed to get too low, however, there is a risk of running out of energy at a time of excess energy demand. Careful stewardship of the reservoir is therefore a key concern of plant operators.

In order to optimally manage a hydroelectric power plant, operators rely on various models to predict future stream flows into the plant reservoir. Inputs into these hydrological models include forecasted weather data, such as air temperature, wind speed and direction (especially when snow fall is a factor), precipitation, and humidity. Since streams flowing into hydroelectric plants can be fed by sources sometimes several hundred kilometers away (and therefore many days upstream), these hydrological models may be very complex and require forecasts over a wide geographic and temporal range. The final predictions made by these models are only as accurate as the data fed into them, making accurate weather forecasts vital to efficient and reliable energy production.

Many factors must be considered when estimating the benefits to hydropower from improved weather

Table A-2 Benefits to Hydropower on the Missouri River from Improvements in Forecasting Ability

Baseline Scenario/forecast knowledge	Average annual hydropower benefits (USD millions)	% increase in benefits from perfect forecast knowledge
Zero forecast skill	359.8	7.1%
1. Current climate knowledge	363.2	6.1%
2. Current climate, perfect snow water content	364.5	5.8%
3. Current climate, perfect and soil moisture content	366.6	5.2%
Perfect forecast skill	385.5	—

Source: Maurer and Lettenmaier (2004).

forecasting, including the current forecasting power, the incremental improvement in forecasting power, variability of stream flow, and characteristics of the river system. Chief amongst these characteristics is the ratio of reservoir capacity to total annual stream flow. If reservoir capacity is very large compared to stream flow, then the ability to predict future stream flow is of little consequence, as there would be little opportunity cost to storing or using excess water.

Given limited resources, and the complexity of the systems involved, one would best approach the estimation of net benefits to the hydropower industry from improved weather forecasts through the use of benefits transfer. Maurer and Lettenmaier (2004) provide one of the few estimates of the benefits of improved stream flow predictability—via improvements in climate variable forecasting—for hydropower along the Missouri River System. They find that the difference between no forecasting ability and perfect forecasting ability leads to only a 1.8% improvement in hydropower benefits for this region. This is mainly due to the fact that the reservoirs along the Missouri River System can store up to three times the annual inflow to the reservoirs, making climate forecasting fairly inconsequential. However, when conducting the same analysis for a hypothetical system

identical to the Missouri River System but with a storage capacity only 1.2 times the annual flow, they find the improvement from no predictive power to perfect predictive power results in a 7.1% increase in hydropower benefits. In addition to the ‘no predictive power’ baseline, the authors also calculate the increase in benefits to hydropower from perfect forecast skill from three other baselines: (1) current climate knowledge and no knowledge of snow water content and soil moisture, (2) current climate knowledge, perfect knowledge of snow water content and no knowledge of soil moisture, and (3) current climate knowledge, and perfect knowledge of snow water content and soil moisture. The benefit from each of these scenarios is given in Table A-2.

In general, how one should adapt these values to apply benefits transfer depends on both the current state of forecasting ability and the degree of improvement from hydromet investments. One should also adjust the magnitude of the expected benefits based on the ratio of reservoir storage capacity to annual flow, as this ratio can be key to determining the importance of weather forecasting to hydropower benefits. In the case of the Missouri River System, a 60% decrease in this ratio leads to a nearly 300% increase in incremental benefits to perfect forecasting power.

Annex 4: The Value of Hydromet Forecasts to the Transport Sector

The highly integrated and multilayered nature of transport sectors means that more accurate forecasts, predictions of seasonal and daily trends, and real time warnings of extreme events, afford benefits that are extensive in that they can impact an entire economy. Transport affects consumers and demand, producers and supply, access to and the functioning of markets, and even ambient environmental quality conditions. Further, transport is multimodal, involving sea, air, and land resources. Outdated hydromet systems are likely to yield inaccurate or untimely forecasts and render the sub-population using the transport sector without critical information to reduce susceptibility to extreme events and to minimize losses.

Avoidable losses from improved hydromet systems in the transport sector include:

- reduction in deaths or damage to property from accidents that are avoided because of more accurate forecasting and weather based closings and evacuations;
- avoided losses in profits by better-managed shipping and delivery according to weather variability;
- avoided cancellations in transport service usage; and
- better management of capacity within the transport sector and travel decisions made by individuals.

The direct benefits of hydromet modernization to the transport sector are highest for users of the road surface, rather than to the firms and consumers that depend on road-delivered items. The latter may still be significant beneficiaries though because improved forecasts affect output and input

planning and delivery for firms who have materials moved through road-based systems. For example, Leviäkangas et al. (2007) find that the most important and immediate benefits from improved hydromet systems are information and warning services offered to road users that cause them to alter their driving and travel plans in ways that avoid harm to them and to property.

Given that hydromet information is needed for an efficient transport sector, the simplest measure of the benefits of hydromet modernization is the overall contribution of the transport industry to GDP and the protection of this economic value afforded by improved weather information. The overall contribution of the transportation industry to GDP in many economies is as high as 50% (Han and Fang 2000). However, as Hallegatte (2012) notes, the benefit-cost ratios of improved weather information in the transport sector, and therefore the importance of hydromet systems to GDP, are heavily dependent on the state of the current hydromet system and infrastructure of a county. While the benefit of hydromet modernization is undoubtedly positive using a GDP protection measure, this benefit may be lower in developing economies compared to developed ones.

Economic Value of Hydromet Modernization to the Transport Sector. The preferred approach to estimating the economic value of better forecasts in the transportation sector is to evaluate the monetary value of decisions that all agents in the sector make with better information. The corresponding economic value of better decisions is the accompanying improvement in incomes and profits that follow from having improved climate information. Unfortunately, nearly all of the current evaluations of the economic value of hydromet services and transport are based on developed country data, primarily because the radar infrastructure already exists and data recording is extensive and reliable.

The study by Leviäkangas (2007) is an exception; it was carried out for Croatia and is illustrative of the scope and scale of benefits one can expect from modernization of hydromet systems in a developing economy that lacks fully modern infrastructure and has lower per capita income than developed countries.

Leviäkangas et al. (2007) use a method for evaluating Croatian hydromet modernization that is based on Value of Statistical Lives Saved (VOSL) through better weather information. While this VOSL is not identical to those traditionally estimated, their measure is one that has been commonly used in transport sector evaluation, as discussed in Advisors to the High Level Group on Infrastructure Charging (AHLG) (1999) and Tiehallinto (2006). It differs from the traditional measure by not including changes in risk and predicted losses under these risks, and is therefore essentially a willingness to pay measure for greater safety augmented by profits and costs avoided by reduced accidents. Their measure is defined as:

$$\text{VOSL} = \text{WTP} + \text{NLP} + \text{HLC} + \text{ADM} + \text{PDV}$$

where

WTP = an estimate of human willingness to pay plus lost consumption

NLP = net lost production to firms

HLC = hospital care cost

ADM = administrative cost

PDV = property and material damage cost

The elements of this VOSL are constructed for each transport sub-sector based on expert opinion. A value from a Finnish hydromet study is used as a baseline VOSL. This is then adjusted according to specific differences between the Finnish economy and the Croatian economy using a price level coefficient (PLC) that relies on purchasing power parities. It is defined by comparing GDP adjusted

for purchasing power and using the ratio of Croatian adjusted GDP to Finnish adjusted GDP. In their study, $\text{PLC} = 0.41$. The implicit assumption here is that human and environmental values are similar across countries since the modern hydromet system will be similar, and therefore the only adjustment made is due to per capita and country wealth differences.

Current State of Croatian Hydromet Services.

The Meteorological and Hydrologic Service of Croatia (DHMZ) maintains a working hydromet system, but it is outdated with a preponderance of older technology and lack of professionally trained meteorologists. The DHMZ oversees 41 surface meteorological stations, 116 climatological stations, and 336 precipitation stations spread across the country. However, only 34 are automated meteorological stations. There are also only 2 radiosonde stations, one pilot balloon station, and 3–5 radar stations. Moreover, these stations are understaffed. The DHMZ employs a staff of roughly 400, with about 8% of these classified as research staff. Only about 6% have graduate degrees (2% are postgraduate level).

Estimated Benefits to Road Surface Users.

Leviäkangas et al. (2007) find that critical aspects of hydromet modernization to road users are related to localized real time information, such as travel warnings or websites that provide updated road conditions, with precipitation amounts, ice and snow coverage, and humidity tabulated by location and road network. They find that the benefits of improved meteorological information to the entire road transport network for Croatia is as high as 17 million EUR annually assuming a modern hydromet system is put in place. In Croatia improved meteorological information enhances winter road maintenance by allowing more efficient employment of scarce resources in advance of storms. Leviäkangas et al. find that this benefit amounts to 340,000

Table A-3 Estimated Annual Total Benefits (Euros) Per Capita and Per Driver from Weather Forecasts and Warnings Under Two Hydromet Assumptions

Sub-Sector	Current Hydromet System		Modernized Hydromet System	
	Per Capita	Per Driver	Per Capita	Per Driver
Road Transport	0.69–1.38	1.55–3.20	1.12–2.23	2.5
Rail Transport	Not Assessed		0.03	
Maritime and Inland Waterway Transport	0.96–1.76		Not Assessed	
Aviation Transport	2.72		3.35	

Source: Based on Leviankangas et al. (2007) and Energy Institute (2012).

EUR annually in cost savings to local and federal governments, as well as additional profits to firms who are able to adjust production and delivery of services optimally as storms arrive. It is worth noting that rail transport is very similar to road transport in terms of the types of benefits from hydromet improvements.

Maritime and Inland Water Transport. The maritime transport industry is arguably the transport sub-sector that is most dependent on hydromet products and services. Important hydromet services include nowcasting and early warning systems for storms and tides, daily condition forecasts, medium term forecasts (4–10 days), and seasonal forecasts (1–6 months). Several studies have found that as much as 20% of all the delays in maritime service delivery are weather related, and as much as half of these delays could have been entirely eliminated by more sophisticated weather monitoring and reporting systems (Smith 1990 and Thornes and Davis 2002). Extreme events also have potential to cause great damage to ships and cargos, leading to very large financial losses. Thus, while meteorological information is critical to on-board decision-making (such as routes and sail dates), other maritime activities are also affected. The most important decision impact areas of

maritime transport are: safety of human beings, protection of environmental values through avoidance of accidents such as spills or sinking, and more reliable material delivery.

Aviation Transport. The aviation sub-sector is similar to the maritime sub-sector in terms of the economic value of hydromet modernization. The main benefits are improved safety from ground radar and storm radar monitoring, more efficient scheduling that reduces fuel and time costs, and increased reliability and demand for services. Leviankangas et al. find for Croatia that accident savings due to weather information services amount to over 5 million EUR per year, a number that will likely grow considerably as the Croatian economy continues to expand (in fact, in developed countries these benefits can be as high as 6–7 million EUR annually due to better flight management, irrespective of safety benefits). Greater forecast accuracy and up-to-date weather information also afford considerable fuel savings through improved decisions concerning flight altitude, routes, and alternate airports which can be fine-tuned to changing weather and wind conditions. In Croatia, improved flight and fuel decisions yield benefits of approximately 1 million EUR per year. Other smaller impacts are also attributable to better forecasts, including reduction

in CO₂ emissions that follow more informed scheduling and fuel management.

Estimation of Total Benefits of Hydromet Modernization. Table A-3 provides a summary of the discussion in this annex and an estimate of the annual per capita total benefits to the transport sector from hydromet modernization. These estimates were taken from Leviäkangas et al. (2007) and based on their estimation of the VOSL equation discussed above. In addition, the 2013 population of Croatia, 4,480,043 and a current estimate of the total number of drivers of all types (commercial and personal use) of 2,000,000 in 2012 (Energy Institute 2012) are used to convert benefits to a per capita value and, in the case of road transport, to a per driver value.

The first set of columns in Table 8 under the heading “Current Hydromet System” presents estimates assuming the current (pre-investment) hydromet system, while the second set of columns under the heading “Modernized Hydromet System” presents

estimates assuming forecast information representative of a fully modern hydromet system. The relative differences between a developing country hydromet investment and the current system described for Croatia can in principle be used to adjust the figures in Table A-3 to any local developing country hydromet system status.

Based on the VOSL measure discussed above, the elements in the Table A-3 are equal to the benefits added together for accident avoidance, plus benefits in terms of profits from better resource management, i.e., better resource allocation, as well as savings in rescue and mitigation activities that accompany better forecasts. Aviation benefits the most, followed by road and then rail, at least for the assumed hydromet modernization. The main benefits for the road category are increased road safety and operational maintenance; the main benefits for rail are time savings and track management; the main benefits of maritime and inland waterways are scheduling and accident reduction; and the main benefits for aviation are enhanced safety.

Annex 5: Economic Analysis of Improving Climate Data and Information Management Project—Jamaica

Introduction

This annex presents an economic analysis of the Improving Climate Data and Information Management Project for Jamaica. The project has four objectives: (i) to upgrade Jamaica's hydromet system so as to enhance climate monitoring, weather forecasting and early warning systems; (ii) to assess the expected consequences of climate change for selected sectors of the economy using climate scenarios, and to enhance climate resilient planning and decision-making in these sectors; (iii) to put in place a comprehensive risk information platform; and (iv) to improve public knowledge, attitudes and practices towards climate change.

The analysis identifies the types of benefits generated by the project and, where feasible, presents first-cut estimates of the magnitudes of the benefits. These estimates form the basis of a cost-benefit analysis. This cost-benefit analysis is likely underestimating the magnitude of benefits, because a number of the benefits generated by the project cannot be quantified given available data. In particular, benefits associated with the second and fourth objectives listed above are difficult to quantify.

Benefits of Improved Climate Data and Information Systems

Types of Benefits. A wide range of benefits have been attributed to upgrading hydromet systems and improving dissemination of information about meteorological conditions and hazards. The benefits

are associated with improved forecasts of extreme meteorological events such as hurricanes, storms, floods, and droughts, as well as improved forecasts of routine weather and climate conditions.¹² Improved forecasts of extreme meteorological events and effective dissemination of information about their effects and appropriate responses to them can substantially reduce economic losses caused by the events. Improved forecasts of routine climate can result in increased enterprise profits (or reduced costs) and improved decision-making by households. Table A-4 presents examples of these benefits, emphasizing the benefits associated with forecasts of routine climate, which are often less apparent. The examples illustrate the pervasive importance of climate forecasts to economic activity.

Existing Estimates of Benefits. A large number of studies have attempted to estimate the economic benefits of climate forecasts. Surveys of these studies can be found in Nichols (1996), Anaman et al. (1998), Stern and Easterling (1999), Houston et al. (2004), Katz and Murphy (2005), Weiher et al. (2005), Teisberg and Weiher (2009), and Rogers and Tsirkunov (2010).

A small subset of this literature focuses on estimating the benefits of improvements in hydromet systems. Instead of estimating the total benefits of existing, or hypothetical perfect, climate forecasts, this body of literature estimates the incremental benefits associated with improved forecasts resulting from upgrading hydromet systems. This distinction between total and incremental benefits is emphasized by Freebairn and Zillman (2002). Much of the work on incremental benefits has been associated with evaluations of investments in upgrading hydromet systems in countries of the former

¹² For brevity, the term “climate” is used to refer to both “weather” and “climate” when a distinction between the two is not important. Weather refers to atmospheric conditions over a period of days or weeks, whereas climate refers to atmospheric conditions over a longer period of time, such as months, seasons, years or decades.

Table A-4 Examples of Benefits from Improved Climate Forecasts

Sector	Period of Forecast ^a	Benefits from Forecast
Agriculture	Short-Range Forecasts	Information on daily precipitation is vital to pesticide application decisions, as heavy rains can wash away recently applied pesticides.
	Medium-Range Forecasts	Accurate medium term precipitation forecasts inform farmers whether or not they need to irrigate, and how much.
		Ideal seeding rates are sensitive to the weather conditions in the days and weeks following planting.
		Timing of planting and harvesting decisions can be improved.
	Seasonal Forecasts	Having more accurate data on the seasonal climate can aid farmers in determining which crops will yield more value.
		Crop insurance programs can benefit from reduced uncertainty of weather patterns.
		Agencies can anticipate food shortages earlier with better seasonal forecasts.
Household	Short-Range Forecasts	Weather forecasts are used to make everyday decisions, such as what to wear, or whether or not to take an umbrella.
	Medium-Range Forecasts	Early warnings of major storms can signal the need for a household to stock up on essentials in case of power outages or road closures, or for potentially life-saving evacuations in cases of extreme events.
	Seasonal Forecasts	Seasonal forecasts can inform households on many decisions, ranging from whether or not and how much insurance to purchase, to what type of household fortifications and improvements to undertake.
Energy	Short-Range Forecasts	Daily weather patterns have an effect on peak energy use patterns, and more accurate weather forecasts can inform power plants when to increase or decrease electricity production.
	Seasonal Forecasts	Hydro-electric generators benefit from improved streamflow forecasts.
Recreation and Tourism	Short-Range Forecasts	Golf course management, recreational fishing, and other outdoor and marine-based activities benefit from more accurate temperature and precipitation forecasts.
	Seasonal Forecasts	Tourist resorts make staffing and investment decisions based on expected tourists in a given season, which can be very sensitive to seasonal weather.
Transportation and Shipping	Short-Range Forecasts	Routing decisions of trucks, ships, and airplanes can be improved with better forecasts of daily weather conditions.
	Medium-Range Forecasts	Cargo ships can use better forecasts to minimize costs and delivery delays in maneuvering around unexpected storms.
Water Resource Management	Seasonal Forecasts	Improved forecasts can lead to more efficient reservoir operations and savings from avoiding groundwater pumping to augment reservoirs.
Fisheries	Medium-Range Forecasts	Fishing vessels rely on weather forecasts to determine when to set sail, how long to stay at sea, and where to navigate to avoid adverse weather conditions.
Emergency Response	Short-Range Forecasts	Adequate warning before extreme weather disaster can significantly reduce losses of life. First responders and emergency rescuers can pre-position emergency response assets to places where they will be most effective for rescue operations.

Source: Authors.

^a Short-range forecasts are forecasts beyond 12 hours and up to 72 hours. Medium-range forecasts are forecasts beyond 72 hours and up to 240 hours. Seasonal forecasts are descriptions of averaged weather parameters over the next 3 to 6 months, excluding individual events.

Soviet Union. Salient examples of such evaluations can be found in SEEDRMAP (2008), World Bank (2008), Rogers et al. (2009, 2010).

A related body of literature has examined the benefits of early warning systems. This literature examines the benefits from improved forecasts of extreme meteorological events and effective dissemination of information about these events and appropriate responses to them. Detailed studies have been conducted on early warning systems for floods, cyclones, and El Nino in various Asian countries (Bangladesh Water Development Board 2006 and Subbiah et al. 2008), as well as on early warning systems for floods, hurricanes and tornadoes in the U.S. (Carsell et al. 2004, Simmons and Sutter 2005, Teisberg and Weiher 2009, and Lazo and Waldman 2011).

For the purposes of this analysis, the literatures on incremental benefits and early warning systems are the most relevant, given that a salient objective of the project is to improve Jamaica's hydromet system and the dissemination and application of the information it generates, including improved early warning systems. The literature on total benefits, rather than incremental benefits, would be the more relevant if the objective of the analysis were to estimate the benefits of having a hydromet system, i.e., the benefits of going from a scenario with no hydromet system to a scenario with a hydromet system. The objective of this analysis is, instead, to estimate the benefits of going from the current hydromet system to an improved hydromet system.

Factors that Influence the Magnitude of Benefits. It bears emphasis that the magnitudes of benefits associated with improved forecasts are very much setting specific. The value of a forecast

depends on a multitude of factors. Among the most important are (e.g., Mjelde et al. 1989, Stern and Easterling 1999, Blench 1999, Houston et al. 2004, Teisberg and Weiher 2009, and Lazo and Waldman 2011):

- time frame (span) of the forecast, e.g., forecasts can be made of tomorrow's weather or next summer's average weather;
- lead time of the forecast, i.e., the length of time between the issuance of a forecast and the time of the event forecasted;
- spatial resolution of the forecast;
- set of weather parameters forecast, e.g., rainfall, temperature, etc.;
- perceived and actual accuracy of the forecast—perceived accuracy can differ from actual accuracy, especially if past forecasts have been wrong;
- timely dissemination of the forecast in a format that is understandable and useful to users, i.e., households, enterprises, or government agencies; and
- ability of users to benefit from modifying their decisions/actions in light of the information contained in the forecast, e.g., a very accurate flood forecast with limited lead time might enable households to flee a flood zone, but might not give them time to move their belongings to higher ground.

These factors imply that benefits estimates derived using benefits transfer¹³, as is done here, should be viewed as indicative rather than exact.

On an economy-wide level, the value of improved forecasts varies across sectors. Some sectors of the economy are more climate sensitive than others, and are therefore more likely to benefit from

¹³ Benefits transfer is a commonly used technique that makes use of benefits estimates derived in one setting to estimate benefits in another, similar setting.

Table A-5 Major Damage-Causing Meteorological Events, 2000–2010

Event	Year	Damage (Billion \$J, Current Prices)	Damage (USD Million, Constant 2010 Prices)	Damage as Percentage of GDP (Official Exchange Rates)
Tropical Storm Nicole ^a	2010	20.60	235.75	1.76%
Tropical Storm Gustav ^a	2008	15.50	262.35	2.01%
Hurricane Dean ^{1a}	2007	23.80	519.08	4.09%
Hurricanes Dennis and Emily ^a	2005	5.98	170.84	1.54%
Hurricane Wilma ^a	2005	3.60	102.85	0.93%
Major Drought ^b	2005	0.38	16.74	0.15%
Hurricane Charley ^a	2004	0.44	14.74	0.14%
Hurricane Ivan ^a	2004	36.99	1,239.14	12.18%
Floods 5/21/02–6/4/02 ^c	2002	2.47	130.95	1.35%
Hurricane Michelle ^a	2001	2.52	150.49	1.65%
Floods 12/29/00–1/4/2001 ^c	2001	0.20	11.94	0.13%
Major Drought ^b	2000	0.25	16.74	0.19%

^a Planning Institute of Jamaica (2011).

^b Government of Jamaica Ministry of Agriculture & Fisheries (2010).

^c United Nations and IDB (2007).

the improved forecasts. The sectors generally considered to be climate sensitive are: agriculture, aviation, construction, surface and water transportation, water resources, energy, fisheries, forestry, health, and tourism and recreation (Houston et al. 2004 and World Bank 2008).

The Jamaican Context. Jamaica's location in the tropics makes its climate especially variable. This variability implies a greater need for, and larger benefits from, high quality climate forecasts. This observation is reinforced by Jamaica's status as one of the most susceptible countries to extreme meteorological events. It is estimated to have the second highest economic risk exposure to two or more natural hazards: 96.3% of the national population is exposed to two or more hazards, as is 94.9% of the national territory and 96.3% of the country's GDP (GFDRR 2010). The dominant natural hazards are hurricanes, tropical storms, floods, and droughts. As revealed in Table A-5, extreme

meteorological events have imposed large economic costs virtually every year of this century. The Strategic Program for Climate Resilience (SPRC) identifies water resources, agriculture, tourism, human health, and human settlements as the most affected sectors in the Jamaican economy (Planning Institute of Jamaica 2011).

The benefits of improving a country's hydromet system depend both on the current state of the system and on the nature and magnitude of the improvements being evaluated. Evidence indicates that Jamaica's hydromet system has deteriorated over time; the original network of 23 climatological stations has dwindled to 6 functioning stations and the 20-year old Doppler weather radar in place is now obsolete and subject to periodic malfunctions. This suggests that there is a scope for sizable benefits to be realized from even modest investments in the hydromet system such as those proposed in this project. Given the onerous debt

burden carried by the Jamaican government (currently at 140% of GDP), it is unlikely that these investments will be made by the Jamaican government on its own.

Overview of Analytical Approach

Estimates are developed for the three broad classes of benefits described above:

benefits from improved forecasts, and associated early warning systems, of extreme meteorological hazards;

- i. benefits to enterprises of improved forecasts of routine climate and dissemination of these forecasts; and
- ii. benefits to households of improved forecasts of routine climate and dissemination of these forecasts; and
- iii. particular emphasis is placed on deriving estimates of the first class of benefits, given Jamaica's high susceptibility to extreme meteorological hazards.

Estimates of all three classes of benefits are derived using benefits transfer given the absence of benefits studies for Jamaica itself. Because of uncertainty about the suitability of the estimates transferred, the values of these estimates are varied as part of a sensitivity analysis. The uncertainty about the estimates stems from the factors listed above.¹⁴ Throughout the analysis, a concerted effort is made to be conservative when estimating benefits, especially those benefits that are subject to a high degree of uncertainty, and to avoid double counting of benefits.

Forecasts are of value only if they are disseminated in a timely manner and in a format that users can understand and make use of. Users must also trust the accuracy of forecasts if they are to act on them (e.g., Stern and Easterling 1999, Chapter 4; Blench 1999, World Bank 2008, Chapter 2). In the case of early warning systems, providing information to users on how to best respond to the warnings is often also critical to the effectiveness of such systems, especially in poor communities (Subbiah et al. 2008; World Bank 2008, Chapter 2; Webster 2012). Without effective mechanisms for timely dissemination of understandable and usable forecasts, as well as information on appropriate responses, the potential benefits associated with upgrading hydromet systems will not be fully realized. The benefits estimates presented here assume that these mechanisms are in place, or are put in place, as indicated in the objectives of the Project Concept Note.

Improved Forecasts of Extreme Meteorological Hazards and Early Warning Systems.

The benefits associated with extreme meteorological hazards are measured in terms of the expected reduction in economic losses resulting from improved forecasts and associated early warning systems. Data on losses in Jamaica from extreme hazards during this century, broken down by sector, are used to derive projections of future expected losses in the without-project scenario.¹⁵ The anticipated economic losses due to increase in the frequency of natural hazards caused by climate change are estimated using predictions developed by the Caribbean Climate Risk Insurance Facility (2010) for Jamaica. These predictions are aggregate losses from

¹⁴ This is particularly true given the absence of precise, quantitative information on the improvements in forecast quality and dissemination that would be generated by the project.

¹⁵ An alternative approach to estimating expected future losses would be to make use of information on return frequencies of different types of extreme meteorological events, together with estimates of the losses that each event would generate. Due to the difficulty in estimating the losses caused by each of a large set of possible extreme events, this approach was not pursued. A further difficulty with this approach is estimating, with any degree of accuracy, the return frequencies of extreme events for a country as small in area as Jamaica.

natural hazards (specifically winds and floods) between 2009 and 2030 for various climate change scenarios.

Reductions in expected losses with the project are based on estimates of percentage loss avoided drawn from existing studies conducted elsewhere that examine the benefits of improved forecasts and associated early warning systems. To the extent possible, the estimates of percentage loss avoided used in this analysis are drawn from studies conducted in countries of similar profile as Jamaica; hazard susceptibility and geographies and economies. This study assumes that the percentage losses avoided remain constant over time, due to lack of information on how they might change.

The approach taken to estimate reductions in expected losses is similar to that employed in previous economic analyses of investments in hydromet systems, however, with some important differences. Unlike the situation faced in some countries, fairly detailed estimates of economic losses from extreme meteorological hazards are available for Jamaica, and these loss estimates are central to this analysis.¹⁶ Because of the absence of percentage loss avoided estimates for Jamaica, this study relies on benefits transfer to derive estimates of the reduction in expected losses with the project. These benefits estimates presented here could be refined by collecting information from Jamaican sector experts on the likely percentage of losses that would be avoided with the project. Such expert-opinion-based estimates have been developed for evaluations of other, much larger, investments in hydromet systems (e.g., World Bank 2008 and Rogers et al. 2009, 2010).

Improved Forecasts of Routine Climate. To estimate the benefits to Jamaican enterprises of improved forecasts of routine climate (the second class of benefits listed above), this study relies on benefits transfer, making use of available estimates in the literature. Useful, transferable estimates are only available for a very small number of sectors. Accordingly, the estimates for this class of benefits are very conservative—they capture only a small subset of the benefits to all climate-sensitive sectors of the Jamaican economy.

The benefits to Jamaican households of improved forecasts of routine climate are based on the findings of a detailed survey of U.S. households' willingness to pay for improved forecasts (Lazo and Chestnut 2002). As discussed in Section 5 (under Stated Preference Methods), Lazo and Chestnut (2002) used a stated value approach to estimate the value to U.S. households of improving the quality of one-day and multi-day weather forecasts from their then-current levels to maximum possible levels. They find that, on average, U.S. households were willing to pay USD 12 to USD 17 per year to improve the quality of the forecasts, with a best estimate of USD 16.

The willingness-to-pay estimates derived from these so-called contingent valuation studies are sensitive to income levels. They are also sensitive to individuals' preferences, which can plausibly differ in a systematic manner across countries and cultures. Therefore, to transfer Lazo and Chestnut's benefit estimate to Jamaica, the USD 16 per household estimate is first adjusted for differences in per capita income between the U.S. and Jamaica, and then expressed in constant (2010) U.S. dollars, yielding a value of USD 2.066. Absent information about

¹⁶ In contrast, the benchmarking method employed to evaluate hydromet investments in countries of the former Soviet Union (World Bank 2008, Rogers et al. 2009, 2010) derives estimates of losses for some sectors of the economy using available data on the weather-sensitivity of those sectors, the economic structure of the country, and the state of the country's hydromet systems.

the precise nature and magnitude of the improvements in forecasts that would be achieved by the Jamaica investment being evaluated, or about the nature of differences in household preferences for such improvements between the U.S. and Jamaica, it is conservatively assumed that the value to Jamaican households of the improvement achieved by the project is one-half of this value, or USD 1.033 per year. Multiplying by the 881,078 households in Jamaica yields a total annual benefit estimate of USD 910,154. Given projections of population and income growth over time, this estimate can be adjusted to reflect changes in population and income over the time horizon being considered.

The benefits of improved forecasts of routine climate are likely to increase as climate becomes more variable and as the number of enterprises and households rises. However, there has been no attempt yet in the literature to quantify this increase in benefits resulting from more variable climate. Absent estimates of this potential increase and projections 20 years out of per capita income and the number of households in Jamaica, this study assumes that the benefits of improved routine forecasts do not increase in magnitude over time. This is a conservative assumption that underestimates benefits.

To assess the robustness of the conclusions drawn from this analysis, a variety of sensitivity analyses are conducted using:

- alternative values of the transferred benefits estimates;
- alternative time horizons (5, 10, 15 and 20 years);
- alternative real discount rates (4%, 10% and 12%);
- alternative assumptions about the increase in expected losses from extreme meteorological hazards over time due to climate change; and
- alternative assumptions about adaptation by households and enterprises that reduces damages from more frequent, or more intense, extreme meteorological hazards even without the project.

Estimated Benefits of Improved Forecasts of Extreme Hazards and Early Warning Systems

The method used to estimate the benefits of improved forecasts of extreme meteorological hazards and associated early warning systems consists of several steps. First, the average annual economic loss from extreme meteorological hazards in Jamaica over the period 2000–2010 is calculated by sector. These average losses form the basis for predictions of expected losses from extreme events in future years given alternative assumptions about the severity of climate change. The percentage of losses that would be avoided with the project in each sector is then estimated using benefits transfer from existing studies. Expected annual benefits are then calculated by multiplying expected annual sector losses for each year by the estimate of percentage loss avoided for that sector. These sector benefits are aggregated for each year. Finally, the discounted present value of annual benefits is calculated for 5-, 10-, 15-, and 20-year time horizons.

In previous studies of improving hydromet systems, losses from meteorological events were not available, so total damages were used (World Bank 2008, Rogers et al. 2009, 2010). However, since sector losses are available for Jamaica for many recent meteorological events, these values were used to improve the expected damage and benefits estimates. By disaggregating expected damages by sector, it is possible to obtain a more accurate estimate of total expected benefits compared to a case where a single, damage mitigation multiplier were applied. Every sector in the economy is affected by meteorological hazards in a different

way, depending on the type of meteorological event, sector's exposure to weather, and value of its assets. In addition, this study estimates incremental benefits, rather than total benefits, and thus must take into consideration each sector's current ability to mitigate damages, and how effective an improvement in the hydromet early warning system will be. For instance, a hotel in Jamaica's bustling tourism industry is likely to be a sturdy structure that takes little damage from hurricane force winds, regardless of how much advanced warning is given. In contrast, a home of a Jamaican farmer can be much more vulnerable to strong winds and rain, and a small increase in the amount of preparation time for the farmer to fortify his property and move his possessions could result in a significant decrease in damages.

In order to calculate average annual economic losses from extreme meteorological hazards, this analysis considers an 11-year period from 2000–2010.

Extreme hazards are separated into four categories: hurricanes, tropical storms, floods, and droughts. Table A-5 (above) lists the 12 extreme meteorological events that occurred in Jamaica over this period. Accurate estimates of losses do not exist for smaller events that also caused damages during this time period, and they are therefore excluded from this study. The loss estimates in this study are therefore conservative.

The losses (or damages) from each event are decomposed into 10 sectors (see Table A-6). Both direct losses, which are losses of physical and natural capital, as well as indirect losses, which consist mainly of losses of profits, are included. Where available, these decompositions are taken from existing damage assessments. For events for which damage decompositions are not available, as is the case for Hurricane Charley (2004) and the January, 2001 flooding event, this study computes average damage percentages by sector and hazard

Table A-6 Average Damages from Hurricanes by Sector (USD Million, Constant 2010)

Sector	Michelle	Ivan	Dennis & Emily	Wilma	Dean	Percent of Damages
Agriculture, Livestock, & Fisheries	32.33	286.42	22.15	7.11	204.90	25.6%
Emergency Operations	0.07	9.30	0.73	0.07	13.15	1.1%
Environment	-	125.77	1.97	-	2.62	6.0%
Government & Institutions	1.24	27.03	0.03	91.39	17.56	6.4%
Health	0.85	25.40	1.58	1.29	6.51	1.6%
Housing	13.46	373.98	5.82	1.03	130.02	24.3%
Industry	-	108.37	1.39	-	44.27	7.1%
Tourism	2.00	53.29	0.07	-	0.95	2.6%
Transportation	94.21	109.21	122.37	-	44.65	17.1%
Utilities	6.53	121.00	14.63	3.03	30.38	8.1%
Total	150.67	1,239.77	170.75	103.92	495.01	100.0%

Source: Planning Institute of Jamaica (2004, 2005a, 2005b, 2007, 2008, 2010 and 2011), Economic Commission for Latin America and the Caribbean (2001 and 2002), Government of Jamaica Ministry of Agriculture & Fisheries (2010), and United Nations and IDB (2007). Note: Damage totals here may be slightly different from totals in Table A-5 due to missing sectors and small discrepancies in totals between different sources.

type for events for which damage decompositions are available, and then applies these average percentages to events for which damage decompositions are not available. For example, in the case of Hurricane Charley, the average percentage of damages experienced by each sector for all other hurricanes over the period 2000–2010 is calculated. These percentages are then used to decompose total damages from Hurricane Charley. Table A-6 illustrates this procedure. An analogous procedure was used for the other hazard types.

Due to the nature of extreme meteorological hazards, it is impossible to reliably forecast their number and severity years into the future. To derive estimates of expected losses from such hazards in the future, it is assumed that absent changes in climate or the values of assets at risk, expected annual losses from extreme hazards in future years are equal in magnitude to average annual losses over the period 2000–2010. These “baseline loss estimates” are then adjusted to reflect the likely increase in the frequency and intensity of extreme meteorological hazards due to climate change, and the expected increase in the values of assets at risk. The adjustments are made using estimates developed by the Caribbean Catastrophe Risk Insurance Facility (CCRIF).

Using 2009 as a base year, CCRIF estimates that by 2030, economic losses in Jamaica from extreme hazards will increase by 28%, 46%, or 76%, given no changes in climate, “moderate” changes, or “high” changes, respectively (CCRIF 2011). Assuming losses increase linearly, this translates to loss increases of 1.3%, 2.2%, or 3.6% of the baseline, per year. These percentages are applied to the baseline loss estimates described above to obtain estimates of expected future losses.

After computing expected annual losses by sector, for each hazard type, the reduction in losses with

the project are estimated by applying estimates of the percentage loss avoided (PLA) with the project. The percentage loss avoided will generally vary across sectors and hazard types. Because of this heterogeneity, a separate PLA estimate is applied to each sector for each hazard type.

As noted earlier, given the lack of Jamaica-specific information, the analysis relies on PLA estimates derived in similar studies of other regions. The estimates used are drawn from a variety of studies as indicated in Table A-7 below. If only one PLA estimate is available for a given sector/hazard, this value is treated as the *high value*. The low value is taken to be 50% of the high value. Where multiple PLA estimates are available, the lowest value is taken to be the low value, and the highest value is taken to be the high value. The middle value is the average of the two values.

Given the scope of this study and limitations of the existing literature, for some sectors/hazards it was necessary to use PLA estimates from countries that are dissimilar to Jamaica. Specifically, for some sectors/hazards, estimates developed as part of an evaluation of a hydromet system upgrade in Tajikistan (Rogers et al. 2009) are used. The PLA values taken from this study are reduced by 50% to be conservative and to acknowledge differences in the climate, geography, economic structure, and hydromet systems of the two countries.

Multiplying the PLA estimates by the expected annual losses for each sector/hazard yields estimates of the expected annual benefits associated with improving forecasts of extreme hazards and early warning systems. However, these estimates implicitly assume that forecasts are perfectly accurate. In reality, forecasts are subject to uncertainty, especially those made in the tropics. It is therefore important to acknowledge the likelihood of incorrect forecasts and the costs associated

Table A-7 Estimates of Percentage Losses Avoided with Project

Sector	Hurricane/Tropical Storm			Flood			Drought		
	Low	Middle	High	Low	Middle	High	Low	Middle	High
Ag., Livestock, & Fisheries	13.0% ^d	19.5%	26.0% ^a	10.0% ^a	40.0%	70.0% ^a	1.8% ^d	2.7%	3.6% ^c
Emergency Operations	1.8% ^d	2.7%	3.6% ^c	1.8% ^d	2.7%	3.6% ^c	Not Applicable		
Environment	1.8% ^d	2.7%	3.6% ^c	Not Applicable			Data Not Available		
Government & Institutions	7.5% ^d	11.3%	15.0% ^a	5.0% ^a	10.0%	15.0% ^a	Not Applicable		
Health	1.8% ^d	2.7%	3.6% ^c	1.8% ^d	2.7%	3.6% ^c	Not Applicable		
Housing	5.0% ^b	7.5%	10.0% ^b	2.0% ^a	19.0%	36.0% ^a	Not Applicable		
Industry	1.8% ^d	2.7%	3.6% ^c	1.8% ^d	2.7%	3.6% ^c	Data Not Available		
Tourism	1.8% ^d	2.7%	3.6% ^c	1.8% ^d	2.7%	3.6% ^c	Data Not Available		
Transportation	3.0% ^d	4.5%	6.0% ^c	0.0% ^a	6.9%	13.7% ^b	Not Applicable		
Utilities	2.1% ^d	3.2%	4.3% ^c	0.0% ^a	6.9%	13.7% ^b	Data Not Available		

^a Subbiah et al. (2008).

^b Bangladesh Water Development Board (2006).

^c Rogers et al. (2009).

^d Calculated as 50% of the high value.

with these forecasts (e.g., the costs of evacuating people unnecessarily).¹⁷ Following Subbiah et al. (2008), it is assumed that the cost of an incorrect forecast is equal in magnitude to the benefit of a correct forecast.¹⁸ Assuming that average forecast accuracy is 80%, this implies that estimates of expected annual benefits need to be multiplied by a correction factor of 0.6 (= 80% - 20%). This correction factor reflects the fact that benefits of

improved forecasts are only enjoyed 80% of the time, while costs of incorrect forecasts are incurred 20% of the time.

Table A-8 presents estimates of the resulting expected annual benefits for the year 2014 (the first year of the project) assuming, hypothetically, that potential annual benefits are fully realized in that year.

¹⁷ This correction is appropriate given that there is no evidence that the PLA estimates drawn from the literature account for forecast error.

¹⁸ This assumption is made given the absence of information on the costs of incorrect forecasts.

Table A-8 Expected Annual Benefits of Improved Forecasts of Extreme Hazards and Early Warning Systems, Baseline Year (USD Million, Constant 2010)

	Hurricanes	Tropical Storms	Floods	Droughts	Total
Low Value	7.17	1.06	0.24	0.03	8.49
Middle Value	10.75	1.59	1.28	0.04	13.66
High Value	14.34	2.12	2.33	0.05	18.84

Source: Authors. In reality, it is unlikely that potential annual benefits will be fully realized in the first year of the project. Absent further information, it is assumed that 20% of potential annual benefits are realized at the end of the first year of the project, and an additional 20% of benefits are realized at the end of each of the four subsequent years. From 2018 onwards, 100% of potential annual benefits are realized in each year.

Note: These estimates incorporate a correction factor that reflects the costs of inaccurate forecasts.

The estimates in Table A-8 assume that as a result of experiencing more, or more intense, extreme meteorological hazards, Jamaican households and enterprises do not adapt to the hazards, reducing the damages they cause. For example, households can respond to more extreme hurricanes by living in dwellings made of concrete rather than wood. It is unclear to what extent Jamaican households and enterprises have already undertaken such adaptation measures, and to what extent these measures would increase over the next 5–20 years (the time horizon of this analysis). The available empirical literature is quite silent on this issue. A recent study of adaptation to tropical cyclones using data from across the world (Hsiang and Narita 2012) indicates that only about 3% of the losses from incremental changes of countries' current tropical cyclone climates are "adapted away" in the long run. As part of the sensitivity analyses conducted here, a scenario is considered that assumes that damages in each year are reduced by 10% due to such adaptation measures. (Accordingly, the benefits estimates presented in Table A-8 would each be reduced by 10%). The assumption that damages in all sectors are reduced by 10% in each year due to adaptation is a liberal one given the aforementioned long-run estimate of 3% for a global

cross-section of countries. The assumption guards against potentially overstating the benefits of the proposed project as a result of ignoring the consequences of adaption measures that are undertaken even in the absence of the project.

Estimated Benefits of Improved Forecasts of Routine Climate

Benefits to Enterprises. A large number of studies have been conducted examining the value of climate forecasts to enterprises in various climate-sensitive sectors of the economy.¹⁹ A small subset of these studies examines the value of improved forecasts. The results of these studies are typically very setting specific and provide estimates that are difficult to transfer to other settings. However, a few studies do provide estimates that can be transferred with some degree of confidence. But these studies cover only a very small number of climate-sensitive sectors. Specifically, Adams et al. (2003) report increases of 0.3–2% in the value of crops produced in five Mexican states as a result of improved ENSO forecasts. Houston et al. (2004) and Yeh et al. (1982) report increases of 0.04–0.1% in GDP from the U.S. electricity, gas and sanitary services sector from improved short-term

¹⁹ See Houston et al. (2004) and Weiher et al. (2005) for a compendium of estimates.

Table A-9 Expected Annual Benefits of Improved Forecasts of Routine Climate for Selected Sectors (USD Million, Constant 2010)

	Low Value	Middle Value	High Value
Agriculture, Forestry & Fishing	1.94	7.43	12.93
Electricity and Water Supply	0.19	0.33	0.46

Source: Authors.

Note: Estimates were calculated only for sectors for which benefits transfer was feasible.

forecasts of temperature and long-term forecasts of precipitation.²⁰

Table A-9 presents the results of applying these estimates to the corresponding sectors of the Jamaican economy. The table shows the benefits in these sectors for the year 2014 assuming, hypothetically, that potential annual benefits are fully realized in this year. When incorporating these benefits in the calculation of project benefit-cost ratios, it is assumed that the benefits are realized in 20-percentage-point increments over the first five years of the project (as is done for the benefits associated with extreme hazards). It bears emphasis that these estimates capture only a fraction of the benefits to enterprises of improved forecasts of routine climate given the large number of sectors omitted.

Benefits to Households. A number of survey-based studies have estimated the benefits to households of weather forecasts.²¹ Far less common are studies that examine the benefits to households of *improvements* in forecast quality, which is the relevant measure for this analysis. A notable exception

is a very detailed 2002 study by Lazo and Chestnut that estimates the value to U.S. households of improving the quality of one-day and multi-day weather forecasts from their then-current levels to maximum possible levels.²² Lazo and Chestnut find that, on average, U.S. households were willing to pay USD 12 to USD 17 per year to improve the quality of the forecasts, with a best estimate of USD 16. To transfer this benefit estimate to Jamaica, the USD 16 per household estimate is first adjusted for differences in per capita income between the U.S. and Jamaica, and then expressed in 2010 US dollars. Absent information about the precise nature and magnitude of the improvements in forecasts that would be achieved by the project being evaluated, or about the nature of differences in household preferences for such improvements between the U.S. and Jamaica, it is conservatively assumed that the value to Jamaican households of the improvement achieved by the project is one-half of this value, or USD 1.033 per year.²³ Multiplying by the 881,078 households in Jamaica (Statistical Institute of Jamaica 2011) yields a total annual benefit estimate of USD 910,154. As with the benefits to enterprises, these benefits to households are

²⁰ These values are in line with values employed by Hallegatte (2012) to estimate the benefits of improved hydromet services in developing countries. He assumes value added gains between 0.1% and 1% in weather-sensitive.

²¹ See Lazo et al. (2009) and Houston et al. (2004) for surveys of these studies, which focus on developed countries, and World Bank (2008) for a study conducted in Azerbaijan and Serbia.

²² The accuracy of one-day forecasts was assumed to increase from 80% to 95%; for multiday forecasts, 14-day forecasts were assumed to become as accurate as current 5-day forecasts. The study focused on valuing improved forecasts of day-to-day weather rather than extreme events, hence double counting of benefits should not be an issue.

²³ This value is plausible given the results of studies conducted in Azerbaijan and Serbia, countries with GDP per capita similar to Jamaica's, indicating that households are willing to pay USD 12–16 per year to support national hydromet services—a total, rather than incremental, benefit measure (World Bank 2008, p 71).

assumed to be realized in 20-percentage-point increments over the five years of the project.

Estimated Costs of the Project

The total cost of implementing the project is USD 6.8MM. It is conservatively assumed that this entire cost is borne at the beginning of 2014. Annual costs of maintaining and operating the improved hydro-meteorological system are not anticipated to increase significantly. The new RADAR will eliminate the relatively high maintenance costs associated with the current 20-year old system, and the installation of automated monitoring and information transmission systems will free up staff resources for other purposes.

Benefit-Cost Ratios for the Project

Given the above estimates and assumptions, benefit/cost ratios for the project are calculated for 5-, 10-, 15-, and 20-year time horizons, using discount rates of 4%, 10%, and 12%. These benefit-cost ratios are presented in Tables A-10a-10c for the three different climate change scenarios, and for the low, middle and high estimates of benefits.

Examining the tables it can be seen that the benefit-cost ratio is well above one in all cases. For the most conservative case—no climate change, “low” benefits estimates, 5-year time horizon and 12% discount rate—the benefit-cost ratio is 3.5 (left-hand panel of Table A-10a). The ratio rises to 10.7 when the project time horizon is extended to 20 years, with all else remaining constant. For the “middle” benefits estimates (middle panel of Table A-10a), the benefit-cost ratios are nearly twice as high, and for the “high” benefits estimates they are approximately two-and-a-half times higher. Comparing the values in Tables 7a, 7b, and 7c reveals modest variations in benefit-cost ratios across the three climate change scenarios. This can be attributed to the relatively short time

horizons being considered in the context of climate change.

The benefit-cost ratios for the “middle” and “high” benefits estimates may seem unusually large, however previous studies of the returns to investments in hydromet systems have, in a number of cases, yielded very large benefit-cost ratios. Subbiah et al. (2008) report a benefit-cost ratio of 40 for development of a cyclone early warning system in Bangladesh, and benefit-cost ratios of 0.9 to 558 for flood early warning systems in Sri Lanka and Bangladesh, respectively. An evaluation of post-civil-war investment in Mozambique’s hydromet system estimated a benefit-cost ratio of 70 (World Bank 2008, p. 5). The large benefit-cost ratios obtained in this study reflect, in large part, Jamaica’s high susceptibility to extreme meteorological hazards.

The benefit-cost ratios presented in Tables A-10a-c assume that Jamaican households and enterprises do not adapt to more frequent or more intense extreme meteorological hazards, reducing the damages they cause in the future, even in the absence of the proposed project. As discussed in paragraph 36, to evaluate the sensitivity of the benefit-cost ratios to such adaptation, an alternative scenario is considered in which the estimated expected damages from extreme meteorological events are reduced by 10% in each year as a result of adaptation. The benefit-cost ratios for this alternative scenario are presented in Tables A-11a-c. As can be seen, the ratios are only reduced by a modest amount, and each is still well above one. For the most conservative case considered in paragraph 42, the benefit-cost ratios diminish from 3.5 to 3.2 and from 10.7 to 9.9.

Additional Observations and Caveats

The analysis presented is best viewed as an effort to develop first-cut estimates of the benefits and

costs of the proposed project. A number of observations about the estimates and about the appropriate time horizon for the analysis are worth noting.

Given the nature of the equipment being installed as part of the project, a 20-year time horizon may well be too long. The current 20-year-old RADAR system is considered outdated and unreliable, and this will likely be true of the new RADAR system 20 years hence. The 10- and 15-year horizons are likely to be more reasonable.

The incremental costs, both capital and recurring, of upgrading the hydromet system itself are likely to be adequately captured by the project implementation costs. As noted above, the new equipment will likely result in reduced maintenance costs and in staff resources being freed up for other purposes.

However, the costs of ensuring that the benefits of improved forecasts are fully realized may not be adequately captured in the project implementation costs. For the benefits to be fully realized, citizens, enterprises, civil society organizations, and government agencies must be effectively informed about the forecasts in a timely manner. No less importantly, they must be educated on how best to interpret and make use of the improved forecasts. This is not an inconsequential task, especially given that forecasts are imperfect.

Another cost that is only implicitly considered in the analysis is the cost of additional measures taken to reduce losses from extreme meteorological events given improved forecasts. Examples are the costs of building additional emergency shelters or the costs borne by homeowners of protecting their dwellings given earlier warnings of storms. The

analysis assumes that these avoidance (or prevention) costs are netted out of the percentage loss avoided estimates that are employed. It is unclear from the literature whether this is always true. In some cases the costs may be small enough, in relative terms, that omitting them is inconsequential.

As has been emphasized, only a subset of the benefits of the project has been quantified. This is especially true of the benefits to enterprises of improved forecasts of routine climate. The benefits are only quantified for two sector aggregates (agriculture, forestry and fishing, and electricity and water supply), but there are other climate-sensitive sectors of the Jamaican economy that are likely to reap significant benefits from improved forecasts.

One class of benefits that has not been mentioned is a reduction in deaths as a result of improved warnings of extreme events. Given the relatively small number of deaths due to extreme meteorological events in Jamaica, benefits of this type are likely to be relatively small.²⁴

An important component of the project is developing higher-resolution climate change scenarios for Jamaica in order to enhance climate-resilient planning and decision-making. The benefits associated with this effort are very difficult to quantify, but they are likely to be substantial. It is not implausible to posit that the unquantified benefits of the project outweigh any unquantified costs of the project.

Key Conclusions

The first-cut estimates of the benefits and costs of the proposed project indicate that in all of the alternative scenarios considered, the benefits of the project exceed the costs by a considerable margin.

²⁴ The largest number of deaths that have been reported in this century are for Hurricane Ivan, with 17 direct deaths and 14 indirect deaths (Planning Institute of Jamaica 2004).

Even with the most conservative set of assumptions, including an assumption that no climate change takes place and that the benefits of the project only last for 5 years, the benefit-cost ratio for the project is 3.2, i.e., the expected benefits from the project are 3.2 times higher than the expected costs of USD 6.88 MM (in present value terms).

The benefits of the project are experienced by a very broad swath of the Jamaican economy, and stem in large part from a reduction in expected damages from extreme meteorological hazards, such as storms, hurricanes and floods. Given the onerous debt burden carried by the Jamaican government (currently at 140% of GDP), it is unlikely that the investments embodied in

the project will be made by the Jamaican government on its own.

Extending the period over which the benefits of the project accrue from 5 years to a more plausible 10 years or 15 years, increases the benefit-cost ratio to 6.6 and 8.7, respectively. Assuming no climate change takes place and otherwise maintaining the most conservative set of assumptions.

Thus, the estimates derived indicate that the project is economically desirable even in the absence of climate change. Acknowledging climate change, and the concomitant increase in the frequency or severity of extreme meteorological events, simply increases the benefit-cost ratios for the project, increasing its estimated economic desirability.

Table A-10a Project Benefit-Cost Ratios Assuming No Climate Change for Alternative Benefits Estimates

Low				Middle				High			
		Discount Rate				Discount Rate				Discount Rate	
Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%
5 Year	4.5	3.7	3.5	5 Year	8.7	7.1	6.7	5 Year	12.9	10.6	9.9
10 Year	11.1	8.0	7.2	10 Year	21.4	15.3	13.8	10 Year	31.7	22.6	20.4
15 Year	16.8	10.7	9.4	15 Year	32.3	20.6	18.0	15 Year	47.7	30.4	26.6
20 Year	21.7	12.5	10.7	20 Year	41.5	24.0	20.4	20 Year	61.3	35.4	30.2

Source: Authors.

Table A-10b Project Benefit-Cost Ratios Assuming Moderate Climate Change for Alternative Benefits Estimates

Low				Middle				High			
		Discount Rate				Discount Rate				Discount Rate	
Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%
5 Year	4.6	3.8	3.5	5 Year	8.8	7.2	6.8	5 Year	13.1	10.7	10.1
10 Year	11.5	8.2	7.4	10 Year	22.0	15.7	14.1	10 Year	32.5	23.2	20.9
15 Year	17.6	11.1	9.7	15 Year	33.4	21.2	18.5	15 Year	49.3	31.3	27.4
20 Year	22.9	13.1	11.1	20 Year	43.4	24.9	21.2	20 Year	63.9	36.7	31.2

Source: Authors.

Table A10-c Project Benefit-Cost Ratios Assuming High Climate Change for Alternative Benefits Estimates

Low				Middle				High			
		Discount Rate				Discount Rate				Discount Rate	
Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%
5 Year	4.7	3.9	3.6	5 Year	9.0	7.4	6.9	5 Year	13.4	10.9	10.2
10 Year	12.1	8.6	7.7	10 Year	22.9	16.3	14.6	10 Year	33.7	24.0	21.6
15 Year	18.8	11.8	10.3	15 Year	35.3	22.3	19.4	15 Year	51.9	32.8	28.6
20 Year	24.8	14.0	11.9	20 Year	46.5	26.4	22.4	20 Year	68.1	38.8	32.9

Source: Authors.

Table A-11a Project Benefit-Cost Ratios Assuming No Climate Change and Reduced Damage Due to Mitigation for Alternative Benefits Estimates

Low				Middle				High			
		Discount Rate				Discount Rate				Discount Rate	
Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%
5 Year	4.2	3.4	3.2	5 Year	8.2	6.7	6.3	5 Year	12.2	10.0	9.4
10 Year	10.3	7.4	6.6	10 Year	20.1	14.3	12.9	10 Year	29.8	21.3	19.2
15 Year	15.6	9.9	8.7	15 Year	30.2	19.3	16.8	15 Year	44.9	28.6	25.0
20 Year	20.1	11.6	9.9	20 Year	38.9	22.4	19.1	20 Year	57.6	33.3	28.4

Source: Authors.

Table A-11b Project Benefit-Cost Ratios Assuming Moderate Climate Change and Reduced Damage Due to Mitigation for Alternative Benefits Estimates

Low				Middle				High			
		Discount Rate				Discount Rate				Discount Rate	
Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%
5 Year	4.3	3.5	3.3	5 Year	8.3	6.8	6.4	5 Year	12.3	10.1	9.5
10 Year	10.6	7.6	6.8	10 Year	20.6	14.7	13.2	10 Year	30.5	21.8	19.6
15 Year	16.2	10.3	9.0	15 Year	31.3	19.9	17.3	15 Year	46.3	29.5	25.7
20 Year	21.2	12.1	10.3	20 Year	40.6	23.3	19.8	20 Year	60.0	34.5	29.3

Source: Authors.

Table A-11c Project Benefit-Cost Ratios Assuming High Climate Change and Reduced Damage Due to Mitigation for Alternative Benefits Estimates

Low				Middle				High			
		Discount Rate				Discount Rate				Discount Rate	
Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%	Time Horizon	4%	10%	12%
5 Year	4.4	3.6	3.3	5 Year	8.5	6.9	6.5	5 Year	12.6	10.3	9.6
10 Year	11.1	7.9	7.1	10 Year	21.4	15.2	13.7	10 Year	31.7	22.5	20.3
15 Year	17.3	10.9	9.5	15 Year	33.0	20.8	18.2	15 Year	48.7	30.8	26.8
20 Year	22.9	12.9	11.0	20 Year	43.3	24.0	20.9	20 Year	63.8	36.3	30.9

Source: Authors.

References

- Adams, Richard M., Kelly J. Bryant, Bruce A. McCarl, David M. Legler, James O'Brien, Andrew Solow, and Rodney Weiher. 1995. "Value of Improved Long-Range Weather Information." *Contemporary Economic Policy* 13(3):10–19.
- Adams, Richard M., Laurie L. Houston, Bruce A. McCarl, Mario L. Tiscareno, Jaime Matus, and Rodney F. Weiher. 2003. "The Benefits to Mexican Agriculture of an El Niño-southern Oscillation (ENSO) Early Warning System." *Agricultural and Forest Meteorology* 115(3–4): 183–94.
- Advisors to the High Level Group on Infrastructure Charging (AHLG). 1999. Calculating Transport Accident Costs. April 27 1999.
- American Meteorological Society. 2011. Strategic Implementation Plan for Generating and Communicating Forecast Uncertainty Information.
- Anaman, K. A., Lellyett, S. C., Drake, L., Leigh, R. J., Henderson-Sellers, A., Noar, P. F., Sullivan, P. J., and Thampapillai, D. J. "Benefits of meteorological services: evidence from recent research in Australia." *Meteorological Applications*, 5: 103–115, 1998.
- ASEAN. 2012. Advancing Disaster Risk Financing and Insurance in ASEAN Member States: Framework and Options for Implementation, GFDRR, ISDR, Volume 2: Technical Appendices.
- Bangladesh Water Development Board. 2006. Bangladesh: Early Warning Systems Study. Ministry of Water Resources, Government of the People's Republic of Bangladesh.
- Blench, R. 1999. Seasonal Climatic Forecasting: Who Can Use It and How Should It Be Disseminated? *Natural Resource Perspectives*, No. 47. London: Overseas Development Institute.
- Carsell, Kim M., Nathan D. Pingel, and David T. Ford. 2004. Quantifying the Benefit of a Flood Warning System. *Natural Hazards Review*, August, pp. 131–140.
- Carriquiry, Miguel A., and Daniel E. Osgood. 2012. "Index Insurance, Probabilistic Climate Forecasts, and Production." *Journal of Risk and Insurance* 79(1): 287–300.
- Diamond, Peter A., and Jerry A. Hausman. 1994. "Contingent valuation: Is some number better than no number?" *The Journal of Economic Perspectives* 8(4): 45–64.
- Dutton, J. 2002. Opportunities and Priorities in a New Era for Weather and Climate Services. American Meteorological Society.
- Economic Commission for Latin America and the Caribbean. Jamaica: Assessment of the Damage Caused by Flood Rains and Landslides in Association with Hurricane Michelle, October 2001. 7 Dec. 2001.
- Economic Commission for Latin America and the Caribbean. Jamaica: Macro-socio-economic Assessment of the Damage Done by Flood Rains and Landslides May 2002. 22 July 2002.
- Energy Institute. 2012. Energy Efficiency Policy and Measures in Croatia, Energy Institute Hrvoje Pozar, Zagreb, available at: http://www.odyssee-indicators.org/publications/PDF/croatia_nr.pdf.
- Freebairn, J.W. and Zillman, J.W. "Economic benefits of meteorological services." *Meteorological Applications*, 9(1): 33–44, 2002.

- Global Facility for Disaster Risk Reduction (GFDRR). 2009. Improving Weather, Climate, and Hydrological Services Delivery in Central Asia (Kyrgyz Republic, Republic of Tajikistan, and Turkmenistan).
- Global Facility for Disaster Risk Reduction (GFDRR). 2010. Disaster Risk Management in Latin America and the Caribbean Region: GFDRR Country Notes, Jamaica.
- Golnaraghi, M. 2009. The role of the WMO in natural hazard awareness and disaster risk reduction education, presentation at the 2nd Conference of the OECD International Network on the Financial Management of Large-scale Catastrophes, September 24, Bangkok, Thailand.
- Government of Jamaica Ministry of Agriculture & Fisheries. 2010. Introducing Innovative Agriculture Weather Risk Management Mechanisms for Small Farmers. Pre-feasibility Assessment for St. Elizabeth and Portland Parishes. 17 May 2010.
- Hallegatte, S. 2012. A Cost Effective Solution to Reduce Disaster Losses in Developing Countries.
- Han, Xiaoli, and Bingsong Fang. 2000. Four Measures of Transportation's Economic Importance. *Journal of Transportation and Statistics* 3, no. 1: 15–30.
- Houston, Laurie L., Richard M. Adams and Rodney F. Weiher. 2004. The Economic Benefits of Weather Forecasts: Implications for Investments in Rusohydromet Services. Report prepared for the World Bank under NOAA contract OG 1330-04-SE-0052.
- Hsiang, S.M. and D. Narita. 2012. "Adaptation to Cyclone Risk: Evidence from the Global Cross-Section." *Climate Change Economics*, 3(2).
- Jewson, Stephen, and Rodrigo Caballero. 2003. The use of weather forecasts in the pricing of weather derivatives. *Meteorological Applications* 10(4): 377–389.
- Katz, Richard W., Allan H. Murphy, and Robert L. Winkler. 1982. "Assessing the value of frost forecasts to orchardists: a dynamic decision-making approach." *Journal of Applied Meteorology* 21(4): 518–531.
- Katz, Richard W. and Jeffrey K. Lazo. 2011. "Economic Value of Weather and Climate Forecasts," in *The Oxford Handbook of Economic Forecasting*, edited by Clements, Michael Peter, and David F. Hendry. Oxford University Press.
- Katz, R.W. and A.H. Murphy. 2005. *Economic Value of Weather and Climate Forecasts*. Cambridge University Press.
- Kokko, H., and Vaisala, O. 2005. Integrated Hydro-meteorological Monitoring Solutions and Network Management. In 21st International Conference on Interactive Information Processing Systems (IIPS) for Meteorology, Oceanography, and Hydrology. Helsinki, Finland.
- Lazo, Jeffrey K. and Lauraine G. Chestnut. 2002. Economic Value of Current and Improved Weather Forecasts in the U.S. Household Sector. Report prepared for Office of Policy and Strategic Planning, National Oceanic and Atmospheric Administration by Stratus Consulting Inc.
- Lazo, Jeffrey K., Robert S. Raucher, Thomas J. Teisberg, Carolyn J. Wagner, and Rodney F. Weiher. 2008. "Primer on Economics for National Meteorological and Hydrological Services."
- Lazo, J.K., Morss, R.E., Demuth, J.L., 2009: 300 Billion served: Sources, perceptions, uses, and values

of weather forecasts. *Bulletin of the American Meteorological Society*, 90, 785–798.

Lazo, Jeffrey K. and Donald M. Waldman. 2011. “Valuing Improved Hurricane Forecasts.” *Economics Letters* 111:43–46.

Leviäkangas, Pekka, Raine Hautala, Jukka Räsänen, Risto Öörni, Sanna Sonninen, Martti Hekkanen, Mikael Ohlström, Ari Venäläinen and Seppo Saku. 2007. Benefits of Meteorological Services in Croatia VTT TIEDOTTEITA, 2420.

Macauley, Molly K. 1997. “Some Dimensions of the Value of Weather Information: General Principles and a Taxonomy of Empirical Approaches,” *Proceedings of the Workshop on the Social and Economic Impacts of Weather*, Boulder, Colorado, April 2–4, 1997.

Malik, Arun S., and Stephen C. Smith. 2012. “Adaptation To Climate Change In Low-Income Countries: Lessons From Current Research And Needs From Future Research.” *Climate Change Economics* 3(2).

Maurer, Edwin P., and Dennis P. Lettenmaier. 2004. “Potential Effects of Long-Lead Hydrologic Predictability on Missouri River Main-Stem Reservoirs.” *Journal of Climate* 17(1): 174–186.

Meyer, R. 2006. Why we Under Prepare for Hazards. In *On risk and disaster: Lessons from hurricane Katrina*, edited by R.J. Daniels et al. University of Pennsylvania Press, pp. 153–173.

Mjelde, James W., Steven T. Sonka, Bruce L. Dixon, and Peter J. Lamb. 1988. “Valuing forecast characteristics in a dynamic agricultural production system.” *American Journal of Agricultural Economics* 70(3): 674–684.

Mjelde, James. W., Steven T. Sonka and Derrell S. Peel. 1989. The Socioeconomic Value of Climate and Weather Forecasting: A Review. Research Report 89–01, Midwestern Climate Center, Climate & Meteorology Section, Illinois State Water Survey.

Millner, A. 2009. “What is the true value of forecasts?.” *Weather, Climate, and Society* 1(1): 22–37.

Mills, B. 2010. “Societal and Economic Research and Applications (SERA) for Adaptation to Atmospheric Hazards in Canada.”

Morss, Rebecca E., Olga V. Wilhelmi, Mary W. Downton, and Eve Gruntfest. 2005. “Flood risk, uncertainty, and scientific information for decision making: lessons from an interdisciplinary project.” *Bulletin of the American Meteorological Society* 86(11): 1593–1601.

Morss, Rebecca E., Jeffrey K. Lazo, Barbara G. Brown, Harold E. Brooks, Philip T. Ganderton, and Brian N. Mills. 2008. “Societal and economic research and applications for weather forecasts.” *Bulletin of the American Meteorological Society* 89(3): 335–346.

National Research Council. 1998. *The Atmospheric Sciences Entering the Twenty-First Century*. National Academy Press.

National Research Council 2006. *Completing the Forecast: Characterizing and Communicating Uncertainty for Better Decisions Using Weather and Climate Forecasts*. National Research Council, Washington, D.C.

Nichols, J.M. 1996. “Economic and social benefits of climatological information and services: a review of existing assessments.” *World Meteorological*

Organization, Geneva, Switzerland. WMO/TD-No. 780. 38.

Pielke, R., and Richard E. Carbone. 2002. "Weather impacts, forecasts, and policy." *Bulletin-American Meteorological Society* 83(3): 393–406.

Pollner, John, Jolanta Kryspin-Watson, and Sonja Nieuwejaar (undated). Disaster Risk Management and Climate Change Adaptation in Europe and Central Asia. Global Facility for Disaster Risk Reduction (GFDRR).

Quirk, James P. 1976. *Intermediate Microeconomics*. Vol. 1. Chicago, Ill: Science Research Associates.

Regnier, E. 2008. "Public evacuation decisions and hurricane track uncertainty." *Management Science* 54(1): 16–28.

Rogers, D, M. Smetanina, and V. Tsirkunov. 2009. Improving Weather, Climate and Hydrological Services Delivery in Republic of Tajikistan.

Rogers, D, M. Smetanina, and V. Tsirkunov. 2010. Improving Weather, Climate and Hydrological Services Delivery in Central Asia.

Rogers, D. and Vladimir Tsirkunov. 2010. Costs and Benefits of Early Warning Systems. Global Assessment Report on Disaster Risk Reduction. ISDR and World Bank.

Sene, Kevin. 2010. *Hydrometeorology: forecasting and applications*. Springer.

Simmons, Kevin M. and Daniel Sutter. 2005. WSR-88D Radar, Tornado Warnings, and Tornado Casualties. *Weather and Forecasting*, 20: 301–310.

Smith, D. A. 2008. "The Social Value of Meteorological Information." SERA North: Economics of Weather, Climate and Climate Change.

Smith, K. 1990. Weather Sensitivity of Rail Transport. Proceedings of the WMO Technical Conference Geneva: WMO. Pp. 236–244.

Solow, Andrew R., Richard F. Adams, Kelly J. Bryant, David M. Legler, James J. O'Brien, Bruce A. McCarl, William Nayda, and Rodney Weiher. 1998. "The value of improved ENSO prediction to US agriculture." *Climatic Change* 39(1): 47–60.

South Eastern Europe Disaster Risk Mitigation and Adaptation Programme (SEEDRMAP). 2008. Strengthening the Hydrometeorological Services in South Eastern Europe.

Stern, Paul C. and William E. Easterling (eds.). 1999. *Making Climate Forecasts Matter*. Panel on the Human Dimensions of Seasonal-to-Interannual Climate Variability; Committee on the Human Dimensions of Global Change, National Research Council.

Stewart, Thomas R., Richard W. Katz, and Allan H. Murphy. 1984. "Value of weather information: A descriptive study of the fruit-frost problem." *Bulletin of the American Meteorological Society* 65(2): 126–137.

Suarez, Pablo, James Hansen, Miguel Carriquiry, Ashok Mishra, and Daniel Osgood. 2008. "Integrating seasonal forecasts and insurance for adaptation among subsistence farmers: The case of Malawi." *World Bank Policy Research Working Paper Series*.

Subbiah, A.R., Lolita Bildan and Ramraj Narasimhan. 2008. Background Paper on Assessment of

the Economics of Early Warning Systems for Disaster Risk Reduction. Report prepared for The World Bank Group, Global Facility for Disaster Reduction and Recovery (GFDRR).

Teisberg, Thomas J. and Rodney F. Weiher. 2009. Background Paper on the Benefits and Costs of Early Warning Systems for Major Natural Hazards. Report prepared for The World Bank Group Global Facility for Disaster Reduction and Recovery (GFDRR).

Thornes, J. E. and B. Davis. 2002. Mitigating the Impact of Weather and Climate on Railway Operations in the UK. Proceedings of the 2002 ASME/IEEE Joint Rail Conference, Washington DC, April 23–25, 2002.

Tiehallinto, 2006. Tieliikenteen ajokustannusten yksikköarvot 2005 [Unit values of external costs in road transport 2005]. Helsinki: Finnish Road Administration.

Tsirkunov, V. 2011. Strengthening Weather and Climate Service Delivery in the Developing Countries –First Lessons and Future Challenges. World Bank GFDRR. Available on http://www.gfdr.org/sites/gfdr.org/files/WCDIS_First_Lessons_Future_Challenges.pdf.

UNISDR. 2008. Strengthening the Hydro-Meteorological Services in South Eastern Europe, available on <http://www.unisdr.org/we/inform/publications/7650>.

UNISDR. 2009. Mitigating the Adverse Financial Effects of Natural Hazards on the Economies of Central Asia, A Study of Catastrophe Risk Financing Options.

United Nations and IDB. Information on Disaster Risk Management Case Study of Five Countries: Jamaica. 13 Dec. 2007.

Webster, Peter J. 2012. Improve Weather Forecasts for the Developing World. *Nature* 493(3): 17–19.

Weiher, Rodney, Laurie Houston, and Richard Adams. 2005. Socio-Economic Benefits of Climatological Services. United States Contribution to the Update of WMO TN 145-No.424. Draft.

World Bank. 2008. Weather and Climate Services in Europe and Central Asia. Working Paper No. 151.

World Bank. 2012. Project Appraisal Document, Modernization of the National Meteorological Services for Improved Climate Adaption Project (United Mexican States).

World Bank/GFDRR. 2013. Building Resilience: Integrating Climate and Disaster Risk into Development, The World Bank Group Experience.

WRMA. 2002. “The Weather Risk Management Industry: Survey Findings for November 1997 to March 2001”, Report prepared for Weather Risk Management Association by PriceWaterhouseCoopers, Available at <http://www.wrma.org>.

Zillman, J.W. and Freebairn, J.W. (2001). Economic framework for the provision of meteorological services. *WMO Bulletin* 50 (3).

Zeng, Lixin. 2000. Pricing Weather Derivatives, *Journal of Risk Finance* 1(3):72–78.

Publications from the LCSEN Environment & Water Resources Occasional Paper Series

- Developing Offsets for Loss of Biodiversity: Experience from Colombia (2014)
- Environmental Health Costs in Colombia: The changes from 2002 to 2010 (2014)
- Metropolitan Pollution Management in Brazil: Taking Stock (2014)
- Developing a Program for Contaminated Site Management in Low and Middle Income Countries (2014)
- Restoring the Coastal Environment in Cartagena, Colombia (2014) (Available in English and Spanish)
- Climate Change Impacts on Water Resources and Adaptation in the Rural Water Supply and Sanitation Sector in Nicaragua (2013) (Available in English and Spanish)
- Climate Change Impacts on Water Resources Management: Adaptation Challenges and Opportunities in Northeast Brazil (2013)
- El Futuro del Riego en el Perú: Desafíos y Recomendaciones (Volumen I: Informe de Síntesis y Volumen II : Informe Principal) (2013)
- Empowering Women in Irrigation Management: The Sierra in Peru (2012)
- Environmental Health in Nicaragua: Addressing Key Environmental Challenges (Originally Published in 2010, Republished in 2012) (Available in English and Spanish)
- Expanding Financing for Biodiversity Conservation: Experiences from Latin America and the Caribbean (2012) (Available in English and Spanish)
- Overcoming Institutional and Governance Challenges in Environmental Management. Case Studies from Latin America and the Caribbean Region (2012)
- Policy and Investment Priorities to Reduce Environmental Degradation of the Lake Nicaragua Watershed (Cocibolca) (Originally Published in 2010, Republished in 2012) (Available in English and Spanish)
- Uncertain Future, Robust Decisions; The Case of Climate Change Adaptation in Campeche, Mexico (2012)

To find copies of these publications, please visit our web site:
<http://www.worldbank.org/LAC/LCSENSeries>

**LATIN AMERICA &
CARIBBEAN REGION**

**Environment &
Water Resources**

OCCASIONAL PAPER SERIES



WORLD BANK GROUP

1818 H Street, NW
Washington, DC 20433 USA
Telephone: 202 477-1234
Facsimile: 202 477-6391
Internet: www.worldbank.org
E-mail: feedback@worldbank.org